Core-Sparse Monge Construction for Full Optimal Binary Search Trees*

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Abstract

We give an exact construction for optimal binary search trees with arbitrary nonnegative integer weights on keys and on the gaps between them. For total mass $W = O(n)$, a reduction to the core-sparse Monge product of Gawrychowski, Gorbachev, and Kociumaka takes $O(n^{3/2}\log^{5/2}(n+1))$ arithmetic operations and returns the complete tree. Terminal-penalty layers preserve Monge structure; a mixed-weight subtree bound controls the number of layers, and a dyadic boundary argument bounds every intermediate core. Centering around separate constant key and gap baselines gives $O(n(1+\sqrt{R})\log^3(n+1))$ arithmetic work when their total nonnegative deviation R is polynomial in n.

A common lower-bound certificate also gives faster exact construction on specified families. We give the certificate once and derive the family guarantees from it. Counterexamples delimit bounded-window relaxations, endpoint-response composition, explicit density representations, and additive separable transfers. A contiguous odd ramp with total mass exactly n has $\Omega(n^{3/2})$ intermediate core entries with explicitly computed densities. Thus the core exponent is necessary, up to a remaining logarithmic factor. A separate positive ruler family has quadratic support at deviation $R = n^2$. An adversarial integer-mass search and independent exact checks are reported in an appendix. The unrestricted subquadratic problem remains open.

1 Model and prior work

Weights are attached to fixed positions in key order; they need not be sorted. Keys have fixed inorder positions $0,\dots,n-1$ and nonnegative weights $p_0,\dots,p_{n-1}$. The $n+1$ gaps have weights $q_0,\dots,q_n$. A key at depth d contributes $p(d+1)$; a gap at depth d contributes qd. Charging an extra comparison at each gap adds the tree-independent constant $\sum q_i$.

For a half-open key interval $[i,j)$, write

$$W(i,j) = \sum_{r=i}^{j-1} p_r + \sum_{r=i}^{j} q_r, \qquad C(i,i) = 0.$$

The exact recurrence is

$$C(i,j) = W(i,j) + \min_{i\le r<j}\{C(i,r) + C(r+1,j)\}.$$

Knuth's monotone-root optimization computes the table in $O(n^2)$ time [1]. The central obstacle here is the number of states, not merely the cost per state. We retain both kinds of weights throughout.

Larmore gives subquadratic exact construction under restrictions on long runs of very low frequency items, together with an approximation tradeoff [2]. Karpinski, Larmore, and Rytter combine exact small intervals with a bounded top-level construction [3]. All running-time bounds below count arithmetic operations and comparisons. They are not bit-complexity bounds for unbounded integer weights.

For uniform key weights a and uniform gap weights b, the external path length equals $I + 2n$, where I is the internal path length. Thus

$$C(T) = (a + b)I + (a + 2b)n.$$

Every minimum-internal-path-length tree is optimal. Such a tree fills every level above the last and is constructible in $O(n)$ time. When $a = b = 0$, every tree is optimal. Later baseline arguments use this exact identity.

2 The main Monge construction

Let $n \geq 1$, let all key and gap weights be nonnegative integers, and put $W = \sum p_i + \sum q_i$ and $L = \lceil \log_2(n + 1) \rceil$. Write $f(W)$ for the first index with $F_{f(W)} > W$, where $F_0 = 0, F_1 = 1$. Set

$$H = \min\{n, f(W) + L\}, \qquad M = (H + 1)W + 1.$$

If every key weight is positive, $H = \min\{n, f(W)\}$ suffices.

Theorem 1 (Integer-mass sparse-product reduction)**.** *Using the core-sparse Monge multiplication and witness algorithms of [10], a full optimal BST can be constructed in*

$$O\Big(H\left[n + \min\{n^2, n\sqrt{WL}\}\right] \log(n + 1)\Big)$$

arithmetic operations. Retaining the witness structures uses $O(H[n + \min\{n^2, n\sqrt{WL}\}])$ *words. In particular,* $W = O(n)$ *gives* $O(n^{3/2} \log^{5/2}(n + 1))$ *exact construction. Intermediate integers have* $O(\log(n + 1) + \log(W + 1))$ *bits.*

The dependence on numerical mass is essential to this statement. The linear-mass case fits ordinary $O(\log n)$-bit arithmetic, but arbitrary large integer weights do not give an unrestricted strongly subquadratic bound. Knuth remains a fallback when the displayed bound is unfavorable.

2.1 A full mixed-weight height bound

Lemma 2 (A sufficient mixed-weight subtree bound)**.** *If S is an internal subtree at edge depth d in an optimal full OBST of total outcome mass W, then*

$$W(S) \leq W/F_{d+1}, \qquad F_1 = F_2 = 1.$$

Proof. For a subtree Z, $W(Z)$ includes every key and external gap in it; Z may be a single gap. A one-level depth change changes cost by the outcome weight for both kinds of outcomes. Write $v(L, R)$ for a key and its two subtrees. Reflection reduces the straight and bent cases to the following two transformations.

In the straight case, with internal grandchild X, rotate

$$v(y(X, B), D) \longrightarrow y(X, v(B, D)).$$

Key y and all outcomes of X rise one level; key v and all outcomes of D fall one level. Every outcome of B keeps its depth. If $O = p_v + W(D)$, the exact change is

$$O - p_y - W(X).$$

Optimality implies $O \geq p_y + W(X) \geq W(X)$.

For the bent case write $X = x(B, C)$ and perform the double rotation

$$v(y(A, x(B, C)), D) \longrightarrow x(y(A, B), v(C, D)).$$

The changes of edge depth, and hence of comparison length, are:

Outcome or subtree	x	B	C	v	D	y	A
Depth change	-2	-1	-1	$+1$	$+1$	0	0

Every outcome in a listed subtree has the indicated change. Therefore

$$C_{\text{new}} - C_{\text{old}} = p_v + W(D) - 2p_x - W(B) - W(C) = O - p_x - W(X).$$

Optimality again gives $O \geq W(X)$. Both transformations preserve inorder positions and every external gap; all other outcomes keep their depths. These are transformations of complete trees.

Going upward from S, write $T_0 = S, T_1, \ldots, T_d = T$. Nonnegativity gives $W(T_1) \geq W(T_0)$, and the grandchild inequality gives $W(T_i) \geq W(T_{i-1}) + W(T_{i-2})$ for $i \geq 2$. Induction gives $W(T_d) \geq F_{d+1} W(S)$. Weak inequalities include zero weights and ties. □

This lemma concerns internal subtrees. A heavy interior external gap can have nearly all the mass at depth two; zero-key subtrees remain permitted. The Fibonacci approach is prior work [5]; sharpness of this particular sufficient bound is not claimed.

2.2 An upper iteration that finishes in a bounded number of levels

Initialize $A_0(i, j) = M(j - i)$ for $i \leq j$, and iterate

$$\begin{aligned} A_h(i, i) &= 0, \\ A_h(i, j) &= W(i, j) + \min_{i \leq r < j} \{A_{h-1}(i, r) + A_{h-1}(r + 1, j)\} \quad (i < j). \end{aligned}$$

This is an h-level partial search tree: explicit decisions incur the usual interval mass, while each key still in an unfinished terminal interval incurs penalty M. If t keys are explicit, the cost is

$$M((j - i) - t) + D, \qquad 0 \leq D \leq hW(i, j).$$

At most $\min\{j - i, 2^h - 1\}$ keys can be explicit, and that number is attainable with inorder labels. Since $M > HW$, every minimizer for $h \leq H$ uses that maximum number. Writing $c_h = 2^h - 1$ gives

$$A_h(i, j) = M \max\{0, j - i - c_h\} + R_h(i, j), \qquad 0 \leq R_h(i, j) \leq hW(i, j). \tag{1}$$

The residual R_h is not asserted to be Monge.

The initial bound dominates every interval optimum: any complete tree on $j - i$ keys costs at most $(j - i)W(i, j) \leq M(j - i)$. The recurrence is monotone, and $A_1 \leq A_0$ because $W(i, j) \leq M$. Thus the iterates decrease and remain at least C. Whenever an optimal full tree fits within h levels, it is feasible without a terminal penalty, so $A_h = C$.

The mixed Fibonacci mass lemma applies here to an *actual optimal full tree*. Every positive-mass internal subtree at depth d satisfies $F_{d+1} \le W$. Zero-mass subtrees can be replaced by balanced trees of height at most L without changing cost. Therefore some optimum has height at most $f(W) + L$; the trivial bound n also applies. When all keys have positive integer weight every internal subtree has mass at least one, so the extra L is unnecessary. These arguments apply to every interval, proving $A_H = C$ throughout the upper triangle.

2.3 Finite Monge products, including the diagonal boundary

For a triangular matrix A with zero diagonal, define its interior density by

$$\Delta_A(i,j) = A(i,j+1) + A(i+1,j) - A(i,j) - A(i+1,j+1) \quad (0 \le i < j < n).$$

A *core* lists nonzero densities. To use a finite matrix product, put $P = 4nM + 4W + 1$ and complete each layer by

$$E_h(i,j) = \begin{cases} A_h(i,j), & i \le j, \\ P(i-j), & i > j. \end{cases}$$

We prove by induction that E_h is Monge. This is immediate for E_0. Given the previous completion, multiply its first n columns by its last n rows in the min-plus sense:

$$B_h(i,j) = \min_{0 \le r < n} \{E_{h-1}(i,r) + E_{h-1}(r+1,j)\}.$$

For $i < j$, every invalid root has a term at least P, whereas a valid pair costs at most nM. Thus the finite product gives the restricted root minimum exactly. The product is Monge. Adding $W(i,j)$, a row term plus a column term, leaves density unchanged in squares entirely above the diagonal. The only interior squares affected by resetting a diagonal entry to zero have $j = i + 1$. Their new density is

$$\min\{A_{h-1}(i,i+1), A_{h-1}(i+1,i+2)\} - q_{i+1} \ge 0.$$

For the first step both old singleton entries are M; thereafter each is its singleton interval mass. The completion adds exactly the n diagonal densities $P + A_h(i,i+1)$; all densities below them vanish. This proves Monge preservation, including the boundary cases. All entries are integers.

2.4 A bound on every intermediate core

Let K_h be the number of nonzero interior densities of A_h. A direct telescoping identity is

$$\sum_{i<j} \Delta_A(i,j) = A(0,n) - \sum_{i=0}^{n-1} A(i,i+1). \tag{2}$$

For $h \ge L$, a balanced full tree is feasible, so $A_h(0,n) \le LW$. Nonnegative integrality then gives $K_h \le LW$. This final-stage argument alone would not bound the early layers, whose terminal penalties are large.

We first make the geometric step explicit, including non-power-of-two sizes.

Lemma 3 (Dyadic boundary count)**.** *In the integer density grid $[0,n)^2$, retain cells with $i < j$ except those on m specified diagonals $j - i = d_t$. Embed the grid in $[0,N)^2$, where N is the least power of two at least n. Subdivide aligned squares into four children, discard squares without retained cells, and accept a square as soon as all its cells are retained. At every dyadic side length s, the accepted squares number at most $8(m+2)n/s$.*

Proof. The enclosing square contains the main diagonal, so it is not accepted. Every accepted square has a parent of side $t = 2s$ containing both a retained and a forbidden cell. That parent either crosses $i = n$ or $j = n$, contains a cell of $i = j$, or contains a cell of a removed diagonal. These exhaust the reasons for subdivision. In a fixed row of side-t parent squares, a diagonal's column indices form t consecutive integers, so it meets at most two parents. Thus each of the $m + 1$ diagonals meets at most $2N/t$ parents. Each outer boundary crosses at most N/t parents; if it is grid-aligned it crosses none. Each charged parent has at most four accepted children. Their number is at most

$$4\{2(m + 1) + 2\}N/(2s) \le 8(m + 2)n/s.$$

The boundaries at n account for the padding when n is not a power of two. Forbidden unit cells are discarded. Every retained cell reaches an accepted ancestor or is itself accepted, proving that these squares partition the retained domain. □

For $1 \le h < L$, the penalty term in (1) has density only on $j - i = c_h$. Account for that line using at most n cells, and apply Lemma 3 with $m = 1$ to the remainder. There are at most $24n/s$ accepted tiles of side s.

On a tile $[a, b) \times [c, d)$ the penalty density vanishes. Its density sum telescopes to

$$R_h(a, d) + R_h(b, c) - R_h(a, c) - R_h(b, d) \le 2hW.$$

All densities in this tile are nonnegative integers. Negative residual densities can occur on the removed line and are not used. The number of nonzero entries in the tile is at most $\min\{s^2, 2hW\}$. Therefore

$$K_h = O\left(n + \sum_{s \text{ dyadic}} \frac{n}{s} \min\{s^2, 2hW\}\right) = O(n + n\sqrt{hW}),$$

capped by $O(n^2)$. Splitting the geometric sum at $s \asymp \sqrt{hW}$ introduces no extra logarithm. For $W = 0$, only the removed-line entries can remain. Combining the early and late bounds, including the n completion entries, yields

$$\delta(E_h) = O\Big(n + \min\{n^2, n\sqrt{WL}\}\Big) \qquad (0 \le h \le H). \tag{3}$$

For completeness, the early-layer argument gives the explicit bound

$$\delta(E_h) \le 2n + 96n\sqrt{2hW} \qquad (1 \le h < L,\ W > 0),$$

since $\sum_{s \text{ dyadic}} \min(s, t^2/s) \le 4t$ for $t \ge 1$. For $W = 0$ only the exceptional and completion diagonals can contribute. Every layer also has at most $n(n + 1)/2$ completed core entries. The constants are conservative; the sharper geometric count is retained in the proof. This controls representation size without charging the large value of M as though it represented many separate core entries.

2.5 Compact computation and full reconstruction

The precise published interface is as follows. Theorem 1.2 of Gawrychowski, Gorbachev, and Kociumaka [10], p. 74:3, states:

> There is a (deterministic) algorithm that, given the condensed representations of a $p \times q$ Monge matrix A and a $q \times r$ Monge matrix B, in time $O(p+q+r+(\delta(A)+\delta(B)) \log(1+\delta(A)+\delta(B)))$ computes the condensed representation of $A \otimes B$.

This quotation is reproduced under the paper's CC BY 4.0 license. The following definitions and guarantees specify the interface used here.

- Definition 2.5 uses precisely our density sign and defines the core as triples $(i, j, \Delta_A(i, j))$ with nonzero density. Its size $\delta(A)$ counts triples, regardless of their magnitudes.
- Definition 3.2 stores the topmost row, leftmost column, and core. This is exactly the representation used here.
- Theorem 1.1 bounds the product core by $2(\delta(A) + \delta(B))$.
- Theorem 4.7, p. 74:11, adds within the same construction time a witness structure occupying $O(p + q + r + \delta(A) + \delta(B))$ space, with $O(\log(2 + \delta(A) + \delta(B)))$ query time. Definition 2.3 specifies the smallest minimizing index, so ties are covered.
- Fact 4.1 supplies logarithmic-time entry queries after preprocessing from the condensed representation.

Here the operands have dimensions $(n+1) \times n$ and $n \times (n+1)$. Every entry is a finite integer: the lower-triangle padding is $P(i - j)$, not infinity. Invalid roots have value at least $P > nM$, whereas a valid root pair costs at most nM. Thus the reduction uses neither extended-real entries nor an unstated triangular version of the published theorem.

Use them on the two contiguous submatrices of E_{h-1}. To obtain the next completion, retain the product core strictly above the first superdiagonal, where adding interval mass has zero density. Recompute the $O(n)$ diagonal and first-superdiagonal core entries, discard lower entries, and recover the top row and left column. Filtering and $O(n)$ logarithmic entry queries take $O((n+\delta)\log(n+1))$ work, including coordinate sorting if needed. The product's temporary core is also bounded linearly by the two input cores, by Theorem 1.1 of [10]. No step first enumerates a dense interval table.

Retain the product witness oracle for every layer. Starting from $(0, n, H)$, retrieve a root and recurse on its two intervals at layer $H - 1$. There are n nonempty calls. An unfinished nonempty interval at layer zero would incur at least M, but the optimum is at most $LW < M$, since $H \geq L$. Thus none survives; when $H = n$, completeness also follows from depth alone. Witness recovery costs an additional $O(n \log(n + 1))$. Equation (3), summed over H layers, proves Theorem 1, including dimensions, representation conversion, storage, and reconstruction. Since $H \leq n$ and the penalties are polynomial in n, W, the stated bit-length bound follows as well.

2.6 A linear-mass family with $\Omega(n^{3/2})$ intermediate core

A contiguous ramp surrounded by zero-weight keys supplies a lower bound on the representation size. We first identify the exact optimization problem inside its height-limited intervals.

Lemma 4 (Perfect-slot representation for zero gap weights)**.** *If every gap weight is zero and $k \leq c = 2^h - 1$, the optimal height-at-most-h key cost equals the minimum over all ordered k-slot subsets of the perfect c-key tree, assigning the input keys to those slots in order. The comparison length of its slot r is $h - \nu_2(r)$, $1 \leq r \leq c$.*

Proof. Every tree of height at most h occupies an ancestor-closed subset of the perfect tree's slots, proving one inequality. Conversely, start from any k selected slots and delete all unselected keys by ordinary BST deletion. For a node with two children, move its inorder successor to that node and splice out the successor from its old position; otherwise splice out the node directly. Every remaining key keeps or decreases its depth. Inorder order is preserved. With nonnegative key weights and

zero gap weights, the resulting feasible tree costs no more than the selected-slot assignment. This proves the opposite inequality. The ruler formula for a perfect tree's inorder depths follows by its recursive halving. □

Let $v_u = \nu_2(u+1)$ for zero-based ruler positions u, and let D bound these coefficients. They alternate between zero and a positive integer. For any consecutive segment of length ℓ,

$$\sum v_u \le \ell + D, \tag{4}$$

because there are at most $\ell/2^d + 1$ multiples of 2^d in the segment, and $v_u = \sum_{d=1}^{D} [2^d \mid u+1]$. Define the odd-ramp subsequence score

$$f_m(a,b) = \max_{a \le t_1 < \cdots < t_m < b} \sum_{r=1}^{m} (2r-1) v_{t_r}.$$

Lemma 5 (Rightmost-zero deletion)**.** *If $b - a = m + s$ and $m > 2s + D$, an optimum for $f_m(a,b)$ deletes the rightmost s zero coefficients and retains every other position.*

Proof. For $s = 0$ every position is retained. For $s > 0$, examine the right endpoint. If its coefficient is zero, some optimum omits it: replace a selected final zero by an earlier omitted position. The inserted coefficient is nonnegative, and the selected coefficients it passes receive larger odd multipliers.

If the final coefficient is positive, every optimum selects it. Otherwise there is a selected zero, since the window has fewer than m positive positions. Let u be the last selected zero. Every zero after u is omitted, and the final positive coefficient uses one more omission. As zeros alternate, the distance from u to the final position is at most $2s-1$. By (4), the selected coefficients after u have sum at most $2s - 1 + D$. Replace that selected zero by the final positive coefficient. The latter receives multiplier $2m - 1$; each selected coefficient after u loses two in its multiplier. The gain is at least

$$(2m-1) - 2(2s - 1 + D) > 0,$$

a contradiction. Thus retain the final positive coefficient and omit its preceding zero while omissions remain. Removing this pair changes (m, s) to $(m-1, s-1)$ and preserves the required inequality. An initial zero omission only strengthens it. Continue until $s = 0$. □

The rule yields an exact density, rather than only a count. Suppose $b - a = m + s$, $s \ge 1$, b is odd, and $m > 2(s+1) + D$, so it applies to all four windows below. Appending the positive coefficient v_b deletes the additional zero at $b - 2s - 1$. The s retained positive coefficients after that zero each lose two in their multiplier. Hence

$$f_m(a, b+1) - f_m(a,b) = (2m-1)v_b - 2\sum_{r=1}^{s} v_{b-2r}.$$

After deleting the left endpoint, the same sum has $s-1$ terms. Subtracting these two increments gives

$$f_m(a,b+1) + f_m(a+1,b) - f_m(a,b) - f_m(a+1,b+1) = -2\nu_2(b - 2s + 1) < 0. \tag{5}$$

Theorem 6 (Linear-mass intermediate-core lower bound)**.** *Let $m = 2^t$, $t \ge 5$, $n = m^2$, $z = (n-m)/2$, and $h = 2t - 1$. Set every gap weight to zero, and set*

$$p_{z+r-1} = 2r - 1 \quad (1 \le r \le m), \qquad p_i = 0 \text{ otherwise.}$$

Then $W = n$, and the terminal-penalty layer A_h has at least

$$\frac{nm}{16} - \frac{9n}{64} - \frac{m}{4} \;\geq\; \frac{n^{3/2}}{32}$$

strictly positive interior densities. Their values have the explicit formula in (6) below.

Proof. Put $c = 2^h - 1 = n/2 - 1$. Every interval of at most c keys is fully exposed in layer h, so Lemma 4 applies. An interval $[i, j)$ containing the entire ramp has $z - i$ leading zero keys and $j - z - m$ trailing zero keys. Its positive keys can occupy any m slots in

$$[a, b) = [z - i,\ c - j + z + m).$$

The preceding and following zero keys fit exactly when the positive slots lie in this window. Since the interval's mass is $W = n$,

$$A_h(i, j) = hn - f_m(a, b).$$

Write $s = c - (j - i)$; then $b - a = m + s$. The four original intervals defining a density correspond to the four score windows based at $(a - 1, b - 1)$. Equation (5) gives, when b is even,

$$\Delta_{A_h}(i, j) = 2\nu_2(b - 2s) \geq 2. \tag{6}$$

Specifically, take $s = 1, \ldots, m/4$ and $a = 1, \ldots, c - m - s - 1$ with $b = a + m + s$ even; then set $i = z - a$ and $j = c - b + z + m$. These are valid, distinct density cells, and all four intervals contain the ramp and have at most c keys. The ruler coefficients are bounded by $D = h - 1 = 2t - 2$. For $t \geq 5$,

$$m > 2(m/4 + 1) + D,$$

so the deletion rule applies to all the windows used. At each s there are exactly $\lfloor (c - m - s)/2 \rfloor$ permitted values of a. With $u = m/4$ even, their sum is

$$\sum_{s=1}^{u} \left\lfloor \frac{c - m - s}{2} \right\rfloor = \frac{u(c - m - 1)}{2} - \frac{u^2}{4} = \frac{nm}{16} - \frac{9n}{64} - \frac{m}{4}.$$

For $m \geq 32$ this is at least $nm/32$. □

All counted cells use fully exposed intervals and avoid the penalty diagonal. Their growth is therefore independent of the numerical size of the terminal penalty. Consequently a uniform $O(n \operatorname{polylog} n)$ bound on every intermediate explicit core is impossible even at $W = n$. The polynomial exponent $3/2$ in the upper bound is necessary for this representation; a factor $\sqrt{\log n}$ remains between the core bounds. The lower bound concerns a layer's density list, not the complexity of constructing an optimal tree by a different representation. The original positive ruler family below supplies the complementary quadratic-support result at centered deviation $R = n^2$.

3 Centering around large constant baselines

The total-mass bound can be pessimistic when large constant key and gap weights dominate smaller variations. We retain the original weights in the solver and center only the analysis of its matrix representation. Let

$$a = \min_i p_i, \quad b = \min_j q_j, \quad u_i = p_i - a, \quad v_j = q_j - b, \quad R = \sum_i u_i + \sum_j v_j.$$

All quantities remain nonnegative integers. The two baselines may differ arbitrarily; the deviations may occur at arbitrary key and gap positions. Write $L = \lceil \log_2(n + 1) \rceil$ as before.

Lemma 7 (Normalized height budget)**.** *For $W > 0$, let s be the smallest positive singleton mass $p_i + q_i + q_{i+1}$. Put $z = 0$ if every singleton mass is positive and $z = L$ otherwise. Every interval has an optimal tree with height at most*

$$H = \min\{n, f(\lfloor W/s \rfloor) + z\}.$$

For $W = 0$, use $H = L$.

Proof. Every positive-mass nonempty interval contains a positive-mass singleton: choose a positive key, or an adjacent key within the interval if a gap supplies its positive mass. Thus every positive internal subtree has mass at least s. Lemma 2 gives $W \geq F_{d+1}s$ for such a subtree at depth d in an optimal tree. The positive part has the resulting bounded depth; zero-mass subtrees can be balanced, adding at most L levels without changing the cost. If all singleton masses are positive no nonempty zero subtree exists. The global W, s, L bounds also apply to every subinterval. The trivial height n and the all-zero balanced case complete the argument. □

Use this H, $M = (H+1)W + 1$, and the same finite completion and penalty iteration. All previous preservation arguments apply. Reconstruction is complete because any unfinished nonempty terminal costs at least M, whereas an optimal tree of height at most H costs at most HW.

3.1 The exact baseline layer and its exceptional diagonals

Let $I(k)$ be the minimum unweighted internal path length for k keys. For $k \geq 1$, with $d = \lfloor \log_2 k \rfloor$,

$$I(k) = dk - 2^{d+1} + d + 2, \qquad I(0) = 0.$$

The familiar baseline identity is $(a+b)I(T) + (a+2b)k$ for a full k-key tree. It was already used by the minimum-path-length certificate; the following application concerns every partial-tree layer.

Lemma 8 (Centered partial-tree formula)**.** *With $c_h = 2^h - 1$, define*

$$F_h(k) = \begin{cases} (a+b)I(k) + (a+2b)k, & k \leq c_h, \\ M(k - c_h) + (a+b)h(k+1) - ac_h, & k > c_h. \end{cases}$$

Then the full-weight layer admits

$$A_h(i,j) = F_h(j-i) + U_h(i,j), \qquad 0 \leq U_h(i,j) \leq hR(i,j),$$

where $R(i,j) = \sum_{i \leq r < j} u_r + \sum_{i \leq r \leq j} v_r$. The baseline's density at interval length $k \geq 1$ is

$$F_h(k+1) + F_h(k-1) - 2F_h(k) = \begin{cases} a+b, & k = 2^t - 1, \quad 1 \leq t < h, \\ M - b, & k = 2^h - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The penalty forces exactly $\min(k, c_h)$ explicit keys. If $k \leq c_h$, a full minimum-path-length tree fits in h levels. Otherwise a partial tree exposing c_h keys must fill every position in the first h levels. Its explicit keys have baseline cost $a\sum_{t=0}^{h-1}(t+1)2^t$; unfinished keys cost $ah(k - c_h)$, and all $k+1$ gaps cost $bh(k+1)$. This gives the second branch of F_h. Nonnegative deviations cannot reduce the baseline optimum. Conversely, a baseline-optimal partial tree charges every deviation at most h times, proving the residual bounds. The displayed densities follow by taking second differences of the two branches. In particular, changes in the full-tree slope occur only at $k = 2^t - 1$. □

The residual U_h need not be Monge. The actual layer is Monge, and on any square whose density cells avoid the exceptional diagonals, its density equals the residual density. Baseline magnitudes do not increase the number of exceptional diagonals.

3.2 A weighted packing argument

Remove all density cells on $j - i = 2^t - 1$ within the domain, and the main diagonal, accounting for $O(nL)$ cells. Tile the remainder with maximal aligned dyadic squares that avoid those lines and the outer boundary. For a side-s tile $T = [x, x + s) \times [y, y + s)$, let $J_T = [x, y + s]$ be its spanning input interval, including both gap endpoints.

Lemma 3 gives $O(nL/s)$ tiles at scale s.

Lemma 9 (Per-position packing)**.** *At a fixed side length s, every input position belongs to the spanning intervals of $O(n/s)$ accepted tiles. Consequently*

$$\sum_{T:\,\mathrm{side}(T)=s} R(J_T) = O(nR/s). \tag{7}$$

Proof. Fix an input position and charge each tile to a boundary met by its parent. A tile charged to a diagonal at distance $d \geq s$ lies within $O(s)$ of that diagonal. For its spanning interval to contain the fixed position, its row start lies in an interval of length $O(d + s)$. Row starts are spaced by s, with only $O(1)$ eligible column starts per row near that diagonal. Thus it contributes $O(1 + d/s)$ covers. The distances $d = 2^t - 1$ are geometric, so summing these bounds gives $O(n/s)$. All distances $d < s$ lie together in an $O(s)$-wide strip near the main diagonal and contribute only $O(1)$ covers of the fixed position. Outer-boundary tiles number $O(n/s)$ in total. Charging each nonnegative deviation to its covering intervals proves (7). The outer-boundary case is covered by Lemma 3, including arbitrary n. □

On an ordinary tile the density sum telescopes to

$$U_h(x, y + s) + U_h(x + s, y) - U_h(x, y) - U_h(x + s, y + s) \leq 2hR(J_T).$$

The true densities are nonnegative integers. The number of nonzero entries on a tile is therefore bounded both by s^2 and by that density sum. Using the tile count and (7), the number across all tiles of side s is $O(\min\{nLs, hnR/s\})$. Splitting the dyadic geometric sum at $s \asymp \sqrt{hR/L}$ gives $O(n\sqrt{hRL})$; for $R = 0$ every ordinary density vanishes. No cancellation of negative residual densities on the removed lines is assumed.

Theorem 10 (Centered core construction)**.** *For arbitrary nonnegative integer mixed weights, the normalized-height penalty construction has completed layer cores bounded by*

$$K_c = O\Big(n + \min\{n^2, nL + n\sqrt{HRL}\}\Big).$$

Using the published compact product and witness algorithms, a complete optimum can be constructed in $O(HK_c \log(n + 1))$ arithmetic operations. Published succinct witnesses use $O(HK_c)$ words; the implemented ordinary maps can require $O(HK_c \log(n + 1))$ words.

Proof. Add the $O(nL)$ removed cells and n completion densities to the ordinary tile bound, and use $h \leq H$. The initial layer has only $O(n)$ core entries. The quadratic cap is immediate. All products, conversion steps, and root queries operate on the actual condensed matrices, so the already established product reduction applies. The previous $O(n+n\sqrt{WL})$ bound is also available; either bound may be used when smaller. □

Corollary 11 (Arbitrarily large baselines with polynomial deviation)**.** *If R is polynomial in n, exact full construction takes*

$$O\Big(n(1 + \sqrt{R}) \log^3(n + 1)\Big)$$

arithmetic operations, independently of the magnitudes of a, b in the operation count. In particular, $R = O(n)$ gives $O(n^{3/2}\log^3(n+1))$, and $R = O(n^{2-\varepsilon})$ gives $O(n^{2-\varepsilon/2}\log^3(n+1))$ for fixed $0 < \varepsilon \le 2$.

Proof. If $a + b = 0$, then $W = R$, and $s \ge 1$ whenever $W > 0$. Otherwise every singleton has mass at least $a + 2b \ge 1$, and

$$W = na + (n+1)b + R \le n(a+2b) + R, \qquad W/s \le n + R.$$

Thus $H = O(\log(n + R + 1)) = O(\log(n+1))$ for polynomial R, including the possible additional L and the all-zero case. Substituting in the theorem gives the result. □

For example, $p_i = n^c + u_i$ and $q_j = n^d + v_j$, with fixed nonnegative integer c, d and bounded nonnegative integer deviations, satisfy the linear-R condition even when W is superlinear. Intermediate integers then still use $O(\log n)$ bits. Arbitrary larger baselines retain a bit-cost dependence on their representation lengths. These families permit nonuniform weights on both keys and gaps. The earlier total-mass result has a smaller logarithmic factor when $W = O(n)$.

4 A common certificate and its constructive corollaries

The following pattern applies to all the exact shortcuts in this section. Every candidate and bound uses the original fixed-position key and gap objective, or a stated tree-independent cost transfer.

Theorem 12 (Certificate pattern)**.** *Let $\mathcal{T}$ be the full set of feasible ordered search trees. Suppose an algorithm constructs a candidate T of cost V and a lower bound $B \le \min_{S\in\mathcal{T}} C(S)$. If $V = B$, the candidate is optimal. More generally, suppose T minimizes cost in a class $\mathcal{F}$ and $B_{\text{out}} \le \min_{S\notin\mathcal{F}} C(S)$. Then $V \le B_{\text{out}}$ certifies optimality. If candidate construction and bound evaluation take $t(n)$ operations including reconstruction, the accepted route takes $O(t(n))$ operations. Exact fallback gives $O(t(n) + n^2)$ worst-case work.*

Proof. In the first case $B \le \text{OPT} \le V = B$. In the second, every tree inside $\mathcal{F}$ costs at least V, and every tree outside costs at least $B_{\text{out}} \ge V$. Adding the work of the two possible stages gives the running-time statement. □

Each family corollary below supplies a candidate construction and proves that one of these comparisons succeeds for every member of the family. A failed comparison leaves an exact fallback available; it does not imply that the candidate itself is suboptimal. The intervening lemmas supply the bounds and construction algorithms used by those corollaries.

4.1 A sharper single-key contraction

A key is a *cherry* when its two children are external nodes. Replacing a cherry with one gap is exact: if the key and its gaps have weights p, x, y, their total is $s = p + x + y$. A cherry rooted at depth d contributes $s(d+1)$, whereas the replacement gap contributes sd.

Theorem 13 (Single-key rule)**.** *Let a key of weight p have adjacent gap weights x, y. If its immediate predecessor and successor keys have weights a, b and*

$$\min(a, b) \;\ge\; 2p + x + y,$$

there is an optimal tree in which this key is a cherry. At the left endpoint, it suffices that the successor weight is at least $p + x$; at the right endpoint, it suffices that the predecessor weight is at least $p + y$. Consequently contraction to a gap of weight $s = p + x + y$ satisfies $\text{OPT}(\textit{original}) = \text{OPT}(\textit{contracted}) + s$.

Proof. Consider the key at depth d in any tree. If its left subtree is nonempty, the immediate predecessor has depth $d + l$, $l \geq 1$. If its right subtree is nonempty, the immediate successor has depth $d + r$, $r \geq 1$. Set the corresponding distance to zero for an empty internal side.

When $l \geq 1$, move the predecessor to the chosen key's position, replacing the predecessor's old position by its left child. Insert the chosen key as a cherry at the leftmost gap of the former right subtree. Inorder order is preserved. All other outcomes either retain their depths or become shallower. The predecessor saves la, and the cost change is at most

$$-la + (r + 1)p + (r - l + 1)x + y. \tag{8}$$

The mirrored successor exchange has change at most

$$-rb + (l + 1)p + x + (l - r + 1)y. \tag{9}$$

If both sides are nonempty and $l \geq r$, substituting $a \geq 2p + x + y$ in (8) gives

$$(r + 1 - 2l)(p + x) + (1 - l)y \leq 0.$$

Use (9) when $r \geq l$. If only the left side is nonempty, $r = 0$ and $a \geq p + y$ makes (8) nonpositive. The other case is symmetric. The stated interior condition implies these one-sided conditions. At a global endpoint only the appropriate one-sided case is possible. With both sides empty the key is already a cherry.

Applying the exchange to an optimum proves the existence claim. Contraction subtracts exactly s. Conversely, every contracted tree expands to a valid original tree at additional cost s, proving equality. □

The weaker condition $\min(a, b) \geq p + x + y$ is false. For $p = (24, 14, 24)$ and $q = (1, 0, 9, 0)$, the middle key has local mass 23 and neighbors of weight 24. Contracting it forces cost 142, while the unrestricted optimum costs 130.

Maintain a doubly linked list of remaining keys and weighted gaps. Initially queue every key; after a contraction, only its predecessor and successor need rechecking. At most $3n$ queue entries are processed. Saving the contracted tree at the new gap permits reconstruction. Thus this pass takes $O(n)$ operations and space, and a Knuth solve on the remaining m keys gives total time $O(n + m^2)$. No confluence or minimum-core-size property is claimed.

4.2 Whole-block contraction

Let $B = [i, j)$ be a nonempty proper interval, including its endpoint gaps q_i, q_j. Put $W = W(i, j)$ and $C = C(i, j)$, with costs measured from a stand-alone root of depth zero.

Theorem 14 (Block rule)**.** *If both boundary keys exist and their weights satisfy*

$$p_{i-1} \geq W + C, \qquad p_j \geq W + C,$$

there is an optimum in which all of B forms one subtree. For a prefix or suffix block, it suffices that the sole boundary key has weight at least C. Contracting the block to one gap of weight W then gives

$$\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{contracted}) + C.$$

Proof. In any tree, let v be the lowest common ancestor of gaps q_i, q_j, at depth d. Its key lies in B, and every outcome of B descends from v. Hence the original contribution of B is at least $(d+1)W$.

Within the subtree rooted at v, delete the keys of B from the inward fringes of its two children. On the left, a deleted key has no retained right child and can be suppressed in favor of its left child; on the right the process is symmetric. Let the retained outside trees be A, D. All their retained outcomes are at depths no greater than before. If A is nonempty, its maximum is boundary key $i-1$; if D is nonempty, its minimum is boundary key j.

Suppose both are nonempty, and their boundary keys had original depths $d+l, d+r$. If $l \geq r$, remove the maximum key from A, replacing its position by its left child, and make that key the new root at v. Its left child is the remainder of A and its right child is D. Put an optimal tree for B at the leftmost external position of D. This position has depth at most $d+r+1$. No outside outcome other than the promoted boundary moves deeper, and that boundary saves lp_{i-1}. The increase in the contribution of B is at most

$$C + (d+r+1)W - (d+1)W = C + rW.$$

Since $l \geq r \geq 1$ and $p_{i-1} \geq W + C$, the saving is at least $l(W+C) \geq rW + C$. The mirrored construction applies for $r \geq l$.

If only one retained outside tree is nonempty, promote its inward boundary key to v and attach the optimal block as the other child. The block increase is at most C, while that boundary saves at least its own weight. The interior hypothesis suffices, and for a global prefix or suffix the weaker stated hypothesis suffices. If both outside trees are empty, B already occupies the subtree at v and can be optimized in place.

The construction never increases cost. Apply it to an optimum. A block at depth e contributes $C + eW$, while its replacement gap contributes eW. Contraction and expansion now prove the equality. □

A balanced k-key tree has height $h_k = \lceil \log_2(k+1) \rceil$ and cost at most $h_k W$. Thus the interior test

$$\min(p_{i-1}, p_j) \geq (1 + h_k)W$$

and endpoint test $p_{\text{boundary}} \geq h_k W$ certify a block without first solving it. These are sufficient conditions, not characterizations.

4.3 An adaptive linear selection pass

Build the maximum Cartesian tree of the key-weight sequence, breaking ties deterministically. Its subtree intervals form a laminar family: intervals are disjoint or nested. Compute their sizes and masses by one traversal and prefix sums. Test each proper interval with the balanced-tree bound above. This inspects only $O(n)$ candidates. The restriction to Cartesian intervals is an algorithmic choice, not a claim that all useful blocks occur there.

For a threshold b, select the maximal certified intervals whose sizes are at most b. Different selected intervals have a surviving separator key between them. Contracting one therefore leaves every other selected interval's weights and boundary-key weights unchanged, so the block theorem applies successively. Write their sizes as $k_1, \ldots, k_s$, and $m = n - \sum_t k_t$. The exact number of nonempty interval-DP states is

$$S(b) = \tau(m) + \sum_t \tau(k_t), \qquad \tau(u) = \frac{u(u+1)}{2}.$$

We choose a threshold minimizing $S(b)$ without running these DPs. For every certified interval I of size k, find its maximal proper certified descendants J. Immediately before I becomes eligible, these descendants are its selected blocks. Its activation therefore changes the core size and block-state total by

$$\Delta m = -\left(k - \sum_J |J|\right), \qquad \Delta S_{\text{blocks}} = \tau(k) - \sum_J \tau(|J|).$$

These sums are computed bottom-up. Bucket activation events by integer size $1, \ldots, n$ and scan the buckets, maintaining m and the block-state total. There are $O(n)$ events and buckets. Equal-size intervals are disjoint; evaluate a threshold after processing its entire bucket. The empty selection $b = 0$ is included.

Corollary 15. *The selected threshold and blocks are found in $O(n)$ arithmetic operations and space. Solving the blocks and surviving core by Knuth's method and expanding their saved trees takes*

$$O\left(n + m^2 + \sum_t k_t^2\right)$$

time and $O(n + m^2 + \max_t k_t^2)$ space when block tables are released after each solve. The chosen state count is no larger than $n(n+1)/2$. Worst-case time remains $O(n^2)$.

The minimization is only over these size thresholds in this candidate hierarchy, not over every possible partition. It does not minimize wall-clock time. Preprocessing, table allocation, root scans, and reconstruction still count as work.

For perspective, a certified partition with $m = O(n^{2/3})$ surviving keys and every $k_t = O(n^{1/3})$ gives $O(n^{4/3})$ time, since $\sum k_t^2 \le (\max k_t)n$. This is a conditional structural consequence, not an unrestricted bound.

4.4 Recursive construction with a bounded preprocessing depth

The recursive constructor combines the proved components. At each call it applies the single-key pass, compares the order-free lower bound of Section 5.1 with a feasible bisection-tree cost, and returns immediately on equality. Otherwise it selects blocks as above. It recurses only if their combined table-state estimate is at most three quarters of the full estimate; small components and unsuccessful decompositions fall back to Knuth's method.

Suppose a component has $m > 8$ keys after the single-key pass and is successfully decomposed. Every child component of size s satisfies $s(s+1) \le \frac{3}{4}m(m+1)$, hence $s < 0.92m$. Child sizes sum to m. Consequently the recursion has logarithmic depth, and the total number of keys processed on any recursion level is at most n. Linear selection and reconstruction cost $O(n \log n)$ in total. The heap-based equality checks cost $O(n \log^2 n)$ over all levels. If the terminal DP sizes are r_t, the construction therefore takes

$$O\left(n \log^2 n + \sum_t r_t^2\right)$$

arithmetic operations. Its worst case is still quadratic. This threshold prevents repeated expensive scans after arbitrarily small reductions. Correctness follows by contraction, exact equality certificates, and induction on the strictly smaller recursive components; reconstruction restores the original key indices and gap subtrees.

4.5 A certified shortcut through nearly complete shapes

Let $I(T)$ be the sum of internal-node depths, with root depth zero. The external path length is $I(T) + 2n$, by induction on the two subtrees. Consequently uniform key weight a and gap weight b give cost

$$(a + b)I(T) + (a + 2b)n.$$

For an arbitrary instance set $a = \min_i p_i$, $b = \min_j q_j$, and let H' be the order-free mixed-node Huffman bound of Section 5.1 on residual weights $p_i - a, q_j - b$. These weights remain nonnegative. Write $I_*(n)$ for the minimum possible unweighted internal path length, and let C_* be the best weighted cost among all trees with $I(T) = I_*(n)$.

Corollary 16 (Minimum-path-length certificate)**.** *For $n \geq 1$, if*

$$C_* \;\leq\; (a + b)(I_*(n) + 1) + (a + 2b)n + H',$$

then a tree attaining C_* is a globally optimal full OBST.

Proof. Split each outcome weight into its uniform baseline and residual. Every tree outside the minimum-I class has integer $I(T) \geq I_*(n) + 1$, and its residual cost is at least H'. Its full cost is therefore at least the right-hand side. Within the class, C_* is minimal by definition. Equality is sufficient, including when $a + b = 0$. □

Optimize the class without interval tables. Put $h = \lfloor \log_2(n + 1) \rfloor$, $S = 2^h$, $m = S - 1$, and $r = n - m$. Every minimum-I tree has the complete upper skeleton with m keys, and expands exactly r of its S external slots into single-key cherries. Indeed, a missing shallower slot could otherwise receive a deepest internal leaf, strictly reducing I. Inorder relabeling preserves search-tree feasibility in this unweighted shape argument. Thus

$$I_*(n) = (h - 2)2^h + 2 + hr.$$

Process skeleton slots $j = 0, \ldots, S - 1$ in inorder. State (j, e) records the number of processed slots and expansions, so the next input index is $i = j + e$. An unexpanded slot charges hq_i; an expanded slot charges $(h+1)(p_i+q_i+q_{i+1})$. Except after the last slot, also charge the next skeleton key at comparison length

$$d_j = h - v_2(j + 1),$$

using p_i after an unexpanded slot and p_{i+1} after an expansion. Here v_2 counts trailing zero bits. Exactly r expansions are required, so state (j, e) is retained only when

$$\max(0, r - (S - j)) \leq e \leq \min(r, j).$$

With $s = \min(r, S - r)$, each stage has at most $s + 1$ states and each state has two outgoing choices. This computes C_* and backpointers in $O(n(1 + s))$ time and space. A linear Cartesian-tree pass converts the chosen comparison lengths into the labeled tree. No key or gap weights are discarded in these transitions.

There is an exact early rejection test. Let F be the cost obtained by assigning independently sorted key and gap weights to the shallowest depths of a minimum-I shape. This is feasible for the order-free residual relaxation, so $H' \leq F - [(a + b)I_* + (a + 2b)n]$. Therefore $C_* > F + a + b$ proves the certificate cannot succeed. Otherwise the sorted lists are reused in a two-queue Huffman implementation: nonnegative weights and nondecreasing internal weights make the sequence of

merged weights nondecreasing. This adds linear work after sorting. The queue optimization is standard and is not claimed as novel.

The implementation attempts this DP only when $s \le \lfloor \log_2 n \rfloor + 1$. Including the residual bound, preprocessing, and reconstruction, a successful attempt costs $O(n \log n)$ arithmetic operations and space. A skipped or failed attempt falls back to Knuth, or to the preceding recursive construction. Both versions remain exact and have quadratic worst-case time. The test is a certificate, not an assumption that near-uniform inputs have balanced optima.

Optimize the baseline rather than fix it. The largest subtractable baseline need not give the strongest certificate. Set $a_0 = \min p_i$, $b_0 = \min q_j$, and use $a = a_0 - \delta$, $b = b_0 - \delta$ for an integer $0 \le \delta \le \lfloor (a_0 + b_0)/2 \rfloor$. The residual weights remain nonnegative and $a + b \ge 0$. Although one baseline term may become negative, it is only an algebraic decomposition; the input probabilities are unchanged. Let H_δ denote the residual order-free optimum. The exclusion bound is

$$B(\delta) = (a_0 + b_0 - 2\delta)(I_* + 1) + (a_0 + 2b_0 - 3\delta)n + H_\delta.$$

Every tree outside the minimum-I class costs at least $B(\delta)$. Thus any sampled price with $C_* \le B(\delta)$ certifies the same candidate.

For a fixed residual tree, increasing δ adds one unit to every key and gap weight, changing its cost at rate $2I + 3n$. Consequently B is a pointwise minimum of affine functions, hence concave. A minimizing residual tree supplies a supergradient

$$g = 2(I - I_* - 1), \qquad B(x) \le B(\delta) + g(x - \delta).$$

If $g > 0$, smaller prices cannot improve the current value; if $g < 0$, larger prices cannot improve it. If $g = 0$, the value is globally maximal. Binary interval elimination, retaining the best sample, therefore finds the maximum over allowed integer prices when run to completion.

The same linear oracle supplies I. Give every original outcome secondary weight1, order queue items lexicographically by residual mass and number of original outcomes, and sum both merge coordinates. The first gives H_δ and the second gives $2I+3n$. This is the Huffman exchange argument with an infinitesimal positive common perturbation, choosing minimum I among residual-cost ties. The weight order is independent of δ, so no additional sorting is needed. Each price costs $O(n)$.

The fixed-profile ceiling extends to all prices: $B(\delta) \le F + a_0 + b_0 - 2\delta$. Therefore an input rejected by the existing profile test cannot be rescued by this search. The implementation tries prices only after the old certificate fails while passing that test, and permits at most $\lfloor \log_2 n \rfloor + 1$ extra oracle calls. This preserves the $O(n \log n)$ attempted-shortcut bound. Reaching the cap does not establish that the best price was found; every sampled certificate is still valid, and failure invokes the exact fallback.

4.6 Exact bounded-excess construction by dyadic intervals

The candidate class can be expanded beyond minimum internal path length. This is necessary even on some mildly nonuniform inputs. For $n = 255$, set all gap weights to 16 and all key weights to 16, except keys $8j + 1$ which have weight 17. The unique minimum-I tree costs 61,680; Knuth gives optimum 61,656, attained with $I = I_* + 2$. The algorithm below constructs and certifies this optimum. It retains both kinds of weight and never assumes that the restricted optimum is globally optimal.

An exact representation. An ordered full binary tree is a tiling of $[0,1]$ by consecutive aligned dyadic intervals: a gap at depth d has width 2^{-d}. A leaf starting at x must satisfy $x/2^{-d} \in \mathbb{Z}$. Conversely such a tiling is a complete prefix code and determines an ordered full tree. Alignment, not just the Kraft equality, enforces order. On a grid of height H, an interior boundary at integer coordinate Y corresponds to a key with comparison length $H - \nu_2(Y)$. Thus gap and key costs are both local in a left-to-right tiling construction.

Theorem 17 (Bounded-excess construction)**.** *For $n \geq 1$, let $N = n+1$, $h = \lfloor \log_2 N \rfloor$, $S = 2^h$, $r = N - S$, and $s = \min(r, S-r)$. For any nonnegative integer budget k, the minimum full weighted cost among all trees with $I \leq I_* + k$ can be found and reconstructed in*

$$O\big(n(s+k+1)(k+1)^2 2^k\big)$$

arithmetic operations, using $O\big(n(s+k+1)(k+1)2^k\big)$ space for backpointers.

Proof. Measure dyadic widths in units of 2^{-h} and put $w = 2^{h-d}$. For integral depth d, the potential

$$f(d) = d - h - 2 + 2^{h+1-d} = d - h - 2 + 2w$$

is nonnegative, with zeros exactly at h and $h+1$. The minimum external path length is $hN + 2r$. Since external path length equals $I + 2n$ and $\sum_i w_i = S$, every full tree satisfies

$$\sum_i f(d_i) = \sum_i d_i - (h+2)N + 2S = I - I_*.$$

Every depth in the budgeted class therefore has $f(d) \leq k$. This implies $d \leq H = h + k + 2$, and only $O(k+1)$ integral depths are possible. For integral d, direct substitution gives

$$\tfrac{1}{2} - \tfrac{1}{2} f(d) \leq w \leq 1 + f(d).$$

Apply these inequalities to a prefix of i leaves and to its remaining suffix, each having potential at most k. The cumulative position y must obey

$$\max\left(\tfrac{i-k}{2}, i - r - k\right) \leq y \leq \min\left(i + k, \tfrac{S-r+i+k}{2}\right).$$

This interval has width at most both $r + 2k$ and $(S-r)/2 + k$, hence $O(s+k)$.

Set $K = 2^{k+2}$. A state after i leaves stores integral drift $u = K(y-i)$ and spent potential $z = K \sum f(d)$. For depth d, let $W = 2^{H-d}$. Require $(iK + u) \bmod W = 0$ and update

$$u' = u + W - K, \qquad z' = z + K(d-h) + 2(W-K).$$

Retain the position band and $0 \leq z' \leq Kk$. Charge $q_i d$ and, for $i > 0$, charge $p_{i-1}(H - \nu_2(iK+u))$. Start at $(u,z) = (0,0)$ and finish after N leaves with $u = -rK$. Every retained complete path is an aligned full tiling with excess at most k; every tree in the class passes the same tests. The DP therefore optimizes exactly this class. For fixed u, the congruence $z \equiv 2u \pmod K$ leaves at most $k+1$ possible spent potentials. There are $O((s+k+1)2^k)$ positions per stage and $O(k+1)$ choices per state, proving the bounds. The bands admit direct array indexing. Backpointers recover gap depths; the boundary lengths reconstruct the tree in linear time. □

Perfect-size refinement. When $N = S$, use instead $f_0(d) = d - h + \frac{3}{2}(2^{h-d} - 1)$. It is nonnegative on integer depths, vanishes only at h, and also sums to $I - I_*$. The inequality $|2^{h-d} - 1| \leq 2 f_0(d)$ and zero total drift imply $|y - i| \leq k$. Here $H = h + k + 1$, $K = 2^{k+1}$, and $z = 2K \sum f_0$. Its increment is $2K(d-h) + 3(W-K)$, and $z \equiv 3u \pmod{2K}$. The implementation uses this smaller grid at perfect sizes, with the same asymptotic bound at $s = 0$.

Certify against all other trees. Let C_k be the computed candidate, $a = \min p_i$, $b = \min q_i$, and H' the order-free bound on residual weights. Every excluded tree has $I \geq I_* + k + 1$, so it costs at least

$$B_k = (a + b)(I_* + k + 1) + (a + 2b)n + H'.$$

Thus $C_k \leq B_k$ proves global optimality. Sorting and the two-queue bound cost $O(n \log n)$ and are included in the counters. The constructor uses $k = 2$ and attempts the shortcut only when $s \leq \text{bit_length}(n)$. Its attempted work is $O(n \log n)$; a failed certificate or skipped size falls back to Knuth. The total worst case remains quadratic. Arbitrary growth of k is exponential, and wide shape spaces remain quadratic even for fixed k.

4.7 A split-height construction covering every input size

The preceding shape band can be wide even at zero excess. We therefore implement the split-height/Monge mechanism already sketched by Karpinski, Larmore, and Rytter in Section 4.4 of their longer preprint [4]. That section explicitly supplies the fixed-height Monge property and the two-part construction. It is different from Section 4.4 of the six-page conference version [3]. No priority claim is made for this mechanism. The present experiment makes the tiling, matrix padding, reconstruction, and separate unrestricted certificate explicit.

Two scales of dyadic intervals. Fix an external height limit $H = L + U$ and write $K = 2^L$, $B = 2^U$. Precompute all relative costs $C_h(i, j)$ for $0 \leq h \leq L$ and $0 \leq j - i \leq 2^h - 1$, using

$$C_h(i, j) = W(i, j) + \min_r \{C_{h-1}(i, r) + C_{h-1}(r + 1, j)\}.$$

Empty intervals have zero cost; the root range enforces both child height capacities. Monge inequalities at fixed height give monotone minimizing roots as the endpoints move. Choose leftmost ties and bound each root by the adjacent shorter intervals. Search ranges telescope to $O(n)$ for each interval length. Summing over lengths and heights costs $O(n2^L)$.

Divide the unit dyadic interval into B top-grid blocks. Every tree of height at most H consists of lower subtrees contained in individual blocks, together with shallower gaps occupying aligned groups of blocks. Let $D_b(i)$ be the minimum partial cost after consuming i gaps and reaching grid boundary b. Keys through $i - 2$ have already been charged. For $b > 0$, put

$$R_b(i) = D_b(i) + p_{i-1}(U - v_2(b)); \qquad R_0(0) = 0.$$

The boundary key has exactly this comparison length. There are two types of transition:

$$D_{b+1}(j) \leftarrow \min\{D_{b+1}(j), R_b(i) + C_L(i, j - 1) + UW(i, j - 1)\}, \qquad i < j \leq i + K,$$
$$D_{b+w}(i + 1) \leftarrow \min\{D_{b+w}(i + 1), R_b(i) + q_i(U - \log_2 w)\}.$$

The second transition requires a power of two $w \geq 2$, $w \mid b$, and $b + w \leq B$. Width-one gaps are zero-key lower subtrees in the first transition. Start at $D_0(0) = 0$ and finish at $D_B(n + 1)$. Dyadic alignment proves that these paths represent every height-feasible tree and only such trees. Backpointers recover every lower tree and every boundary key.

Linear matrix searches with feasible padding. At fixed b, the first transition is a banded Monge matrix in i, j; row additions and the separable interval mass preserve the inequality. Reachable counts i are contiguous: the coarsest tiling of the prefix has popcount(b) leaves, and successive

refinement gives every count through bK, truncated to n before another transition. Search output columns from $i_{\min}+1$ through $\min(n+1, i_{\max}+K)$, each of which has a feasible input row.

SMAWK requires care outside the band. Let $S=(H+1)(\sum p+\sum q)+1$, larger than every feasible prefix cost. In the transposed matrix with row j and candidate column i, use

$$A(j,i)=\begin{cases} S+j-K-i, & i<j-K,\\ R_b(i)+C_L(i,j-1)+UW(i,j-1), & j-K\le i<j,\\ S+i-j+1, & i\ge j. \end{cases}$$

This padding preserves total monotonicity. For $i_1<i_2$, once i_2 is preferred at a row, it remains preferred as j increases: within the feasible band this is Monge; outside it, distances decrease toward the band and increase away from it, and finite entries beat all padding. When the candidates straddle opposite invalid sides, their cost difference is $i_1+i_2+K+1-2j$, also decreasing. Ties preserve the same non-strict implication. Thus SMAWK finds leftmost minima in $O(n)$ queries per grid boundary. Padding by tied infinities does not supply this argument.

Across all boundaries there are $B-1$ aligned choices of coarse width, so the second transitions and scans also cost $O(nB)$. Choosing $L=\min(H,\lfloor H/2\rfloor+1)$ gives

$$O(n(2^L+2^U))=O(n2^{H/2})$$

work and storage including backpointers. Consequently a fixed extra height $H=\lceil\log_2(n+1)\rceil+c$ gives $O(n^{3/2})$ attempted work at *every* size. The prototype uses sparse dictionaries; the bounded integer state coordinates also admit direct array indexing.

Lemma 18 (Minimum path length with a tall path)**.** *For $1\le R\le n$, the minimum internal path length $J(n,R)$ among n-key trees of external height at least R is obtained by forcing a path of R internal nodes and filling the remaining available positions in increasing order of depth. The excluded class is empty if $R>n$.*

Proof. Every tree in the class contains such a path, contributing $R(R-1)/2$. The remaining nodes form an ancestor-closed subset of its free forest: one side root at each depth $1,\dots,R-1$, and two roots at depth R. The $n-R$ shallowest positions in this forest are ancestor-closed and minimize the depth sum among all subsets of that size. Their addition constructs a feasible tree attaining the bound. Directions of the forced path do not change the forest's depth counts. □

A level-count implementation computes J in $O(R+\log(n+1))$ work.

Corollary 19 (Certificate against taller trees)**.** *Let C_H be the exact height-restricted optimum above, and let Q be the order-free mixed-node Huffman lower bound for residual weights $p-a, q-b$, where $a=\min p$, $b=\min q$. If $H<n$ and*

$$C_H\le(a+b)J(n,H+1)+(a+2b)n+Q,$$

the constructed tree is globally optimal. If $H\ge n$, there is no height restriction and certification is automatic.

Proof. Every excluded tree has internal path length at least $J(n,H+1)$. Its baseline contribution is $(a+b)I+(a+2b)n$ and its residual cost is at least Q. Within the height class, C_H is optimal. □

Sorting and the residual oracle add $O(n\log n)$ work. Failed tests invoke Knuth, so the full constructor remains quadratic in the worst case. The certificate never assumes success from near-uniform weights.

4.8 A variable interval frontier for exact full construction

The height test can fail even when its candidate is optimal. We now use a class guaranteed in advance to contain an unrestricted optimum. The short-interval/top-skeleton architecture is established work of Larmore and Karpinski–Larmore–Rytter [2, 3]; the experiment compares a uniform width envelope with a variable interval contour. Both use the same safe mass guarantee and matrix infrastructure. This is not a priority claim for the architecture or its restricted subquadratic regime.

The height guarantee follows from Lemma 2.

Store a contour instead of a global width. For fixed d, let $E_d(i)$ be the largest $j > i$ satisfying $F_{d+1}W(i,j) \le W$, or i if there is none. Empty intervals always have stored cost zero. Nonnegative weights imply that $E_d(i)$ is nondecreasing and the domain is closed under taking subintervals. A two-pointer scan computes it in $O(n)$ work. Write

$$S_d = \sum_{i=0}^{n} (E_d(i) - i).$$

By Lemma 2, every internal subtree at depth d of an optimal tree belongs to the domain. For comparison, the uniform version sets $\ell_d = \max_i(E_d(i) - i)$ and stores

$$U_d = \ell_d(n+1) - \ell_d(\ell_d+1)/2 \ \ge S_d$$

intervals. Both classes contain an unrestricted optimum.

Lemma 20 (Root-search work on a hereditary contour)**.** *Computing all unrestricted interval optima in this domain takes $O(n + S_d)$ work, with at most $3S_d$ tested roots.*

Proof. Bucket only existing cells by length in $O(n + S_d)$ work. Knuth's recurrence applies because every child and adjacent shorter interval is present. Let $r(i,j)$ be the leftmost optimal root and put $t(i,j) = r(i,j) - i \ge 0$. For length at least two, the root count is $2 + t(i+1,j) - t(i,j-1)$. Sum over the domain. A positive occurrence $t(a,b)$ cancels its negative occurrence whenever $(a, b+1)$ is present. An unmatched positive occurrence requires $b = E_d(a)$ and $E_d(a-1) = E_d(a)$. There is at most one per row, of size at most $E_d(a) - a$. Discard unmatched negative occurrences. Adding the length-one cells proves the $3S_d$ bound. Scanning nonexistent diagonal cells would lose this saving; the buckets are essential. □

Compose the exact interval costs. Use 2^d top-grid blocks and the preceding dyadic transitions, replacing height-limited lower costs by these unrestricted interval costs. A block starting after i gaps can finish after j exactly when $i < j \le E_d(i) + 1$. It contributes $C(i, j-1) + dW(i, j-1)$ and the starting boundary key cost $p_{i-1}(d - v_2(b))$ at $b > 0$. Coarse gaps retain their aligned power-of-two skip transitions. Every optimal tree is represented, while every reconstructed candidate is a valid full tree. Thus the resulting optimum is unrestricted.

The variable band needs another implementation detail. Reachable input rows, and their union of feasible output columns, may have holes. Scan once per top boundary to retain exactly output columns covered by some reachable row. Put $L(j) = \min\{i : E_d(i) + 1 \ge j\}$, a nondecreasing sequence with $L(j) \le j - 1$. In the transposed matrix, use finite true entries when $L(j) \le i < j$, and padding

$$Z + L(j) - i \quad (i < L(j)), \qquad Z + i - j + 1 \quad (i \ge j), \qquad Z = (n + d + 1)W + 1.$$

All finite prefix costs are smaller than Z. Within the band, exact interval costs are Monge. Across padding, opposite-side differences $i_1 + i_2 + 1 - j - L(j)$ are nonincreasing; same-side preferences also move right. Thus the matrix is totally monotone, even after removing arbitrary rows and columns, and every searched column has a feasible minimum. SMAWK, scans, and coarse transitions together cost $O(n2^d)$. Backpointers recover every key and gap contribution.

Theorem 21 (Exact mass-frontier constructor)**.** *The adaptive implementation constructs a full optimal BST in*

$$O\Big(n \log(n+1) + \min_{d \geq 0}\{S_d + n2^d\}\Big)$$

arithmetic work. The uniform-envelope implementation replaces S_d by U_d. At the selected depth, storage is $O(n + S_d + n2^d)$ or $O(n + U_d + n2^d)$ respectively. Both remain quadratic in the worst case.

Proof. The interval and composition arguments establish exactness at every depth. Start with depth zero and the full Knuth table. Evaluate successive depths while $n2^d$ is smaller than the best work proxy so far; each contour costs $O(n)$. No later depth can improve the proxy because its top term alone is too large. At most $O(\log(n+1))$ depths are evaluated. Depth zero uses Knuth directly, omitting a redundant top pass and its lower-order n proxy term. The preceding work and storage bounds now apply. □

This constructor has no posterior certificate failure followed by a second solve. It may select the full table initially. The Python prototype uses dictionaries; bounded integer top coordinates also allow direct arrays. When each key has weight at least $a > 0$ and $W = O(na)$, we have $S_d = O(n^2/\phi^d)$, where $\phi = (1 + \sqrt{5})/2$. Balancing against $n2^d$ recovers the known exponent $1 + \log_{2\phi} 2 \simeq 1.59023$.

A scoped asymptotic separation. Let n range over fourth powers and $L = n^{3/4}$. Give every key and gap weight $A = n^2$, except a central block of L keys and its $L + 1$ gaps, all of weight one. Every weight is positive and $W = \Theta(nA)$. An admitted interval contains at most $m = O(n/F_{d+1})$ heavy keys. Intervals wholly inside the light block number $O(L^2)$. Other starts contribute $O(nm)$ ordinary endpoints; only $O(m)$ outside starts can reach across a block boundary and gain $O(L)$ endpoints, and starts inside add at most $O(Lm)$ heavy endpoints. Consequently

$$S_d = O(n^2/F_{d+1} + L^2).$$

With $d = \lfloor \log_{2\phi} n \rfloor$, the adaptive work is $O(n^{1+\log_{2\phi} 2} + n^{3/2}) = O(n^{1.59024})$.

Every uniform-envelope plan, however, has proxy $\Omega(nL)$. If $2^d \geq L$, the top term suffices. Otherwise $F_{d+1} = O(L^{\log_2 \phi})$, so W/F_{d+1} exceeds the entire light block's $O(L)$ mass. Hence $\ell_d \geq L$ and the uniform table has at least $nL/2$ cells. This gives $\Omega(n^{1.75})$ at every depth. The separation is within the specified architecture, not against every prior OBST algorithm or combinations of our shortcuts. In particular, this light block also offers an opportunity for certified contraction.

4.9 Dominant-gap separators and local mass frontiers

The global mass bound can admit many unnecessary intervals when a single heavy outcome dominates otherwise ordinary regions. We first split at certified outcomes, then apply the frontier constructor with each terminal component's own total mass. The following exchanges retain all key and gap weights. Their priority is not established.

Lemma 22 (Restriction to an ordered interval)**.** *A BST restricted to a contiguous key interval and its endpoint gaps has no greater depth for any retained outcome. If the old root is outside the retained key interval, every retained outcome loses at least one comparison.*

Proof. Delete each comparison at an outside key and follow the branch containing all retained outcomes. Comparisons at retained keys remain. This gives the required BST, with each retained search path a subsequence of its old path. The argument also applies to an empty key interval containing one gap. □

Fix an interior gap g in a current component. Its adjacent keys are $a = g - 1$ and $b = g$. Let L contain the keys strictly left of a, with its endpoint gaps, and let R contain those strictly right of b with its endpoint gaps. Their masses are W_L, W_R, and the component mass is

$$W = W_L + p_a + q_g + p_b + W_R.$$

Theorem 23 (Interior-gap separator)**.** *If*

$$q_g + \min(p_a, p_b) \;\geq\; W_L + W_R, \tag{10}$$

there exists an optimal tree with gap g at depth two. Its top two keys are a, b, in either orientation, and its remaining subtrees are L, R. Writing C_L, C_R for their unrestricted optimal relative costs gives

$$\mathrm{OPT} = C_L + C_R + W + q_g + \min(p_a + W_L,\; p_b + W_R). \tag{11}$$

At equality, other optimal trees need not have this shape.

Proof. Take any tree in which g has depth at least three. Restrict it to L, R using Lemma 22. If its root lies in L, make a the new root, attach restricted L on its left, and make b its right child, with g on b's left and restricted R on its right. Every L outcome deepens by at most one. Every R outcome lost the old root comparison, then gains two, so also deepens by at most one. Key a saves at least one comparison, key b does not deepen, and gap g saves at least one. The cost change is at most $W_L + W_R - p_a - q_g$. For an old root in R, the mirror exchange costs at most $W_L + W_R - p_b - q_g$.

If the old root is a, keep it and promote b to its right child. The L depths do not increase and R depths increase by at most one. As a, b are adjacent and g was deeper than two, b had depth at least two and saves at least one comparison. The change is at most $W_R - p_b - q_g$. The case with old root b is symmetric. Condition (10) makes each bound nonpositive, so the exchange can be applied to an optimum. An interior gap cannot have depth below two; at depth two its bounding ancestor keys must be exactly a, b.

Both orientations leave the same two independent optimization problems. With root a, the cost is $C_L + C_R + W + q_g + p_b + W_R$; with root b it is $C_L + C_R + W + q_g + p_a + W_L$. Minimize these two values after solving each outside interval once. This proves (11) without duplicated recursive branches. □

Replacing the smaller neighboring weight in (10) by the larger one is false. For $p = (100, 1, 50)$ and $q = (1, 1, 1, 1)$, the weakened test accepts $g = 1$ since $101 \geq 53$. Yet the full optimum is 212 and forcing this gap to depth two costs 261. The valid test rejects it since $2 < 53$. This strictly positive counterexample is retained.

Endpoint and key certificates. For a component's first key and left endpoint gap, $2(p_i+q_i) \geq W$ suffices to put that key at the root in some optimum; the right endpoint is symmetric. If its original depth is $h \geq 1$, its endpoint gap has depth $h+1$. Promotion saves $h(p_i+q_i)$ while restricting and reattaching the other interval increases cost by at most $W-p_i-q_i$. Likewise, $2p_r \geq W$ certifies an arbitrary key r as root: promote it and attach the two restrictions, increasing every other depth by at most one while saving at least one comparison on r. These are sufficient conditions, not a general largest-weight root rule.

Avoiding quadratic separator scans. Components remain contiguous intervals of the original input, with no weight aggregation. Store the static scores

$$G_j = 2q_j + p_{j-1} + p_j + \min(p_{j-1}, p_j), \qquad K_j = 2p_j.$$

An interior gap in component $[i,k)$ satisfies (10) exactly when $G_j \geq W(i,k)$, for $i < j < k$. Two range-maximum segment trees find qualifying gaps and keys in $O(\log(n+1))$ work per query, after $O(n)$ construction. Prefix sums give component masses in constant work. Each separator removes one or two keys, and terminal intervals are disjoint. Only terminal slices are copied, so a long chain does not cause quadratic copying or rescanning. The implementation tests endpoints first and builds the indexes only when an endpoint test fails on both sides. All-zero components are returned as balanced trees directly.

Theorem 24 (Local-frontier composition)**.** *For terminal components t with m_t keys and local admissible-interval counts $S_{t,d}$, the constructor is exact and uses*

$$O\left(n\log(n+1) + \sum_t \min_d \{S_{t,d} + m_t 2^d\}\right)$$

arithmetic work. If $m_t \leq b$ for every terminal, the bound is $O(n\log(n+1)+nb)$. Sequential reuse of terminal scratch storage gives $O(n + \max_t\{S_{t,d_t} + m_t 2^{d_t}\})$ space at selected depths d_t. The unrestricted worst case remains quadratic.

Proof. The separator identities reduce the optimum to disjoint exact subproblems. There are $O(n)$ separator calls and $\sum_t m_t \leq n$. The local frontier bounds apply separately, including parameter selection since $\sum_t m_t \log(m_t+1) \leq n\log(n+1)$. Each terminal may select depth zero and use its full Knuth table, so $\sum_t m_t^2 \leq nb$ when all sizes are at most b. Include the reconstructed output and separator stack in $O(n)$ space; completed terminal scratch arrays need not remain allocated. □

Two exact structured regimes. For $p_i = q_i = 2^i$ and $q_n = 2^n$, the total mass is $3\cdot 2^n - 2$ and the rightmost key/gap pair has mass $3\cdot 2^{n-1}$. The right endpoint certificate passes, and the remaining prefix has the same form. The optimal chain is constructed with n endpoint decisions, no range index, and no terminal DP: $O(n)$ arithmetic work. At the final single key the implementation selects the equivalent left endpoint rule. Weights have $O(n)$ bits, so this is not a linear bit-complexity assertion.

For a different positive mixed family, recursively plant a heavy central gap separating two outside intervals, stopping below 32 keys. Set each planted weight to four times its component mass after its children have been planted. Each planted gap satisfies the theorem, and all terminal sizes are below 32. This gives $O(n\log n)$ arithmetic work on the family, including varying positive background weights on both keys and gaps. Neither regime is an unrestricted subquadratic result.

A positive obstruction to the combined constructor. For $n \geq 5$, set every key and gap weight to one except $q_1 = q_{n-1} = M = n^3$. Then $W = 2M + 2n - 1$. The largest gap score is $2M + 3 < W$, endpoint scores are four, and key scores are two, so no separator fires. Indeed the frontier planner selects depth zero and the full table. To see this, write $\tau(k) = k(k+1)/2$ and $h = \lfloor (n-1)/2 \rfloor$. Depth one admits all $\tau(n)$ cells. At depth two, an interval containing one heavy gap has mass $M + 2\ell$, so the cutoff $W/2 = M + n - 1/2$ admits it exactly for length $\ell \leq h$. There are four disjoint sets of h such intervals, and $\tau(n-4)$ intervals avoiding both heavy gaps. Thus

$$S_2 + 4n = \tau(n-4) + 4h + 4n = \tau(n) + 6 + 4h > \tau(n).$$

For $d \geq 3$ with $2^d \leq n$, we have $3 \leq F_{d+1} \leq 2^d \leq n$. The cutoff is below M but above every all-unit interval mass, so $S_d = \tau(n-4)$ and

$$S_d + n2^d \geq \tau(n-4) + 8n = \tau(n) + 4n + 6 > \tau(n).$$

If $2^d > n$, the top term alone exceeds $n^2 \geq \tau(n)$. Every positive depth is therefore worse than the direct Knuth plan. This is a quadratic obstruction to the specified constructor, not a general OBST lower bound. An independent contour check rejected the initial assumption that all one-heavy-gap intervals were admitted at depth two; the correct length cutoff above is retained in the test. The family motivates joint treatment of several comparable heavy outcomes. Eight full instances through 256 keys match Knuth and reconstructed costs, with cubic checks through 32 keys. Four larger instances through 65,536 keys check plans only; at that size the selected table has 2,147,516,416 cells, not a completed full solve.

4.10 Joint heavy-outcome cores

The two-gap obstruction motivates selecting several heavy outcomes together. The following certificate uses their necessary comparison keys as an ancestor-closed core. It preserves the complete mixed-weight objective and solves every intervening interval exactly. Priority of the sufficient condition has not been established.

For $n \geq 1$, choose a nonempty set $\mathcal{H}$ of positive-weight key or gap outcomes. Let A contain every selected key and the adjacent key(s) of every selected gap. Write $m = |A|$, $\theta = \min_{x \in \mathcal{H}} w_x$, and let W_0 be the total weight outside $\mathcal{H}$. Both adjacent keys of an interior gap are ancestors of that gap in every BST. Thus each A-key lies on the search path to some selected outcome, possibly itself. Call a tree *core-closed* if no key outside A is an ancestor of an A-key. Then A forms a top BST and the intervals between consecutive A-keys, including both end intervals, are its payloads. Selected gaps remain singleton core gaps.

Lemma 25 (Projection preserving selected depths)**.** *Every BST can be reduced to a BST on A without increasing any selected outcome's depth. If the original tree is not core-closed, at least one selected outcome becomes strictly shallower.*

Proof. Consider an outside key r whose subtree contains an A-key. If just one child contains A-keys, replace r by that child. The other branch contains no selected outcome: any nonempty subtree containing a selected gap must contain one of its neighboring keys, and a selected empty child gap would make r itself an A-key. All retained selected outcomes save a comparison.

If both children contain A-keys, let a be the largest such key in the left subtree. Replace r by a, restrict the old left subtree to keys below a and their endpoint gaps, and retain the old right subtree. No selected outcome lies strictly between a and r: a selected key would belong to A, and a selected gap would put an adjacent key in A before or at r. All A-keys survive. Selected outcomes

in the left restriction lose zero or more comparisons by Lemma 22; right depths are unchanged. If a is selected, it moves strictly upward. Otherwise it must border a selected gap on its left. Its right gap cannot be selected, since that would put $a+1$ in A, contradicting either the maximality of a or $r \notin A$. The selected left gap loses the comparison at the old a. Thus some selected depth decreases in either case.

These exchanges may discard light keys, which will be reinserted later. Each exchange reduces the number of outside keys and retains A. Repeat until no outside key has an A-descendant, then discard the remaining anchor-free branches. Selected gaps are preserved throughout. This produces the asserted core tree. Arbitrary deletion of nonretained comparisons would not suffice: a nonretained key can separate retained outcomes on both sides; the explicit promotion above resolves that case. □

A first sufficient condition is $\theta \geq (m+L)W_0$, where $L = \lceil \log_2(n+1) \rceil$. Project an optimum and reinsert balanced payloads: the selected cost falls by at least θ, and the new light cost is at most $(m+L)W_0$. The next theorem substantially sharpens the height term using the earlier Fibonacci lemma.

Corollary 26 (Protected-height core certificate)**.** *Let $W_{\mathcal{H}}$ be the selected total weight and*

$$H = \max\{j \in \{1, \ldots, m\} : F_j \theta \leq W_{\mathcal{H}}\}, \qquad F_1 = F_2 = 1.$$

If

$$\theta \;\geq\; (H+L)W_0, \tag{12}$$

there exists a core-closed optimum. If the payload intervals have masses W_t and unrestricted relative optimum costs C_t, then

$$\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{original } A\textit{-key weights},\ (W_t)_t) + \sum_t C_t.$$

Equality in the certificate does not force every optimum to be core-closed.

Proof. In a protected-only core instance, keep selected weights and set every other weight to zero. Every internal subtree contains an A-key and therefore a selected outcome of weight at least θ in that subtree. By Lemma 2, a depth-d internal subtree of a protected-only optimum has mass at most $W_{\mathcal{H}}/F_{d+1}$. Its height, including external leaves, is consequently at most H.

Project a non-core-closed full optimum by Lemma 25, saving at least θ in selected weighted cost. Replace the projected core by a protected-only optimum; its selected weighted cost cannot increase, although individual depths may change. Reinsert balanced payloads. Every unselected key comparison length or gap depth is at most $H+L$, so the new light cost is at most $(H+L)W_0$. The old light cost was nonnegative. Condition (12) makes the exchange nonincreasing, establishing the existence of a core-closed optimum.

At core-gap depth d, an exact payload contributes $dW_t + C_t$. The core and payload relative costs therefore optimize independently, proving the identity. Empty intervals contribute zero relative cost and keep their gap weight. The final full-weight core is optimized without a height restriction: H bounds the protected-only witness, not the output core. □

For k equal selected weights, $H = O(\log(k+1))$, whereas m can be $2k$. The certificate thus handles much less extreme dominance than the size-based version. It does not remove unselected weights from the final objective.

A threshold sweep with no repeated interval scans. The implementation selects all outcomes of weight at least a threshold θ. Key i is in A precisely when $\max(p_i, q_i, q_{i+1}) \geq \theta$. Sort these activation weights and all outcome weights. Sweep thresholds upward, grouping equal values. Before each threshold, activate as unanchored each key whose activation weight is strictly smaller. A disjoint-set structure maintains the contiguous payload runs and their total triangular state count. A singleton adds one; merging runs of sizes a, b adds ab. Union by size bounds each find by $O(\log n)$, even without using an inverse-Ackermann analysis.

At each certified threshold, evaluate

$$\tau(m) + \sum_t \tau(k_t), \qquad \tau(k) = k(k+1)/2.$$

Keep the minimum proxy, accepting only a strict improvement on $\tau(n)$. This optimizes a Knuth-state proxy, not actual frontier runtime. Fibonacci values through index n take $O(n)$ additions. Since $W_{\mathcal{H}}/\theta$ is nonincreasing during the upward sweep, a single decreasing Fibonacci index supplies the height bound in $O(n)$ total index decrements; cap it by m. The entire selection uses $O(n \log(n+1))$ arithmetic work and $O(n)$ arithmetic storage. Failure invokes the preceding separator constructor, adding screening work without a failed quadratic solve.

Kernels, reconstruction, and scoped bounds. Both core and payloads use either exact local frontiers or Knuth. Before each kernel, uniform key weights p and uniform gap weights q are recognized. Such a tree costs $(p+q)I + np + 2nq$, where I is its unweighted internal path length. A recursively balanced tree minimizes I and is built in linear work. This familiar uniform-weight shortcut is not claimed as new.

If $K(s)$ bounds the chosen exact kernel on s keys, successful construction costs

$$O\left(n \log(n+1) + K(m) + \sum_t K(k_t)\right), \qquad m + \sum_t k_t = n.$$

Sequential scratch reuse retains only the payload outputs between solves. The general work and space bounds remain quadratic. A uniform residual background and k selected outcomes give $m \leq 2k$ and work $O(n \log(n+1) + k^2)$. If residual weights lie in $[a, Ca]$ for fixed $a > 0$ and C, each payload's local frontier costs $O(s^\alpha)$, where $\alpha = 1 + \log_{2\phi} 2 < 1.59024$. A certified plan then costs

$$O\left(n \log(n+1) + k^2 + \sum_t k_t^\alpha\right).$$

With payload sizes at most B, the last sum is at most $nB^{\alpha-1}$. These are bounds for the certified plan; minimizing the triangular proxy need not minimize this sharper expression.

For the preceding obstruction, $q_1 = q_{n-1} = n^3$ with unit background, $n \geq 5$, we have $m = 4$, $H = 3$, and $W_0 = 2n - 1$. The certificate passes, so the four-key core and uniform payloads take $O(n \log n)$ total work. The previous separator/frontier constructor used all $n(n+1)/2$ cells. Bounded positive mixed background and sufficiently large equal selected gap weights instead give an $O(n^{1.59024})$ plan. This resolves that scoped obstruction; it is not an unrestricted subquadratic theorem.

4.11 Residual-cost certificates for heavy-outcome cores

The preceding height certificate charges all light search cost against one saved heavy comparison. Every competing tree must itself pay for light searches. Crediting a valid lower bound for that

unavoidable cost gives a stronger test for a fixed selected set. This is an additive exclusion certificate built on Lemma 25; it does not assert a new Huffman algorithm or an established priority claim.

Write the original nonnegative weight vector as $w = w_H + w_0$, where w_H retains selected positive outcomes and is zero elsewhere. Let A be the necessary core keys from Section 4.10, let P be the exact protected-only optimum on that core, and let B_0 be any valid lower bound on the full n-key problem with residual weights w_0. In particular, the order-free mixed-node relaxation of Section 5.1 supplies B_0. Residual weights on both keys and gaps are retained.

Corollary 27 (Residual exclusion certificate)**.** *Every tree that is not core-closed has full cost at least*

$$P + \theta + B_0, \qquad \theta = \min_{x \in \mathcal{H}} w_x.$$

Consequently, if a core-closed witness has full cost $U \le P + \theta + B_0$, some global optimum is core-closed. More strongly, if the exact optimum C_A within the core-closed class satisfies this inequality, then C_A is the unrestricted optimum.

Proof. Project any non-core-closed tree T onto A. Its selected weighted cost decreases by at least θ, and the projected selected cost is at least P. Thus $C_{w_H}(T) \ge P + \theta$. Independently, $C_{w_0}(T) \ge B_0$. Add the inequalities, using linearity of tree cost in the weights. A witness at or below this excluded-class lower bound cannot be worse than every non-core-closed tree. Optimizing within the core-closed class therefore gives a global optimum. Equality allows both classes to contain optima. □

A test before solving the payloads. Solve the protected core, retaining one optimal shape, and attach balanced payloads. This produces a full witness of cost $P + U_0$, where U_0 is its exact residual cost. The inexpensive sufficient condition becomes

$$U_0 - B_0 \le \theta. \tag{13}$$

Only after it passes does the default constructor solve the full-weight core and all payloads exactly. Thus failed witnesses need no exact payload tables. For the same selected set, this test includes the preceding height certificate: $U_0 \le (H + L)W_0$ and $B_0 \ge 0$. The implemented threshold selection is not exhaustive, so this logical implication is not a claim that the new selector always chooses the old selector's successful set.

An optional refinement solves one core-closed candidate even after its witness fails and tests $C_A \le P + \theta + B_0$. For example, $p = (0, 0, 1)$, $q = (0, 4, 1, 1)$, and $\theta = 4$ give $P = 8$, $B_0 = 4$, and excluded-class lower bound 16. The first witness costs 17, while the exact closed candidate costs 14 and is globally optimal. A failed witness is not evidence that the core structure is invalid.

Several candidates without quadratic screening. A single minimum-proxy threshold can include ordinary outcomes whose weights are too small to pay for the exchange. The revised selector keeps two representatives in each dyadic core-size band: the least $\tau(m) + \sum_t \tau(k_t)$ proxy and the highest threshold. It considers only $m \le b = \lfloor \sqrt{n \lceil \log_2(n+1) \rceil} \rfloor$ and only strict proxy improvements over $\tau(n)$. The existing sorted activation sweep finds these representatives in $O(n \log(n+1))$ work. There are $O(\log(n+1))$ candidates, and their squared core sizes sum to $O(b^2)$, because at most two lie in each geometrically increasing band.

Sort the original key and gap profiles once. For a threshold θ, remove the profile entries at least θ and append the same number of zeros. The remaining profile is still sorted. The mixed-node two-queue bound then takes $O(n)$ work per candidate, without another sort. Building and evaluating each balanced witness also takes $O(n + m^2)$ work. All candidate screening therefore

costs $O(n \log(n+1))$ arithmetic operations. Candidates are tried in increasing proxy order until a witness succeeds. If none succeeds, the optional mode refines only the first candidate; limiting refinement to one avoids an $O(\log n)$ factor on failed exact payload solves. Failure invokes the preceding exact constructor.

A successful plan retains the bound

$$O\left(n \log(n+1) + K(m) + \sum_t K(k_t)\right).$$

Both modes keep quadratic general worst-case work and storage. The witness screen can use $O(n \log n)$ storage for protected-core tables; the final kernel can still require quadratic space. Ordinary Huffman coding is classical [9]; the mixed-node relaxation and its limitations are already described in Section 5.1. The result here is the certified use of a residual lower bound against a specified excluded class, rather than a relaxation substituted for the full objective.

A linear-weight separation from the height certificate. Let $n = 2^h \geq 8$, put every original key and gap weight equal to one, and replace q_1, q_{n-1} by an equal selected weight $\theta > 1$. The necessary core has four keys. Its protected optimum is $P = 5\theta$: each selected interior gap needs at least two comparisons, and both cannot have depth two because their adjacent-key pairs are disjoint. Depths two and three are attainable.

Use the core whose root is original key 1, left child is key 0, right child is key $n-2$, and right-right child is key $n-1$. The sole nonempty payload contains $n-4$ keys and hangs at depth two. The residual cost of this balanced-payload witness is

$$U_0 = (2h+3)n - 6h - 3.$$

Indeed, the core's residual cost is $4n-1$, while $I_{\min}(n-4) = (h-2)n - 3h + 5$ gives payload cost $2I_{\min}(n-4) + 3(n-4) = (2h-1)n - 6h - 2$.

For the residual problem all key weights are one, all ordinary gap weights are one, and the two selected gaps have weight zero. The identity $E = I + 2n$ rewrites any residual tree cost as $\sum_j (q_j + 1)d_j - n$. The order-free leaf weights are therefore two ones and $n-1$ twos. Huffman first merges the two ones; the resulting $n = 2^h$ weights are all two and have a perfect code of height h. Hence the exact order-free residual lower bound is

$$B_0 = 2nh - n + 2, \qquad U_0 - B_0 = 4n - 6h - 5.$$

Thus $\theta \geq 4n - 6h - 5$ suffices for core closure. In particular, $\theta = 4n$ passes, while the preceding height condition requires $\theta \geq (h+4)(2n-1)$ and rejects it.

At $\theta = 4n$, the described core is also optimal with the full collapsed weights. The four protected-optimal shapes have residual core costs $4n-1$ or $6n-9$; every other core shape has selected cost at least $6\theta > 5\theta + 4n - 1$. It follows that the full optimum is

$$\mathrm{OPT} = (2h+23)n - 6h - 3. \tag{14}$$

The new constructor uses a constant-size core and a uniform payload, for $O(n \log n)$ work and $O(n)$ space on this strictly positive mixed-weight family. These weights and the construction's intermediate values have $O(\log n)$ bits. This is a separation from the preceding height certificate, not a quadratic lower bound for its fallback frontier or for other algorithms.

4.12 Coupled two-gap skeleton certificates

The preceding certificate adds independent protected and residual minima. We now retain enough geometry to couple these costs at the first possible violation. Select internal gaps $a < b$, with weights M_1, M_2, and let $A = \{a-1, a, b-1, b\}$, $m = |A| \in \{3, 4\}$, and $\theta = \min(M_1, M_2)$. A tree is A-closed when no outside key is an ancestor of an A-key. All weights below may be zero.

Lemma 28 (First violating skeleton)**.** *In a nonclosed tree the selected gap depth sum D is at least $m+2$. If $D = m+2$, the union of the two search paths has exactly $m+1$ keys: the keys in A and one outside key. The paths share only the root.*

Proof. Each key neighboring a gap is an ancestor of that gap. Thus the path union contains A. Nonclosure puts an outside ancestor of an A-key in the same union. If $L \geq 1$ keys are shared, its size is $D - L$, so $D \geq m + 1 + L \geq m + 2$. Equality forces $L = 1$ and union size $m + 1$. □

The one outside key is called the bridge. Its inorder position is in one of the light intervals between keys of A. Enumerating binary-tree shapes on A and the bridge takes at most $C_5 = 42$ shapes per interval. Only nonclosed shapes with selected depth sum $m + 2$ are retained.

Lemma 29 (Unordered mixed forests)**.** *Let $F_k(P, Q)$ be the minimum mixed cost of a k-root binary forest with roots at depth zero, internal-weight multiset P, and leaf-weight multiset Q, where $|Q| = |P| + k$. Weights may be permuted within each type and empty trees are allowed. Sort P ascending. For each weight in that order, merge the two smallest current leaf weights with it, add the resulting weight to the cost, and reinsert it as a leaf. After all merges the accumulated cost is $F_k(P, Q)$.*

Proof. Choose a deepest internal node in an optimal forest; it is a cherry. Exchange weights to put a smallest internal weight there and the two smallest leaf weights at its children, without increasing cost. Contract the cherry to a leaf of their total weight. Its removed cost is exactly that total. Conversely, expanding this leaf in any feasible contracted forest has that added cost. Induction proves the recurrence, including forests consisting only of leaves. Nonnegative weights ensure the generated merge weights are nondecreasing. Two queues therefore take linear work after sorting, for total $O(n \log n)$ work. □

Delete A from inorder to obtain $m+1$ intervals. Two are the selected singleton gaps. For every other region s, write its weights as (P_s, Q_s) and mass as W_s. In a retained skeleton, let h_s be its hole depth. An ordinary region contributes the lower bound

$$h_s W_s + F_1(P_s, Q_s).$$

If the bridge is in region t, there are two holes. Lower their depths to their minimum h. If the bridge comparison length is ℓ, then $h \geq \ell$, since both holes descend from it. All other internal slots in this relaxed region have comparison length at least $h + 1$. Exchanging internal weights therefore places a largest regional weight x at the bridge. The relaxed contribution is

$$\ell x + h(W_t - x) + F_2(P_t \setminus \{x\}, Q_t). \tag{15}$$

This exchange applies after lowering the two hole depths; it does not identify the bridge key in the original ordered tree. Keep the exact costs of all A-keys and both selected gaps. Minimizing the sum of these costs and the regional bounds over retained skeletons gives B_{first}, a valid lower bound for $D = m + 2$.

Let B_0 be the unordered mixed-tree optimum after zeroing the two selected gap weights, retaining all other weights. Each internal gap has depth at least two, so define

$$B_{\text{basic}} = 2(M_1 + M_2) + (m-2)\theta + B_0,$$
$$B_{\text{tail}} = 2(M_1 + M_2) + (m-1)\theta + B_0.$$

The first bounds every nonclosed tree; the second bounds those with $D \geq m+3$.

Corollary 30 (Coupled exclusion certificate)**.** *Every nonclosed tree has cost at least*

$$B = \max\{B_{\text{basic}}, \min(B_{\text{first}}, B_{\text{tail}})\}.$$

If the first class is empty, replace the inner minimum by B_{tail}. An A-closed witness of cost $U \leq B$ certifies that some unrestricted optimum is A-closed. Its exact construction then requires independent exact solutions of the intervening intervals and a constant-size core. The screening work is $O(n \log n)$.

Proof. Lemma 28 partitions excluded trees into the first and tail classes. Each has its stated lower bound, and both have B_{basic}. The witness is no more costly than any excluded tree. An optimum of the closed class is consequently globally optimal. There are constantly many regions, forest computations, and skeletons; Lemma 29 gives the screening bound. □

The implementation first uses midpoint payloads and an exact full-weight core as the witness. Optional refinement computes the exact closed candidate when this witness fails. An uncertified attempt falls back to the preceding exact method; the general worst case remains quadratic.

4.12.1 A family with arbitrary nonuniform key positions

Constant key weights require a scope qualification. For $p_i = c$, the internal/external path-length identity $E = I + 2n$ gives

$$C(T; c, q) = \sum_g (q_g + c) d_T(g) - cn.$$

Such inputs reduce to alphabetic leaf-weight problems. The next family instead has a linear number of nonconstant key weights.

Corollary 31 (Nonuniform-key family)**.** *Let $n = 2^h$, $h \geq 5$, $r = n/16$, and $s = r - 1$. All gaps have weight one except $q_1 = q_{n-1} = M = 3n$. All keys have weight one except any s keys in positions $2, \ldots, n-3$, which have weight two. The midpoint witness passes Corollary 30 for every placement of these keys. Combining it with the existing mass-frontier construction takes $O(n^\alpha)$ arithmetic work, where $\alpha = 1 + \log_{2\varphi} 2 < 1.59024$. For $h \geq 6$, the earlier fixed-pair bound B_{basic} cannot certify even the exact closed optimum on this family.*

Proof. For unit keys write $U_0 = (2h+3)n - 6h - 3$. The middle payload has $n-4$ keys, mass $2n-7$, and optimal relative cost $(2h-1)n - 6h - 2$. The minimum four-key core has selected depth sum five, middle hole depth two, and other contribution thirteen, giving cost $5M + 4n - 1$. For completeness, the distinct other triples (selected depth sum, middle depth, other contribution) among its 14 shapes are $(5,3,12), (6,3,13), (6,3,15), (6,4,14), (7,4,13)$. None is cheaper. Thus the unit-key closed optimum is $5M + U_0$. A midpoint payload attains that baseline cost and has maximum relative key comparison length h, so the nonuniform witness satisfies

$$U \leq 5M + U_0 + (h+2)s. \tag{16}$$

Let $G_k(t)$ be the minimum sum of comparison lengths of t marked internal nodes in a k-root forest. Filling the capacities $k2^{\ell-1}$ at comparison lengths $\ell = 1, 2, \ldots$ gives

$$G_1(s) = (h-5)r + 1, \qquad G_2(s-1) = (h-6)r + 2.$$

In the unit first-violation relaxation the middle two-root forest has $n-5$ internal nodes. Its balanced profile has three leaves at depth $h-2$ and $n-6$ at depth $h-1$. The forest identity $E = I + 2(n-5)$ yields

$$F_2 = (2h-3)n - 6h + 5.$$

There are ten admissible five-key skeleton profiles. Their distinct triples (minimum middle hole depth d, bridge length ℓ, fixed core/endpoint contribution c) are

$$(2,2,16),\ (2,2,15),\ (3,3,13),\ (2,1,15),\ (3,1,14),\ (2,1,16).$$

Their unit costs are $6M + 2d(n-4) + \ell + c + F_2$; the minimum at $(2,1,15)$ is

$$B^{\text{unit}}_{\text{first}} = 6M + (2h+1)n - 6h + 5.$$

With nonuniform keys the bridge takes weight two in the relaxation. The remaining $s-1$ extra units receive hole shift at least two and forest comparison lengths summing to at least $G_2(s-1)$. The bridge adds at least one. Therefore

$$B_{\text{first}} \geq B^{\text{unit}}_{\text{first}} + 2s - 1 + G_2(s-1).$$

The residual unordered optimum is exactly

$$B_0 = 2nh - n + 2 + G_1(s).$$

For the unit baseline, absorbing the key weights leaves two leaf weights one and $n-1$ weights two. Their Huffman tree is a perfect h-level tree with one bottom leaf expanded into the two unit leaves; subtract n to recover mixed cost $2nh - n + 2$. Every tree pays at least $G_1(s)$ for the extra key units. The baseline optimal shape has its top $h-4$ internal levels full, so these same slots attain $G_1(s)$ simultaneously in the unordered problem, establishing equality.

Substitute $M = 3n$ in these bounds and in (16):

$$B_{\text{first}} - U \geq 5n/8 + h + 9 > 0, \qquad B_{\text{tail}} - U \geq 25n/16 + 7h + 8 > 0.$$

The witness certifies every placement. The remaining middle interval has key weights in $[1,2]$ and unit gap weights. Its bounded-positive mass-frontier bound is $O(n^\alpha)$, and the $O(n \log n)$ screening and full reconstruction fit within this bound. This exponent is inherited from that architecture, not a new unrestricted exponent.

Finally, every closed tree costs at least its unit-background closed optimum plus $G_1(s)$. Since $B_{\text{basic}} = 6M + B_0$, its deficit is at least $n - 6h - 5 > 0$ for $h \geq 6$. This is a separation from the earlier certificate for this fixed pair, not a claim about every alternative core its selector could attempt. □

The family's total mass is $O(n)$ and all key weights are at least one. Consequently the earlier global mass frontier already has the same $O(n^\alpha)$ guarantee on the entire instance. Corollary 31 is a certificate separation with arbitrary nonuniform key placements, not an extension of the known subquadratic input regime. Its practical value must be judged by actual construction work and matched measurements.

4.13 Certified pruning around affine weight profiles

The dense ruler example suggests retaining its structure symbolically. Here it bounds the states of a shape DP instead of the entries of a Monge matrix. Perturbations may change the optimal tree; we optimize over all retained shapes and separately certify global optimality. The result covers arbitrary signed perturbations on both keys and gaps, under explicit size conditions. The unrestricted question and priority remain open.

4.13.1 An affine potential and a signed residual bound

Let $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $s = n - (S-1)$, and $z = S - s$. A minimum internal-path-length shape expands s of S skeleton slots. Let T_0 expand the first s slots. Choose nonnegative slopes λ, μ and reference weights $\bar{p}_i = \lambda i$, $\bar{q}_i = \mu i$. Constant intercepts contribute equally to every shape in this class and can be omitted. Define

$$\gamma = \lambda + 2\mu, \qquad F(g) = (2\lambda + 2\mu)g - \lambda \operatorname{popcount}(g).$$

By (27), moving an expansion one slot right costs $\gamma + \lambda\nu_2(g+1) = F(g+1) - F(g)$. Thus the affine cost of a selected slot set A, relative to T_0, is

$$G(A) = \sum_{g \in A} F(g) - \sum_{g=0}^{s-1} F(g). \tag{17}$$

For $\gamma > 0$, it is at least γ times the number of unexpanded slots preceding expanded slots.

The analytical residuals $\delta_p = p - \bar{p}$, $\delta_q = q - \bar{q}$ may be signed. Actual problem weights remain nonnegative. All minimum-path-length shapes have the same key and gap depth multisets. Assigning the largest residual weights to the shallowest lengths, separately by type, gives a valid rearrangement lower bound even for signed residuals. With descending sorts indicated by $\downarrow$, it is

$$\begin{aligned} L_\delta = {} & \sum_{i=0}^{n-1} \delta_{p,i}^{\downarrow} \operatorname{bitlength}(i+1) + h \sum_i \delta_{q,i} \\ & + \text{sum of the smallest } 2s \text{ gap residuals.} \end{aligned}$$

The last term is zero for $s = 0$. Set

$$V = C_\delta(T_0) - L_\delta \geq 0. \tag{18}$$

For a class optimum T^*, comparison with T_0 gives

$$G(T^*) \leq C_\delta(T_0) - C_\delta(T^*) \leq V.$$

Adding a constant to all residuals of either type changes both terms in (18) equally. The bound is therefore independent of intercepts and takes $O(n \log n)$ work including sorting.

4.13.2 A certified corridor with few states

At a slot-prefix state let a be the number of zeros already used and b the number of ones still needed, with $0 \leq a \leq z$, $0 \leq b \leq s$. Increasing F makes the earliest available slots optimal on each side of this prefix. The exact minimum affine regret of any completion through it is

$$K(a,b) = \sum_{t=s-b}^{s-1} \big(F(t+a) - F(t)\big). \tag{19}$$

For $P(t) = \sum_{g=0}^{t-1} F(g)$, this is $P(s+a) - P(s-b+a) - P(s) + P(s-b)$, computable in constant time after a linear prefix pass. Every prefix of T^* satisfies $K(a,b) \le V$. Delete the others and run the existing two-transition shape DP using the *original* weights. The bound preserves a class optimum; unnecessarily retained paths are still feasible original shapes.

Theorem 32 (Affine-regret corridor)**.** *For $\gamma > 0$, put $D = \lfloor V/\gamma \rfloor$. Exact optimization over minimum internal-path-length shapes, including preprocessing and reconstruction, takes*

$$O\Big(n \log(n+1) + \min\{n^2, D \log(n+1)\}\Big)$$

arithmetic work and $O(n + \min\{n^2, D \log(n+1)\})$ auxiliary space.

Proof. Each term in (19) is at least γa, so a retained state must satisfy $ab \le D$. The two boundary segments have $S+1$ states; the total is bounded by

$$S + 1 + \sum_{a=1}^{z} \min\{s, \lfloor D/a \rfloor\} \le S + 1 + \min\{sz, D \operatorname{bitlength}(z)\}. \tag{20}$$

Group the harmonic sum into dyadic blocks for the last inequality. Popcounts and P are computed in a linear pass, each state has at most two transitions, and stored predecessors reconstruct all keys. Sorting for (18) adds $O(n \log n)$ work. □

For $\gamma = 0$ the safe bound is the full shape grid. If $V = 0$, the reference is already class-optimal; a perfect skeleton also bypasses DP. The implementation tries both orientations and uses the smaller certified state bound. The default full solver rejects a bound exceeding $n \operatorname{bitlength}(n)^2$ before starting the shape DP. An unbudgeted class optimizer remains available for the theorem above.

Global optimality still requires the mixed residual-Huffman gate. It also provides an exact preflight rejection: let L_{class} be the same order-free fixed-profile lower bound on the original weights and L_{out} the gate's lower bound for every nonminimum-path-length tree. If $L_{\text{class}} > L_{\text{out}}$, no class candidate can satisfy the gate. The solver goes directly to Knuth before fitting or shape optimization. This rejects an impossible certificate, not the possibility that a class tree is optimal. When preflight passes, its sorted lists and lower bound are reused. An unfiltered mode is retained as a performance control.

4.13.3 Robust fitting and full mixed-weight guarantees

For a list of length N, let $m = \lfloor N/2 \rfloor$. Take the median of the disjoint secants $w_{i+m} - w_i$, $0 \le i < m$, divide by m, round to the nearest integer, and clip at zero. Sorting costs $O(n \log n)$. If $w_i = A + \lambda i + u_i$ with integer $\lambda \ge 0$, the sum of secant errors is at most $E = \sum |u_i|$. Hence its median differs from λm by at most $E/\lceil m/2 \rceil$. The odd unused endpoint remains in every cost and certificate. Fitting quality controls the bound's usefulness; arbitrary fitted slopes still give valid pruning inequalities.

Corollary 33 (Arbitrary perturbations of a positive linear profile)**.** *Let $n \ge 32$, $B = 4(n+1)^3$, and*

$$p_i = B + i + u_i, \qquad q_i = B + i + v_i,$$

with integer perturbations. The implemented full constructor has the following guarantees, including inference of slopes, the global certificate, and complete reconstruction:

1. *If $\sum |u_i| + \sum |v_i| \le n$, time and auxiliary space are $O(n \log^2(n+1))$.*

2. *If* $|u_i| \le 1$ *and* $|v_i| \le 1$ *independently at every position, time and auxiliary space are* $O(n \log(n+1))$.

Both types of weights are positive and may be nonmonotone. The returned tree is globally optimal in the full mixed-weight problem.

Proof. For the first case the forward fits recover $\lambda = \mu = 1$: $E/(m\lceil m/2 \rceil) < 1/2$ for either list at these sizes. Key lengths range from one to $h+1$ and gap lengths from h to $h+1$, even in the relaxed profile assignment. Writing $E_p = \sum |u_i|$ and $E_q = \sum |v_i|$,

$$V \le hE_p + E_q \le hn, \qquad \gamma = 3.$$

Equation (20) gives $O(n \log^2 n)$ states and a bound below the default budget. Choosing the better orientation cannot increase that bound.

For the second case every secant error is at most two, so the forward slopes are again recovered. Let $\rho_p, \rho_q \le 2$ be residual ranges. The total key depth deficit from $h+1$ is

$$Q = n(h+1) - (I_{\min}(n) + n) = 2S - h - 2.$$

The profile's possible key gain is at most $\rho_p Q$. For gaps it is at most $\rho_q \min\{2s, z\}$, using either the deeper set or its complement. Thus $V < 6S$, $D < 2S$, and (20) gives $O(n \log n)$ states, again below the default budget for $n \ge 32$.

It remains to certify unrestricted optimality in both cases. Put $a_0 = \min p$, $b_0 = \min q$, and $R_0 = \sum(p - a_0) + \sum(q - b_0)$. In the first case,

$$a_0 + b_0 \ge 2B - n, \qquad R_0 \le n^2 + (n+1)E_p + (n+2)E_q \le 2n^2 + 2n.$$

In the second, $a_0 + b_0 \ge 2B - 2$ and $R_0 \le n^2 + 4n + 2 \le 2n^2 + 2n$. The residual cost of any class candidate is at most

$$(h+1)R_0 \le 2n(n+1)^2 < 2B - n.$$

Every tree outside the class pays at least one extra unit of internal path length, whose constant-baseline penalty is $a_0 + b_0$. Therefore the existing global gate passes even with zero residual lower bound. In particular, preflight cannot reject these candidates. The computed class optimum is a full optimum. All weights and intermediate costs have $O(\log n)$ bits on these families, so the arithmetic bounds also hold in a word RAM with $O(\log n)$-bit words. □

These guarantees allow perturbations at arbitrary positions and do not require the reference tree to remain optimal. For example, at $n = 24$, $B = 62{,}500$, $p_i = B + i + (-1)^i$, $q_i = B + i$, the left-filled tree costs 13,252,388. Moving its expansion from slot 7 to slot 9 uses two adjacent exchanges and reduces the cost to the exact optimum 13,252,387. The previous one-direction exchange certificate rejects this input; the corridor DP finds and certifies the better tree.

4.14 Exact transfers for joint key/gap signatures

The affine corridor needs a favorable reference slope. The following representation instead compresses constant runs of a local mixed-weight signature, allowing arbitrary values between blocks and no small-residual assumption. Full optimality still requires a separate certificate.

4.14.1 A skeleton word and its exact reward

For $n \geq 1$, let $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $m = S-1$, and $s = n-m$. Encode a minimum-path-length shape by a word on its inorder keys: a skeleton key is one and an added bottom cherry is zero. Exactly the words with m ones and no adjacent zeros are feasible. Initial and terminal zeros are allowed. Each zero selects a distinct bottom slot, so this is a bijection with the $\binom{S}{s}$ slot subsets.

The skeleton key of rank r among the ones has comparison length $h - \nu_2(r)$; a cherry has length $h+1$. A gap has length $h+1$ exactly when adjacent to a cherry, and otherwise length h. Put $c_i = q_i + q_{i+1}$. Since zeros are nonadjacent, no deep gap is counted twice. The full original cost of a feasible word x is therefore

$$C(x) = A - \sum_{i:x_i=1} ((1+\nu_2(r_i))p_i + c_i), \tag{21}$$
$$A = (h+1)\sum p_i + h\sum q_i + \sum c_i.$$

Both weight types remain in this identity. It does not apply to arbitrary taller shapes.

Partition (p_i, c_i) into its k maximal constant runs. In a block of length L with signature (a, c), define

$$U(r) = (2a+c)r - a\,\mathrm{popcount}(r).$$

The sum of $\nu_2(t)$ through r is $r - \mathrm{popcount}(r)$. Consequently every feasible block increasing the skeleton count from r_0 to r_1 earns exactly $U(r_1) - U(r_0)$. Its internal arrangement affects only feasibility, not reward.

Let t be the previous key's bit and u the final bit of a nonempty block. Its possible number of ones is every integer in $[\ell, b]$:

t	u	ℓ	b
1	1	$\lceil L/2 \rceil$	L
0	1	$\lfloor L/2 \rfloor + 1$	L
1	0	$\lfloor L/2 \rfloor$	$L-1$
0	0	$\lceil L/2 \rceil$	$L-1$

Empty ranges are infeasible. These bounds follow by separating zeros with ones, respecting the first and last bits; every intermediate count is attained. Use virtual initial bit $t = 1$ and either final bit.

4.14.2 Sliding transfers and linear-space reconstruction

For a prefix reward profile $D_t(r_0)$, the exact next profile is

$$D'_u(r) = U(r) + \max_{t \in \{0,1\}} \max_{r-b \leq r_0 \leq r-\ell} \{D_t(r_0) - U(r_0)\}. \tag{22}$$

Bounds ℓ, b depend on t, u. Exclude unreachable and out-of-range entries. The inner maximum is a sliding window; a standard monotone deque evaluates all ranks in linear time. Each candidate is inserted and removed at most once. There are four bit pairs. Backward profiles use the analogous window over $r_1 \in [r+\ell, r+b]$ with values $B'_u(r_1) + U(r_1)$, then subtract $U(r)$. Popcounts are tabulated once.

Theorem 34 (Exact joint-signature optimizer)**.** *For arbitrary nonnegative mixed weights, let k be the number of runs of $(p_i, q_i + q_{i+1})$. Exact optimization over all minimum-path-length shapes, including full reconstruction, takes $O(nk)$ arithmetic operations and $O(n)$ auxiliary space for $n \geq 1$.*

Proof. An initial sweep finds the maximum final reward at rank m. For traceback, split the block list in half and compute its left forward and right backward profiles between fixed endpoint ranks and bits. Their sum selects an optimal middle state. Release both profiles before recursively solving the two halves. A one-block leaf explicitly places its required count of nonadjacent zeros using the boundary rules. The complete word yields all comparison lengths and the existing linear Cartesian reconstruction. This uses Hirschberg's established recomputation principle [11].

At recursion level d, an active subproblem has at most $\lceil k/2^d \rceil$ blocks. The sum of active rank-interval widths is at most $m+2^d$, because endpoint rank spans partition the original span. Summing sweep work gives $O(km + k\log(k+1))$, plus $O(n)$ preprocessing and output. Since $m = \Theta(n)$ and $k \le n$, this is $O(nk)$. Only a constant number of profiles is retained at once; releasing them before recursion prevents a stack of profiles. All other summaries and the output occupy $O(n)$ space. □

A perfect shape or a single signature block is constructed directly. The potential and transfer use arbitrary block values; they do not assume that an affine approximation is accurate or that the reference tree stays optimal.

4.14.3 Certification and full mixed-weight guarantees

The mixed residual-Huffman gate remains

$$C_* \le (a_0 + b_0)(I_{\min}(n) + 1) + (a_0 + 2b_0)n + H, \qquad a_0 = \min p, \quad b_0 = \min q.$$

It can also be built from the compressed signatures. Each block contains one key weight. Its gap weights obey $q_{i+1} = c - q_i$, so they alternate between at most two values. Record the block's L keys as one weighted multiplicity and its L left gaps as at most two; finally add q_n once. There are at most k key records and $2k+1$ gap records. Sort these records, expand their multiplicities, and run the two-queue bound. The total work is $O(k\log(k+1) + n)$ and space is $O(n)$. Gaps at block boundaries occur once. Thus an accepted full solve retains the theorem's time and space bounds. Failed gates use exact Knuth; the fixed-profile preflight rejection also remains available.

Corollary 35 (Arbitrary mixed block values above a baseline)**.** *For $n \ge 1$, $M \ge 1$, $0 \le u_i, v_i \le M$, set*

$$B = 4(n+1)^2 M, \qquad p_i = B + u_i, \qquad q_i = B + v_i.$$

If $(u_i, v_i + v_{i+1})$ has k runs, the standalone signature constructor returns a full optimal mixed-weight BST in $O(nk)$ arithmetic operations and $O(n)$ auxiliary space. Constant k gives linear construction and $k = o(n)$ gives subquadratic construction.

Proof. The residual mass above a_0, b_0 is at most $(2n+1)M$. Every class candidate has residual cost at most $(h+1)(2n+1)M < 2B \le a_0 + b_0$. A tree outside the class pays an additional baseline cost of at least $a_0 + b_0$. Hence the gate passes even with $H = 0$, and preflight cannot reject. The exact class optimum is a full optimum. The preceding theorem and compressed certificate give the stated work and space. □

The weights are positive and both types may be nonconstant. Key values can change arbitrarily between blocks, and gap weights can alternate at every position. No small perturbation budget or affine-closeness assumption is imposed on u, v. If M is polynomial in n, all weights and costs have $O(\log n)$ bits, giving the corresponding word-RAM bound. For arbitrarily large M, bit costs remain additional. The large baseline is a real restriction; few signatures alone do not guarantee the full gate. For example, $p = (100, 1, 1, 1)$ and $q = (1, 1, 1, 1, 1)$ have only two signatures, but their best minimum-path-length shape costs 218 and their full optimum is 121. Direct enumeration of all fourteen trees, cubic DP, and Knuth agree; the separate gate correctly rejects the class candidate.

4.14.4 Bounded routing and its space qualification

Generic transfer sweeps and recomputation have substantial constants when $k = n$. The initial matched experiment exposed roughly tenfold regressions against Knuth in this regime. The standalone kernel remains an explicit control. The implemented portfolio first scans signatures and selects it only when $k \le \max\{16, \lfloor n/32 \rfloor\}$; otherwise it calls the existing affine solver and its exact fallback. The empirical threshold is not a correctness assumption or a claim of optimal routing.

On the preceding positive family, portfolio work is still $O(nk)$: the signature route certifies, while rejected block counts satisfy $k > n/32$, making even quadratic fallback $O(nk)$. The portfolio does not preserve the standalone kernel's linear-space guarantee on every such input; the affine or Knuth route may use more space. The separate standalone constructor retains the theorem's time and space bounds. All scans, certificates, and output reconstruction are included in the implementation and matched measurements.

4.15 Global alternating-gap bounds and linear profile construction

The preceding signature representation can have $k = n$ even on a short alternating pattern. We now avoid its rank table by recognizing attainment of a global key-depth lower bound. The crucial gap inequality removes the large-baseline restriction on the recognized family.

4.15.1 A gap lower bound valid for all tree shapes

For $n \ge 1$, set $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, and $s = n + 1 - S$. Suppose q_i alternates between arbitrary nonnegative values a, b. Every minimum-internal-path-length shape has gap cost

$$G(q) = h \sum_{i=0}^{n} q_i + s(a+b).$$

Lemma 36 (Alternating-gap component bound)**.** *Every BST on these keys, including every taller tree, has gap contribution at least $G(q)$. Every minimum-internal-path-length shape attains equality.*

Proof. Fix one parity of the $n+1$ inorder gaps. Give its H leaves unit weight and the other $Z = n + 1 - H$ leaves zero weight. Delete a zero leaf and contract its parent into its sibling. The sibling contains a unit-weight leaf: remaining zero leaves are separated in inorder by unit leaves, and a subtree's leaves are contiguous. An all-zero sibling would contradict that separation. Deleting only zero leaves preserves separation. Each contraction decreases the selected depth sum by at least one. After all Z contractions, a full tree with H unit leaves remains. Its external path length is at least $E_{\min}(H)$.

Here $S/2 \le H \le S$, so the standard balanced-leaf formula gives $E_{\min}(H) = (h+1)H - S$, valid at both endpoints. Thus the original selected depth sum is at least

$$E_{\min}(H) + Z = (h+1)H - S + n + 1 - H = hH + s.$$

A minimum-path-length shape has gaps at depths $h, h+1$. Each of its s bottom cherries has two adjacent gaps, one of each parity, at depth $h+1$. Thus its selected depth sum is exactly $hH + s$. Apply the inequality to both parities and multiply by the nonnegative weights a, b. □

Let $p_{(1)} \ge \cdots \ge p_{(n)}$ be the key weights sorted by value, ignoring their fixed positions only for a lower bound. There are at most 2^{d-1} keys at comparison length d. Consequently the sorted lengths

of every BST dominate $\text{bitlength}(1), \ldots, \text{bitlength}(n)$ componentwise. Rearrangement yields the globally valid bound

$$K(p) = \sum_{r=1}^{n} p_{(r)} \,\text{bitlength}(r), \qquad \text{OPT}(p, q) \geq K(p) + G(q). \tag{28}$$

An ordered tree attaining both components is globally optimal. This compares with all shapes; it does not assume that an optimum is nearly complete. Finding a tree attaining $K(p)$ is the remaining constructive task.

4.15.2 Two key levels reduce to interlaced intervals

Suppose key weights have two distinct levels $\ell < u$, and let H count the high keys. Put $m = S - 1$ and assume $S/2 - 1 \leq H \leq m$. In a minimum-path-length shape, even skeleton ranks have comparison length less than h, odd skeleton ranks have length h, and cherries have length $h + 1$. Attainment of $K(p)$ is therefore equivalent to retaining every high key in the skeleton and placing high keys at every even skeleton rank. Low keys may occupy odd ranks or cherries. The inorder word must have m ones and no adjacent zeros, as in the preceding section.

Read the high/low pattern from left to right. A high key must be retained; a low key can be retained only when its new rank is odd. At a boundary immediately after a high key, the previous bit is one. Encode reachable even ranks as $2A$ and odd ranks as $2B+1$. The virtual initial boundary has $A = \{0\}$, $B = \varnothing$.

Lemma 37 (Interlaced parity intervals)**.** *The sets A, B are integer intervals, possibly empty. When both are nonempty, their lower endpoint difference and their upper endpoint difference each belong to $\{0, 1\}$. Four endpoints and empty flags thus represent the exact reachable state.*

Proof. Let L low keys precede the next high key. The exact transformations are

L	A'	B'
0	$B + 1$	A
1	$(A \cup B) + 1$	A
2 or 3	$A + 1$	$\varnothing$
≥ 4	$\varnothing$	$\varnothing$

For $L = 1$, remove the low key or retain it at an odd rank. For $L = 2$, the low word is 01 or 10; for $L = 3$ it is 010. These require an even incoming rank. A low-only run can retain at most one key, because two retained lows would occupy consecutive skeleton ranks, one even. At most one removed key can lie on either side, so $L \geq 4$ is impossible.

For $L = 0$, each endpoint difference becomes one minus its previous value. For $L = 1$ with both intervals nonempty, their union is $[\min B, \max A]$. The new lower difference is $1 + \min B - \min A \in \{0, 1\}$ and the upper difference is one. If only A is nonempty, the pair becomes $(A + 1, A)$; if only B is nonempty, only $B + 1$ survives as A'. The remaining cases are immediate. This proves the invariant inductively. □

For a terminal low run, write R for all incoming ranks and E for its even subset. The final ranks are R for $L = 0$, $R \cup (E + 1)$ for $L = 1$, $E + 1$ for $L = 2, 3$, and empty otherwise. Save each high boundary's intervals. After selecting final rank m, trace backward: each step has at most two candidate predecessor ranks, and the short words above realize the chosen transition. Concatenating the words and applying the comparison-length Cartesian reconstruction returns the full tree in linear time and space. Parity cannot be discarded: the high-key pattern 010100 has reachable final ranks $\{3, 5\}$, so a single ordinary interval would invent rank four.

Corollary 38 (Full linear construction on attaining two-level patterns)**.** *For arbitrary nonnegative two-level key weights with $S/2-1 \le H \le S-1$ and arbitrary nonnegative alternating gap weights, the recognizer decides whether a minimum-path-length shape attains the stated key profile and constructs a witness in $O(n)$ arithmetic operations and $O(n)$ auxiliary space. Every accepted witness is a full optimal mixed-weight BST, without a baseline assumption.*

Proof. The interval transitions and traceback take constant work per high boundary and linear total input/output work. Only two key multiplicity records are ordered, so computing $K(p)$ also takes linear work. The recognized word assigns the high keys to the shortest H comparison lengths, attaining $K(p)$. Its minimum-path-length shape attains $G(q)$. Equality in (28) certifies global optimality, including comparison with every taller tree. □

One key level is handled directly. At perfect sizes $s = 0$, the implementation constructs and evaluates the unique class shape. Equality in (28) certifies it; otherwise the earlier mixed-Huffman gate may certify it. Failure invokes exact Knuth. Rejected recognition does not imply that every minimum-path-length shape is suboptimal. The standalone solver and routed portfolio both remain exact, with quadratic worst-case fallback. The linear-space guarantee applies to the successful route; general fallback can use quadratic space.

4.15.3 An explicit alternating family with no baseline

Corollary 39 (Alternating keys and gaps)**.** *For $n \ge 1$ with $n + 1$ not a power of two, strictly alternating two-level nonnegative key weights and alternating nonnegative gap weights admit full exact $O(n)$-time, $O(n)$-space construction. The relative weight magnitudes are unrestricted. Equal key levels are handled by the uniform route.*

Proof. Here $s > 0$. Let $A = 1$ if the first key is high and $A = 0$ otherwise. If $s \bmod 2 = A$, remove the first s low keys. Otherwise remove the first $s - 1$ lows and the last low. There are enough lows, and their positions are nonadjacent. Every high key remains. Before the initially removed lows end, retained keys are high. After removing t initial lows, a high key's skeleton-rank parity is $A - t$. The suffix therefore has high keys at even ranks when $t \bmod 2 = A$. In the second construction, removing the final low affects at most one subsequent high, which may occupy an odd rank. Every even skeleton rank is high. The high count lies in the theorem's range, so the recognizer succeeds and (28) proves the full claim. □

This family has $k = n$ joint signatures when the key levels differ. The interval representation therefore escapes the preceding $O(nk)$ bound without invoking a small-residual assumption. Nonperiodic accepted patterns are also possible: choose any minimum-path-length shape, mark all even skeleton ranks high, choose the odd skeleton labels freely, and label every cherry low. Recognition receives only the resulting weights, not this witness.

The perfect-size exception matters. For $p = (100, 1, 100)$ and four zero gaps, the unique minimum-path-length tree costs 401 while the full optimum is 303. Cubic DP and Knuth agree. Thus the corollary cannot include every size and orientation. Arbitrary integer magnitudes also retain their bit costs; polynomially bounded weights yield the corresponding word-RAM bounds.

4.16 Threshold runs and a linear Cartesian certificate

We now allow arbitrary nonnegative gap sequences and arbitrary key values in the certificate. The returned tree is optimal for the full mixed objective; failure still requires an exact fallback. This extends the preceding global alternating-gap argument rather than restricting the problem to alphabetic coding.

4.16.1 Binary indicators and arbitrary weighted layers

For $H \geq 1$, write

$$E(H) = (\lfloor \log_2 H \rfloor + 2)H - 2^{\lfloor \log_2 H \rfloor + 1}.$$

This is the minimum external path length of a full tree with H unit leaves; in particular $E(1) = 0$.

Lemma 40. *Give $H > 0$ of N ordered leaves weight one and the others weight zero. Let R be the number of contiguous zero runs, including endpoint runs. The exact minimum weighted external path length is $E(H) + R$. For $H = 0$ the optimum is zero.*

Proof. Remove every maximal zero-only subtree. Each lies within one zero run, and every run contains at least one such subtree. Each removed subtree's parent has a selected descendant in its other child, by maximality. Suppressing the resulting unary nodes therefore decreases the selected depth sum by at least one per removed subtree. What remains is a full H-leaf unit tree, with cost at least $E(H)$. This proves the lower bound.

For attainment, start with any optimal H-leaf unit tree. Assign each interior zero run to the selected leaf immediately to its left; assign an initial run to the first selected leaf's left and a final run to the last selected leaf's right. Each selected leaf has at most one right run, and only the first can also have a left run. Expand each such leaf to a small ordered tree accommodating the required zero-only subtrees. One assigned run increases its selected depth by one; two runs increase it by two. The zero-only subtrees have arbitrary required sizes and zero cost. All original leaves and their order are preserved, with total cost $E(H) + R$. □

For arbitrary $q = (q_0, \ldots, q_{N-1})$, let $Q = \max_i q_i$. At threshold $0 < t \leq Q$, let H_t count weights at least t and R_t count runs of weights below t. Lemma 40 gives

$$\text{gapcost}(T, q) \geq G(q) := \int_0^Q (E(H_t) + R_t)\, dt \qquad \text{for every tree } T. \tag{23}$$

These integrals are exact finite layer sums. The minimum and sum need not commute: $q = (100, 10, 1, 10)$ has $G(q) = 152$, but alphabetic optimum 153.

Let $q_{[r]}$ denote the rth largest weight. Since $E(r) - E(r-1) = \lceil \log_2 r \rceil + 1$ for $r \geq 2$, the bound becomes

$$F(q) = \sum_{r=2}^{N} (\lceil \log_2 r \rceil + 1) q_{[r]}, \tag{24}$$

$$D(q) = Q - q_0 + \sum_{i=1}^{N-1} \max(q_{i-1} - q_i, 0), \tag{25}$$

$$G(q) = F(q) + D(q).$$

For (25), a zero run starts at position zero for $q_0 < t \leq Q$, or at position $i > 0$ for $q_i < t \leq q_{i-1}$. Equivalently,

$$D(q) = Q - \frac{q_0 + q_{N-1}}{2} + \frac{1}{2} \sum_{i=1}^{N-1} |q_i - q_{i-1}|.$$

Variation terms are familiar in alphabetic-code bounds, including upper bounds reviewed by Bruno et al. [14]. The present lower-bound derivation and equality certificate do not by themselves establish priority.

4.16.2 Dyadic selection avoids sorting the gap weights

Theorem 41. *For arbitrary nonnegative weights, $G(q)$ can be computed in $O(N)$ arithmetic and comparison operations and $O(N)$ auxiliary space.*

Proof. The variation term is one scan. For a current multiset of $M > 1$ weights, let k be the largest power of two strictly less than M. Every rank $k+1, \dots, M$ has coefficient $\log_2 k + 2$ in (24). Use deterministic linear selection to retain the k largest values and charge this coefficient times the discarded sum. Repeat on the retained multiset; the last single weight contributes zero. After the first step the retained sizes halve. The total sizes processed, $M, k, k/2, \dots, 2$, sum to less than $3M$. Standard median-of-five selection [12], partitioning, summation, and copying are linear in each processed size. Their total is therefore linear, with linear auxiliary storage. Equal values at a partition boundary may be allocated to either side to obtain exactly the required cardinality. □

The implementation also uses a cheaper scan-only formula when applicable. Put $h = \lfloor \log_2 N \rfloor$, $S = 2^h$, $s = N - S$, and $\beta = \min_i q_i$. If at least $S/2$ gaps have weight Q and at least s have weight β, then

$$F(q) = (h+1)\sum_i q_i - SQ + s\beta. \tag{26}$$

Indeed, for $\beta < t \le Q$, $S/2 \le H_t \le S$, and $E(H_t) = (h+1)H_t - S$. The part below β is $\beta E(N)$, where $E(N) = hN + 2s$; integration gives (26). Uniform weights and $N = 1$ also satisfy the formula. These checked multiplicities choose an evaluation method, not an optimality certificate.

4.16.3 Checking the key profile without sorting

The preceding key capacity bound remains

$$K(p) = \sum_{r=1}^{n} p_{[r]} \lfloor \log_2 r + 1 \rfloor, \qquad \mathrm{OPT}(p, q) \ge K(p) + G(q).$$

Let H be the number of positive keys in a candidate tree and d_i their comparison lengths. Define $M(H) = \sum_{r=1}^{H} \lfloor \log_2 r + 1 \rfloor$.

Lemma 42. *A candidate's key cost equals $K(p)$ if and only if*

1. *$\sum_{i:p_i>0} d_i = M(H)$; and*
2. *every positive weight at a shallower depth is at least every positive weight at a deeper depth.*

Both conditions are testable in linear work and space.

Proof. The sorted positive comparison lengths dominate the capacity profile $\lfloor \log_2 r + 1 \rfloor$ componentwise. All weights being paired with these lengths are strictly positive. Thus any excess length makes even the rearranged cost strictly larger than $K(p)$. Equality of the unweighted sums forces equality of the complete length profiles. The second condition is precisely equality in the rearrangement step: there is no strict weight inversion between different lengths. A traversal collects the length sum and each positive depth's minimum and maximum weight; adjacent nonempty depth levels suffice for the second test. When $H = 0$ the conditions are vacuous. □

Corollary 43. *Construct the max-Cartesian tree on p, with fixed leftmost priority for ties. If its key cost equals $K(p)$ and gap cost equals $G(q)$, it is a globally optimal full mixed-weight BST. Candidate construction, testing, and successful output take $O(n)$ arithmetic work and auxiliary space. If all keys have positive pairwise distinct weights, this test succeeds if and only if some tree attains $K(p) + G(q)$.*

Proof. The two lower bounds hold for every tree, so their simultaneous attainment proves optimality. Standard monotone-stack Cartesian construction [13] is linear; use Lemma 42 for the key test and Theorem 41 for the gap test. Iterative traversal and reverse traversal construct all output nodes in linear work, even when the candidate has linear height.

For distinct positive weights, any attaining tree has weights ordered by depth, so every parent has strictly greater weight than its children. The inorder max-heap with distinct priorities is the unique Cartesian tree. Hence any attaining tree must be the tested candidate. □

With ties, failure does not prove nonattainment. For $p = (1, 1, 1)$ and $q = (1, 1, 1, 1)$ the balanced tree attains cost 13, whereas leftmost Cartesian tie-breaking produces a chain. Even without this candidate issue, simultaneous attainment is not universal: for $p = (2, 1, 1)$ and $q = (100, 10, 1, 10)$, the mixed bound is 158 and full optimum 160. The standalone solver uses exact Knuth fallback. The portfolio first tries the previous profile certificate, then the Cartesian certificate, then the earlier signature/affine portfolio. The unrestricted worst case remains quadratic. All linear claims here count exact arithmetic operations; they are not bit-complexity or instruction-count bounds.

4.16.4 A positive family with many key and gap levels

Corollary 44. *Start with a perfect skeleton of $S - 1$ keys, $S = 2^h$, and expand any s odd-indexed bottom slots to cherries, where $0 \le s \le S/2$. Give every even slot gap weight $Q > 0$, every unexpanded odd slot an arbitrary weight in $[\beta, Q]$, and both gaps of each expanded slot weight β, where $0 < \beta \le Q$. Give all keys positive weights strictly decreasing between depth layers, with arbitrary weights within each layer. The Cartesian certificate constructs a full optimal tree in $O(n)$ arithmetic work and space, where $n = S - 1 + s$.*

Proof. For $\beta < t \le Q$, every even slot is selected. All selected gaps have depth h. If H are selected, zero runs number $S - H$: every zero slot is odd and separated from the others by selected even slots. Expanding a zero slot into two zero gaps does not change this run count. Since $S/2 \le H \le S$, we have $E(H) + (S - H) = hH$, exactly the candidate cost. For $0 < t \le \beta$, all $N = S + s$ gaps are selected and the minimum-path-length shape attains $E(N)$. The candidate therefore attains every threshold optimum and thus $G(q)$. Its key lengths form the capacity profile, and the assigned weights decrease between depths, so it attains $K(p)$. Strict ancestor inequalities make the Cartesian construction recover this tree, even if some same-depth weights coincide. □

For the explicit tested family let $s = S/4$, $\beta = 1$, $Q = n^2 + 2$, give unexpanded odd slot j weight $j + 2$, and assign

$$p_i = (h + 2 - d_i)(n + 1) + i.$$

All n key weights are distinct, and there are $S/4 + 2$ gap levels. Both alphabets have $\Theta(n)$ values, gaps need not alternate, and there is no large common baseline. The weights have $O(\log n)$ bits. Independent positive rescaling of the key and gap vectors preserves the two equalities; the theorem requires no dominance between their masses. The solver receives only p, q, without the reference tree or slot pattern.

4.17 Shifted alphabetic certificates beyond key-capacity equality

Theorem 45. *Let $m = \min p$, with $m = 0$ when $n = 0$. Define*

$$L_m = K(p - m) + A(q + m) - mn.$$

Then $K(p) + G(q) \le K(p) + A(q) \le L_m \le \mathrm{OPT}(p, q)$. Among all admissible uniform transfers, m gives the largest such separate key/exact-gap bound. An alphabetic optimum T for $q+m$ with original full cost $C(T; p, q) = L_m$ is globally optimal. The bound, candidate, certificate, and complete tree construction take $O(n \log(n+1))$ arithmetic work and $O(n)$ space.

Proof. The transfer identity gives $L(a) = K(p-a) + A(q+a) - an \le \mathrm{OPT}$ for all admissible a. Uniform subtraction preserves ranks and $K(p-a) = K(p) - aM(n)$, so

$$L(a) = K(p) + \min_T \left\{ \sum_i q_i d_i(T) + a \left(\sum_i d_i(T) - E(n+1) \right) \right\}.$$

Every affine term has nonnegative slope. Their minimum is nondecreasing, so the largest admissible $a = m$ maximizes the bound. Taking $a = 0$ gives the claimed chain of inequalities. Equality of a feasible full cost with a lower bound proves optimality. Equivalently, the alphabetic candidate must attain $K(p-m)$ in the original residual key order.

Implement Hu–Tucker compatible merging using a leftist heap of internal items in each gap between consecutive original leaves, and a priority queue of each group's best compatible pair. Each pair is selected among at most four items: two boundary leaves and two heap minima. Removing a leaf melds its incident groups. There are $O(n)$ groups and events, each meld or priority operation takes $O(\log(n+1))$, and stale events are discarded. Recovering depths, alphabetic stack reconstruction, and full original-cost evaluation are linear. Sorting residual keys costs $O(n \log(n+1))$. No interval table is used on acceptance. □

The tie convention exactly reproduces infinitesimal original-leaf weights 2^i without storing those large integers. Active items partition the leaves; at equal mass, individual items compare by their largest original index. Two pairs compare lexicographically by the larger, then the smaller, of their items' largest indices. A shared item cancels; otherwise the largest differing leaf determines the symbolic sum order. Only ordinary indices are stored. The implementation separately counts heap-item, priority-entry, and key-sort comparisons, as well as all events, melds, reconstruction, cost visits, and fixed scans. These are arithmetic/logical counts, with bit costs additional.

The six-key obstruction has $L_1 = 213$, and its shifted alphabetic candidate is the full optimum. Thus normalization repairs the bound when the exact alphabetic optimum replaces G, even though $K + G$ is invariant. Tied alphabetic optima can hide an attaining candidate, and rejection does not prove suboptimality or nonattainment. The standalone implementation uses exact Knuth fallback.

4.17.1 A positive nonuniform family with linear capacity deficit

Corollary 46. *Let $r = 2^h$, $n = 7r - 1$, and repeat the original seven-gap vector $(18, 2, 13, 3, 17, 1, 2)$ exactly r times. Place the six-key witness tree above into every leaf of a perfect r-leaf skeleton. Give its r deepest keys weight 1, and every other key weight 1 plus a positive residual decreasing between depth layers. The shifted alphabetic certificate constructs a full optimal tree in $O(n \log(n+1))$ work. Its original key contribution exceeds $K(p)$ by exactly r.*

Proof. The $6r - 1$ residual-positive keys fill all capacity levels above the last occupied level, with $2r$ keys in that last level. Their weights decrease between levels, so they attain $K(p-1)$. The r residual-zero keys have length $h + 4$, one greater than their original unit-weight capacity slots at length $h + 3$. This gives the deficit r.

The shifted gap block is $(19, 3, 14, 4, 18, 2, 3)$. In every block compatible merging performs $2 + 3 = 5$, $3 + 14 = 17$, $17 + 4 = 21$, and $18 + 5 = 23$. Each stage finishes in all blocks before the next: cross-block pairs do not undercut these costs. Per block remain an original leaf 19 and

internal items $21, 23$. For even r, all original leaves pair at 38 before any competing remaining pair, removing all barriers. Then 21 pairs to 42 and 23 to 46, giving $r/2$ copies of twice $(19, 21, 23)$. Repeat until a single triple remains, with final merges $19 + 21 = 40$ and $40 + 23 = 63$. For $r = 1$ these are already compatible. All same-stage ties give the same depths. Thus the ordered gap depths are $h + (2, 4, 4, 3, 2, 3, 3)$ per block, with alphabetic cost $r(169 + 63h)$. Reconstruction gives the stated tree, which attains both shifted terms. □

For a concrete integer family, if ℓ_i is its key comparison length, set $p_i = 1$ at length $h + 4$, and otherwise

$$p_i = 1 + (h + 4 - \ell_i)(n + 1) + i.$$

This gives $6r$ distinct key weights, six distinct gap weights, strictly positive outcomes, common key minimum 1, and $O(\log n)$-bit weights. No key-versus-gap dominance assumption is needed for the equality proof. The family is outside the earlier original-capacity certificate for every r: its full cost exceeds $K(p) + G(q)$ by at least r.

4.18 Endpoint prices and the limit of additive transfers

The endpoint identities $\ell_0 = d_0$ and $\ell_{n-1} = d_n$ permit transfers that the preceding uniform normalization does not exploit.

Theorem 47. *For $n \geq 3$, every additive weight change whose cost difference is independent of the tree has the following form. Put $L = p_0 + q_0$ and $R = p_{n-1} + q_n$, and choose*

$$p' = (x, p_1 - a, \ldots, p_{n-2} - a, y),$$
$$q' = (L - x, q_1 + a, \ldots, q_{n-1} + a, R - y).$$

Then $C(T; p', q') = C(T; p, q) + an$. Nonnegativity is exactly

$$-\min_{1 \leq j < n} q_j \leq a \leq \min_{1 \leq i < n-1} p_i, \qquad 0 \leq x \leq L, \quad 0 \leq y \leq R.$$

Proof. The depth identity and endpoint equalities prove sufficiency. For necessity write $U_i = \delta p_i + \delta q_i$ and $V_i = \delta p_i + \delta q_{i+1}$. A two-key rotation gives $U_i = V_{i+1}$. Compare a rotation on keys $i+1, i+2$ with and without key i in its outer left subtree: subtraction gives $U_i = 0$ for $0 \leq i \leq n-3$. The symmetric comparison gives $V_i = 0$ for $2 \leq i < n$. The two-key identities then supply $V_1 = 0$ and $U_{n-2} = 0$. All these rotations can be embedded in full trees on the original keys. Consequently all interior gap changes equal one a, all interior key changes are $-a$, and the endpoint key/gap sums are unchanged. The resulting constant is an. □

This characterizes additive transfers on the given outcome coordinates, not arbitrary encodings or all possible lower bounds.

Theorem 48. *Let $F(a, x, y) = K(p') + A(q') - an$. The maximum of this valid lower bound over the entire admissible box is attained on the face $a = a_* := \min_{1 \leq i < n-1} p_i$.*

Proof. Increase a by δ while replacing x, y by $\max(0, x - \delta), \max(0, y - \delta)$. On a segment with $e \in \{0, 1, 2\}$ endpoint residual keys clamped at zero, the other $n - e$ positive keys decrease uniformly until a_*. Thus K has slope $-M(n - e)$. All gaps except those e endpoints increase uniformly. By Lemma 40, their minimum depth sum is $E(n + 1 - e) + e = M(n - e) + n$. Every affine tree term in F therefore has nonnegative slope. Their minimum is nondecreasing; continuity joins the segments. Moving any feasible point to the maximal face cannot reduce F. □

4.18.1 An exact bounded two-price search

Fix $a = a_*$. An alphabetic optimum for q' and a decreasing assignment of p' to capacity lengths provide the value $F(x, y)$. If the assigned key lengths are λ_i and the alphabetic gap depths are d_j, they also give a global supporting plane

$$F(X, Y) \le c + (\lambda_0 - d_0)X + (\lambda_{n-1} - d_n)Y.$$

The plane is an upper bound on the concave *bound function*; the sampled $F(x, y)$ is a lower bound on the full optimum. The alphabetic tree evaluated with the original ordered weights gives a feasible upper bound on that optimum. These directions are distinct.

Maintain the planes and maximize their minimum over $[0, L] \times [0, R]$. The implementation enumerates vertices of the three-dimensional hypograph using integer determinants and Cramer's rule. Feasibility and objective comparisons use exact cross-products; prices and bound values are rational. There is no floating-point tolerance. Retain the largest sampled bound and the smallest original tree cost across calls. For integer original weights, equality of the latter with the ceiling of the former certifies full optimality. Equality between the LP upper bound and best sampled bound instead proves completion of the price maximization. An early full certificate or a call-budget stop need not complete that maximization.

With b alphabetic calls, all preprocessing, exact pricing, and candidate construction take

$$O\big(bn \log(n + 1) + b^5\big) \text{ arithmetic work}, \qquad O(n + b) \text{ space}.$$

Each LP call checks $O(b^3)$ triples against $O(b)$ constraints. The default $b = \max(2, \text{bitlength}(n))$ gives $O(n \log^2(n+1) + \log^5(n+1))$, asymptotically $O(n \log^2 n)$, for the attempted certificate. Rejection uses Knuth; the unrestricted worst case remains quadratic. Supporting slopes have magnitude $O(n)$ and integer intercepts $O(nW)$, so vertex denominators are $O(n^2)$ and the scaled alphabetic weights use only $O(\log n)$ additional bits. Bit costs remain additional. The uncapped diagnostic has no proved small iteration bound here.

4.18.2 A positive family with distinct gaps and light endpoint keys

Corollary 49. *Let $h \ge 3$, $B = 2^h$, $n = B - 1$, and let $\ell_i = h - v_2(i + 1)$ be the perfect tree's key comparison lengths. Initially set $p_i = B + (h + 1 - \ell_i)B + i$, then replace $p_0 = p_{n-1} = 1$ and $p_2 = B$. Set $q_0 = B$, $q_n = B + n$, and $q_j = j + 1$ for $1 \le j < n$. The first endpoint-price call constructs a full optimum in $O(n \log n)$ arithmetic work and $O(n)$ space, with exact cost*

$$\frac{(8h - 11)B^2 + (13 - 7h)B}{2} - 1.$$

There are $n - 1$ distinct key weights and $n + 1$ distinct gap weights, all positive and with $O(\log n)$ bits. The preceding uniform-shift bound is at least three below the optimum for every such size.

Proof. At $a_* = B$, $x = y = 0$, exactly three bottom residual keys are zero. The other $n - 3$ keys attain capacity and decrease between depth layers. The transformed gaps are $B + j + 1$, all in $[B + 1, 2B]$. Unrestricted Huffman merging pairs these B outcomes level by level: at each level every newly generated mass exceeds every unmerged same-level mass since the maximum weight is less than twice the minimum. All leaves have depth h. The perfect alphabetic tree attains that unrestricted optimum and both transformed lower bounds, proving full mixed optimality. Summing the perfect lengths, their squares, and the three bottom-key corrections gives the formula.

The preceding uniform shift uses only $\min p = 1$. Its residual key bound is already attained by the perfect tree. In the last four shifted gaps, the two left weights sum to $2B - 3$, while the rightmost weight is $2B$. Changing the balanced four-leaf subtree to the right-heavy alternative saves three in the gap objective. Thus the old bound is strictly below the perfect full optimum, regardless of which candidate it constructs. □

Twelve complete family constructions reach $n = 262143$, with 262142 distinct keys, 262144 distinct gaps, and full cost 4569830391807. The preceding bound is 4561240391642, a deficit of 8590000165. The largest run uses one alphabetic call, 262143 merges, 7111153 priority comparisons, 8924411 heap-item comparisons, and 1482369 key-sort comparisons. It constructs every key with no LP vertices or interval-DP cells. Seven sizes through 511 match Knuth; three also match cubic. The formula and an independently built perfect tree verify all larger sizes. All scans, depth recovery, events, and output visits are retained in the source-hashed results.

4.19 Root/nonroot coupling beyond the transfer barrier

The four-key obstruction proved in Theorem 63 loses the coupling between one residual interior key and the surrounding gap tree. A partition by the identity of the root retains part of that coupling at the cost of two additional alphabetic constructions. It applies to arbitrary mixed input weights; successful certification, rather than the validity of the bound, is conditional.

Theorem 50. *Let $n \geq 2$, let r index a largest key weight $w = p_r$, and put $v = \max_{i \neq r} p_i$, $P = \sum_i p_i$, and $Q = \sum_j q_j$. Define*

$$N_r = K(p) + A(q) + w - v,$$
$$R_r = P + Q + K(p_{[0,r)}) + K(p_{(r,n)}) + A(q_{[0,r]}) + A(q_{[r+1,n]}).$$

Then

$$K(p) + A(q) \;\leq\; \min(N_r, R_r) \;\leq\; \mathrm{OPT}(p, q).$$

Here K is the free key-capacity bound and A the exact alphabetic gap optimum; empty key subvectors have bound zero.

Proof. If the root is not r, let its weight be $u \leq v$, and let the comparison length of r be $\ell \geq 2$. Exchanging those two weights reduces the key cost by $(w - u)(\ell - 1) \geq w - v$. The exchanged cost is at least $K(p)$, so the original key cost is at least $K(p) + w - v$. Adding $A(q)$ proves N_r. The exchange is solely a lower-bound argument; it does not change the ordered input or construct a different search tree.

If the root is r, the recurrence pays $P+Q$ and the two ordered subproblem costs. Bounding their keys and gaps separately gives R_r. These cases partition all trees. Moreover, the root-constrained free key placement costs at least $K(p)$, and the root-constrained alphabetic tree costs at least $A(q)$. Hence each case bound dominates the original separable bound. □

The implementation first applies Theorem 47 with $a = \min_{1 \leq i < n-1} p_i$ and endpoint residuals $x = \max(0, p_0 - a)$, $y = \max(0, p_{n-1} - a)$. It subtracts an from transformed bounds and evaluates every candidate using the original mixed weights. One alphabetic call constructs the global gap optimum. A single decreasing key sort supplies K and, by scanning its left and right subsequences, both conditional capacity sums. If the global candidate already meets $K + A$, the attempt ends. Otherwise two alphabetic calls construct the gap optima on either side of r, and a second candidate joins them at root r. The smaller original candidate cost is compared with $\min(N_r, R_r) - an$. Equality certifies full optimality.

At most three alphabetic calls process a total of $2(n+1)$ leaves. Sorting, normalization, rebasing, reconstruction, and cost evaluation give $O(n \log(n+1))$ arithmetic work and $O(n)$ space for the complete attempt. The constant-size $n < 3$ case is solved directly. Rejection uses Knuth and retains the unrestricted quadratic worst case. The fixed endpoint choice need not preserve certificates found at another price; the preceding solvers remain separate and unchanged.

For Theorem 63, normalization gives $p' = (0, 0, 1, 0)$ and $q' = (2, 3, 2, 2, 2)$ with additive constant four. Here $K = 1$, $A = 26$, $r = 2$, and both N_r and R_r equal 28. The new original bound is therefore 24, exactly the full optimum, while the maximum over the entire preceding transfer class is 23.

4.19.1 A positive family outside every separable additive transfer

Corollary 51. *Let $h \geq 3$, $N = 2^h$, $n = N - 1$, and let $\ell_i = h - \nu_2(i+1)$ be the perfect tree's key comparison lengths. Put*

$$r = N/4 - 1, \qquad s = N/2 - 1, \qquad C = 4hN, \qquad w = hN^2/2.$$

Initially set $p_i = 1 + (h - \ell_i)N + i + 1$, then replace

$$p_0 = p_2 = p_{n-1} = 1, \qquad p_r = w + 1.$$

Set $q_j = C + j - 1$ for $0 \leq j < N$. The root/nonroot certificate constructs a full optimum in $O(n \log n)$ arithmetic work and $O(n)$ space. Its exact cost is

$$\mathrm{OPT} = \frac{(8h^2 + 6h - 7)N^2 + (12 - 5h)N}{2} + 1 - 3h.$$

Every weight is positive and has $O(\log n)$ bits. There are $n-2$ distinct key weights and $n+1$ distinct gap weights. Nevertheless, the maximum $K+A$ bound over every admissible additive cost-preserving transfer falls short by exactly

$$D = \frac{hN^2 - (2h-1)N}{2} > 0.$$

Proof. The maximal interior transfer is $a = 1$. At endpoint prices zero, residual keys $\widehat{p}_i = p_i - 1$ have largest value w at r, next-largest value $v = (h - \frac{1}{2})N$ at the perfect root s, and three zero bottom keys. All other weights decrease strictly between perfect depth layers. The perfect tree attains the nonroot key bound $K(\widehat{p}) + w - v$: exchanging the weights at r and s attains the free capacity profile.

Transformed gaps are $\widehat{q}_j = C + j$, with maximum less than twice the minimum. Since N is a power of two, unrestricted Huffman merging pairs nodes level by level. Every gap has depth h, and the perfect alphabetic tree attains the unrestricted minimum. Consequently

$$A(\widehat{q}) = h\left(CN + \frac{N(N-1)}{2}\right).$$

If r is the root, its two sides contain $N/4$ and $3N/4$ gaps. The minimum unit-weight total depth, including that root, is $Nh + N/4$. All transformed gap weights are at least C, so the root-r alphabetic cost exceeds $A(\widehat{q})$ by at least

$$\frac{CN}{4} - \frac{hN(N-1)}{2} = w + \frac{hN}{2} > w - v.$$

Its conditional key bound is at least $K(\widehat{p})$. Thus $R_r > N_r$, and the perfect candidate attains N_r, proving full optimality.

For the cost formula, use $\sum_i \ell_i = (h-1)N + 1$, $\sum_i \ell_i^2 = N(h^2 - 2h + 3) - 3$, and $\sum_i (i+1)\ell_i = (N/2)\sum_i \ell_i$. After the three zero-key corrections and the replacement at r, the perfect residual key cost is $(5h-7)N^2/2 + (7-2h)N - 3h$. Adding $A(\widehat{q})$ and subtracting n yields the displayed full cost.

Finally, Theorem 48 reduces all admissible transfers to $a = 1$ and two endpoint prices. At $x = y = 0$, both zero endpoint keys occupy capacity slots of length h, and both endpoint gaps also have depth h. The supporting plane from Section 4.18.1 therefore has both slopes zero and value $K(\widehat{p}) + A(\widehat{q})$. This bounds the entire price objective from above and is attained at zero. The best original separable bound is exactly $K(\widehat{p}) + A(\widehat{q}) - n = \mathrm{OPT} - (w - v)$, giving the stated deficit. □

4.20 A sufficient exchange condition for a wide last level

Let $h = \lfloor \log_2(n+1) \rfloor$, $c = 2^h - 1$, and $s = n - c$. As proved earlier, minimum internal-path-length shapes expand s of the 2^h external slots of a perfect c-key skeleton. Consider adjacent skeleton slots $g, g+1$ containing one expansion. If their first input index is i, moving the expansion from the second slot to the first changes the weighted cost by exactly

$$(1 + \nu_2(g+1))(p_i - p_{i+1}) + q_i - q_{i+2}. \tag{27}$$

Only the intervening skeleton key, added key, and three gaps change their comparison lengths; all later positions are unchanged. The coefficient $1 + \nu_2(g+1)$ lies between one and h.

Theorem 52 (Exchange-certified minimum-path-length shape)**.** *If*

$$\max\{p_i - p_{i+1},\ h(p_i - p_{i+1})\} \le q_{i+2} - q_i \quad (0 \le i < n-1), \tag{28}$$

then filling the last level from the left minimizes weighted cost among all minimum internal-path-length shapes. The reflected condition gives the corresponding right-filled shape. For $s = 0$ the shape is unique without any condition.

Proof. The maximum in (28) bounds the key term in (27), whether its difference is positive or negative. Every adjacent swap of expansion pattern 01 to 10 is therefore nonincreasing in cost. Repeated swaps move every expansion leftward, yielding the claimed shape. Reflection proves the other case. □

In particular, nondecreasing key and gap weights satisfy the condition; both lists nonincreasing satisfy its reflection. Nonmonotone keys are also permitted when the gap increments compensate for their decreases. This proves optimality *within the shape class*. Global optimality still requires the earlier residual-Huffman gate. For candidate value V, minimum weights a, b, and residual mixed-node lower bound H_{res}, the sufficient condition is

$$V \le (a+b)(I_{\min}(n) + 1) + (a + 2b)n + H_{\mathrm{res}}.$$

The exchange constructor constructs the candidate, checks this gate, and otherwise falls back to Knuth. Accepted instances take $O(n \log n)$ time and $O(n)$ auxiliary space, including sorting for the bound. If both weight lists are monotone, scans/reversal and the existing two queues make the whole accepted route $O(n)$. No optional-slot width appears. Sorted geometric controls pass the shape condition but fail the global gate, preventing an incorrect general claim about balanced trees.

5 Negative results and representation barriers

These results identify information lost by particular relaxations or costs imposed by particular representations. Each statement specifies its model of failure. None is an unrestricted computational lower bound.

5.1 A useful relaxation that failed as a general solver

Allow arbitrary assignment of key weights to internal positions and gap weights to external positions. In this relaxed problem, repeatedly take the two smallest current external weights a, b and the smallest remaining internal weight p, replace them by $a + b + p$, and add that sum to the cost.

To prove the rule, take a deepest internal node of an optimum. Its children are leaves. Swapping the smallest internal weight into it and the two smallest external weights into its children cannot increase cost. Contract that cherry and induct. Sorting and a heap give $O(n \log n)$ time. This elementary extension of the Huffman exchange argument is not claimed as novel.

The relaxed cost is a valid lower bound for full OBST. Equality with a feasible ordered tree certifies optimality. It agreed with an independent shape-enumeration and rearrangement oracle on 22,143 exhaustive and 2,700 random instances in the basic harness.

However, using the relaxation on demand in an exact sparse DP was poor on random inputs. At $n = 128$ it visited roughly 4,300–4,600 exact intervals instead of Knuth's 8,256, but spent approximately 3.9–4.1 million weight comparisons constructing bounds and inspected over 100,000 candidate roots. Fewer exact states did not mean less computation.

Nor does the relaxation's interval matrix inherit the required quadrangle inequality. For $p = (9, 1, 12)$, $q = (5, 1, 18, 10)$, its costs H satisfy

$$H(1, 2) + H(2, 3) = 60 > 54 = H(1, 3) + H(2, 2).$$

Thus applying Knuth-style matrix optimization directly to this bound would be unjustified.

Two additional exact counterexamples exclude tempting local certificates. With $p = (3, 5, 18)$, $q = (13, 14, 4, 8)$, the tree with root 2 and left subtree $0(\varnothing, 1)$ costs 127, every single-rotation neighbor costs at least 134, yet the balanced tree costs 125. With $p = (18, 8, 18)$, $q = (17, 11, 1, 13)$, the balanced tree costs 164 and every distinct tree obtained by promoting a descendant to the root of an existing subtree costs at least 169; the tree $0(\varnothing, 2(1, \varnothing))$ costs 157. These are barriers to those local certificates, not complexity lower bounds.

5.2 An ordered depth-chain lower bound

Let l_i be the comparison length of key i and g_i the depth of gap i. For every valid BST, adjacent keys have different lengths and

$$g_0 = l_0, \quad g_n = l_{n-1}, \quad g_i = \max(l_{i-1}, l_i) \quad (0 < i < n).$$

This identity is existing work; see Becker [6], Lemma 1. Let K_d count keys of length d and G_d count gaps at depth d. The root and child-count identities are

$$K_1 = 1, \qquad G_d + K_{d+1} - 2K_d = 0 \quad (1 \le d \le n), \qquad K_{n+1} = 0.$$

Permit any sequence in $\{1, \dots, n\}^n$ with unequal adjacent entries. For arbitrary multipliers $\lambda_0, \dots, \lambda_n$, minimize

$$\sum_i p_i l_i + \sum_i q_i g_i + \lambda_0(K_1 - 1) + \sum_{d=1}^{n} \lambda_d(G_d + K_{d+1} - 2K_d). \tag{29}$$

Call the minimum $L(\lambda)$. Every actual tree is permitted and has zero penalty, so $L(\lambda) \le \mathrm{OPT}$ for every multiplier vector. The expression is a chain with key energy $U_i(b) = p_i b + \lambda_{b-1} - 2\lambda_b$, gap energy $V_i(b) = q_i b + \lambda_b$, and constant $-\lambda_0$. For an interior transition into length b, take the smaller of

$$\min_{a<b} F(a) + V_i(b), \qquad \min_{a>b}\{F(a) + V_i(a)\}.$$

Prefix and suffix minima give $O(H)$ work per key for a domain of H lengths. Using $H = n$ would cost quadratic time even for one bound evaluation.

Theorem 53 (Exact affine-tail compression)**.** *Fix $h \ge 1$, arbitrary $\lambda_0, \dots, \lambda_h$, and a slope $0 \le s \le \min_i p_i$. Extend $\lambda_d = \lambda_h + s(d-h)$ for $d > h$. The minimum in (29) is unchanged if lengths are restricted to $1, \dots, \min(n, h+3)$.*

Proof. For $b \ge h+1$, the unary key energy has slope $p_i - s \ge 0$ and the gap energy has slope $q_i + s \ge 0$. Both are nondecreasing in this tail. If some $l_i > h+3$, replace it by a member of $\{h+1, h+2, h+3\}$ unequal to its current neighbors. At most two values are forbidden. The key energy does not increase. Each adjacent maximum is either unchanged or decreases within the affine tail, so neither gap energy increases. Endpoint gaps obey the same argument. Repeat for all large lengths. The restricted domain is a subset of the original domain, and every original sequence has a no-more-expensive restricted sequence; therefore the minima are equal. □

Taking $h = \lceil \log_2(n+1) \rceil + 2$ gives an $O(n \log n)$ oracle. This is a statement about minimization of the relaxed energy, with no height assumption on the unknown optimal BST. Optimizing only the head multipliers requires folding all tail constraint violations into the derivative for λ_h. The implementation uses exact integer multipliers and costs scaled by 1024. Its subgradient ascent is heuristic; the bound is valid at every iteration regardless of convergence.

A stronger oracle requires exactly one length1 key. For each candidate root, the left and right chains then contain only lengths at least2. One forward and one reverse pass supply their closing costs for every root position, giving all forced-root bounds in $O(nH)$ total time. The projection above preserves the unique root, so tail compression remains valid for each marginal. Retaining the maximum bound seen for each root, then taking the minimum over roots, is also a valid lower bound.

A structural gap. The sequence $l = (3, 1, 2, 3, 2)$ has $g = (3, 3, 2, 3, 3, 2)$, a unique root, and all the level-count equalities. It is not a valid BST depth sequence. For

$$p = (3, 27, 21, 16, 12), \qquad q = (18, 22, 27, 22, 12, 22),$$

it costs 470, while the true optimum is 472. Its penalty is zero for every multiplier vector. Hence neither oracle family can ever exceed 470 on this instance. More optimization iterations cannot remove this obstruction.

5.3 Exact depth ordering and an arbitrary-window obstruction

Call a positive integer sequence *ranked* if every nonempty contiguous interval has a unique minimum. Equivalently, equal labels have a smaller label between them. This is vertex ranking on a path with the label order reversed; vertex ranking and bounded-path variants are existing notions [7, 8]. The following results explain the ordering information absent from the preceding relaxation; no novelty claim is made for this characterization.

Theorem 54 (Depth characterization)**.** *A sequence $l \in \{1, \ldots, n\}^n$, $n \geq 1$, is the inorder key-comparison-length sequence of a BST if and only if it is ranked, $K_1 = 1$, and every level-flow equality holds, with gaps derived from l as above.*

Proof. Necessity follows from the unique shallowest key in each inorder interval and the two children of every internal node. For sufficiency, delete all labels at most d. Let U_d count the remaining nonempty runs, and let A_d count the deleted labels. If the deleted labels form R runs and include e endpoints, then the number of gaps of length at most d is $A_d - R + e$, whereas $U_d = R + 1 - e$. Thus

$$G_{\leq d} = A_d + 1 - U_d.$$

Summing the level-flow equalities from 1 through d, and using $K_1 = 1$, also gives

$$G_{\leq d} = A_d + 1 - K_{d+1}.$$

Hence $U_d = K_{d+1}$. Ranking permits at most one label $d + 1$ per remaining run, so equality forces exactly one. Starting with the unique length1 root, every remaining nonempty interval therefore receives its root at the next level. This constructs the tree with exactly the supplied lengths. □

Ranking has a linear recognizer: scan with an increasing stack, pop larger labels, reject an equal top, and otherwise push. A rejected pair has no smaller intervening label; conversely every such pair causes rejection. Each entry is pushed once and popped at most once. This establishes cheap recognition, not cheap minimization over all valid sequences.

Theorem 55 (Bounded-window obstruction)**.** *For every integer $w \geq 2$ and $t \geq 2$, there is a strictly positive full-OBST instance with $n = w + 3$ keys and an inadmissible depth sequence satisfying all level-flow equalities, a unique length1 root, and ranking on every interval of at most w keys. Its cost is at most $(7/8 + 1/(8t))\,\mathrm{OPT}$. Consequently every lower bound obtained by pricing the level-flow equalities over this window-constrained domain has the same upper limitation, regardless of the multipliers.*

Proof. Take

$$l = (3, 1, 2, 3, 4, \ldots, w + 1, 2).$$

The ascending middle portion has $w - 1$ entries. Its two enclosing length2 labels form the only interval with a repeated minimum, of width $w+1$. Every shorter window passes. Direct substitution into the gap formula verifies $K_1 = 1$ and every level-flow equality. The sequence is invalid: the unique root is key1, but its singleton left subtree has length3, and two keys to its right both claim length2.

Give every gap weight1 and every key weight1 except

$$p_1 = 3B, \qquad p_2 = p_{n-1} = B.$$

The unit-weight outcomes contribute $E = w^2 + 4w + 10$ in the inadmissible sequence, whose cost is $7B + E$. Every actual tree costs at least $8B$. If key1 is the root, the two B-weight keys lie to its right and their comparison lengths sum to at least5, giving $3B + 5B$. Otherwise key1 has length at least2 and each B-weight key has length at least1, giving $6B + 2B$.

Set $B = tE$. The inadmissible cost satisfies

$$7B + E = \left(\frac{7}{8} + \frac{1}{8t}\right) 8B \leq \left(\frac{7}{8} + \frac{1}{8t}\right) \mathrm{OPT}\,.$$

Its level-flow penalty is zero for every multiplier vector, so every priced minimum is at most this cost. □

For $t = 16$ the upper ratio is 113/128, independently of w; as t increases it approaches 7/8. This rules out universal exactness of the specified window-minimum repair, not other local constraints, adaptive nonlocal cuts, or subquadratic full-OBST algorithms. Equivalently, at every $n \geq 5$ the construction passes all windows of up to $n - 3$ keys.

Burying the obstruction below a forced prefix. Let a current instance have mass W and a feasible tree of cost U. Prepend a key of weight $A = W + U + 2$ and a new leftmost gap of weight1. Using the new key as root costs $A + 1 + W + U = 2A - 1$, while every other root costs at least $2A$ from that key alone. Hence the new root is forced and $\mathrm{OPT}_{\mathrm{new}} = A + 1 + W + \mathrm{OPT}_{\mathrm{old}}$. The transformation $l \mapsto (1, l_0 + 1, \ldots, l_{n-1} + 1)$ preserves zero flow and local window ranking, and changes the inadmissible cost by the same $A + 1 + W$. Repeating this D times places the strict gap beneath the true optimal top-D right-chain prefix. Thus fixing any constant number of top levels cannot make the remaining window/flow relaxation exact. The constant relative gap is not claimed after this embedding, because the forcing weights add a potentially large common cost.

5.4 Endpoint responses: insufficiency and representation size

An intact subtree interacts with its ancestors through its relative cost and total mass. If its interior is not yet fixed, one tempting richer summary is the response to added weights at its endpoint gaps:

$$\mathcal{R}(x, y) = \mathrm{OPT}(p, q + xe_0 + ye_n), \qquad x, y \geq 0.$$

For a fixed tree the response is $C(T) + xd_0(T) + yd_n(T)$, so $\mathcal{R}$ is a concave lower envelope of affine planes. We distinguish whether this function contains enough information for composition from how many pieces an explicit representation requires. Neither result below is a general lower bound for optimal-BST construction; priority remains unestablished.

5.4.1 Equal responses need not be interchangeable

Theorem 56 (Insufficient endpoint-value interface)**.** *An interval's endpoint response, length, total mass, and endpoint gap weights do not determine its optimal cost in an arbitrary surrounding search problem. This remains true if its full vector of key weights and its multiset of gap weights are also retained.*

Proof. Consider two four-key intervals with

$$p = (1, 1, 2, 3), \quad q^A = (1, 4, 2, 1, 1), \quad q^B = (1, 4, 1, 2, 1).$$

Both have mass 16 and endpoint gap weights one. Enumerating the 14 four-key shapes gives the following least intercepts for each possible endpoint-depth pair:

Depths	A	B
$(1, 2)$	45	45
$(1, 3)$	42	42
$(1, 4)$	44	45
$(2, 1)$	43	43
$(2, 2)$	36	36
$(2, 3)$	36	37
$(3, 1)$	39	39
$(3, 2)$	36	36
$(4, 1)$	43	42

Each plane with slopes $(1,4),(2,3),(3,2)$, or $(4,1)$ is pointwise dominated on the nonnegative quadrant by a common plane whose slopes and intercept are no larger. Hence both responses are exactly

$$\mathcal{R}(x,y) = \min\{45 + x + 2y,\ 42 + x + 3y,\ 43 + 2x + y,\\ 36 + 2x + 2y,\ 39 + 3x + y\}.$$

This proves equality for all parameters, not just sampled ones.

Prepend keys of weights $(7,1)$, append keys of weights $(1,7)$, and give the four new gaps weight one. The full inputs are

$$P = (7,1,1,1,2,3,1,7),$$
$$Q^A = (1,1,1,4,2,1,1,1,1),$$
$$Q^B = (1,1,1,4,1,2,1,1,1).$$

Using the exact recurrence, the least full costs conditional on each root are

Root	0	1	2	3	4	5	6	7
A	108	108	103	101	101	101	111	110
B	108	108	103	101	100	101	111	110

Thus the optima are 101 and 100. Direct enumeration of all 1,430 eight-key trees independently verifies both rows. Equal summaries with the identical surrounding context therefore have unequal answers. □

In particular, exact prepend/append updates closed on the stated summary cannot exist: identical starting summaries and identical updates would remain indistinguishable. The theorem does not exclude extra shape information, non-exposed depth states, signed-price responses, or richer interfaces. It does not assert a distinction between mixed and alphabetic algorithms; it rejects this summary even on small positive mixed inputs.

The lost information includes internal alternatives with identical boundary depths. Both intervals have minimum cost 36 at depths $(2,2)$. The shape rooted at key 2, with left child 0 (whose right child is 1) and right child 3, also has depths $(2,2)$, but costs 37 in A and 36 in B. The least intercept erases this distinction; the surrounding tree can make the additional alternative valuable.

5.4.2 Quadratic explicit envelopes with linear factorizations

For any $n \geq 1$, the endpoint paths share only the root. If their depths are a, b, their union has $a + b - 1 \leq n$ keys. At most one least intercept per depth pair is needed, giving at most $n(n+1)/2$ affine pieces. The quadratic order can occur already for a single interval.

Theorem 57 (Explicit-facet obstruction)**.** *There are strictly positive mixed-weight instances on $n = 2m+1$ keys whose endpoint response has at least m^2 affine pieces on the nonnegative quadrant. Consequently, explicitly listing all such pieces can require $\Omega(n^2)$ entries.*

Proof. Set $K = 4m$ and $P_i = K8^i$ for $0 \leq i < m$. First consider these m keys with all gaps one and added left endpoint weight x. Write $S_j = \sum_{i=0}^{j} P_i$ and $x_j = 2S_j + 2m$.

An endpoint key and its adjoining outer gap are forced to the root if their combined weight exceeds the mass of all other outcomes. Indeed, each rotation of that key toward the root saves one

comparison on both outcomes and adds at most one to each other outcome. The strict inequality makes every rotation improve cost. At x_j, for every $k > j$,

$$P_k + 1 > S_{k-1} + x_j + k + 1,$$

while in the remaining prefix

$$P_0 + 1 + x_j > (S_j - P_0) + (j + 1).$$

The geometric series and $K = 4m$ imply these inequalities. For example, $S_{k-1} < P_k/7$ and $x_j \le 2S_{k-1}+2m$ for $k > j$, whereas $P_k \ge 32m$. Thus every optimum first uses roots $m-1, m-2, \dots, j+1$, then key 0, giving left endpoint depth $m - j$. All m possible one-sided slopes occur, with strict inequalities persisting in open neighborhoods of the probes.

Mirror this instance about a central key:

$$p = (P_0, \dots, P_{m-1}, R, P_{m-1}, \dots, P_0), \quad q = (1, \dots, 1).$$

Let $X = x_{m-1}$, $S = S_{m-1}$, $B = 2S + 2m + 2 + 2X$, and $R = (2m + 2)B$. For $0 \le x, y \le X$, the mass outside the central key is at most B. A center-root tree costs at most $R + nB < 2R$; any other root charges the central key at least $2R$. Thus the center root is forced. At (x_j, x_k), every optimum has endpoint depths

$$(m - j + 1, m - k + 1).$$

These are m^2 different gradients. Since there are finitely many trees and all minimizing trees at each probe have the same gradient, that probe has an open affine neighborhood. Different gradients require different pieces. □

The weights have $O(n)$ bits each. This is an arithmetic/output-size obstruction to explicit facet enumeration, not a word-RAM lower bound with $O(\log n)$-bit weights, nor a lower bound for a single optimum query. The same example also explains why counting facets cannot rule out implicit representations.

Lemma 58 (Exact hinge factorization)**.** *For the one-sided family above, let*

$$C_0 = \sum_{i=0}^{m-1} (m - i)P_i + m + \frac{m(m + 1)}{2}.$$

Its response for every $x \ge 0$ is

$$H_m(x) = C_0 + mx - \sum_{i=1}^{m-1} \max\{0, x - (P_i - P_0)\}.$$

Within the forced-center box the two-sided response equals

$$R + 2S + 2m + 2 + x + y + H_m(x) + H_m(y).$$

Proof. At x_j, the remaining right subtree of key 0 is a decreasing-key chain, by the same strict endpoint condition. Passing from j to $j + 1$ rotates key 0 above key $j + 1$. At base weights the intercept increases by $P_{j+1} - P_0$, while the endpoint slope decreases by one. At $j = 0$ the intercept is C_0. Every possible endpoint depth occurs among these m trees, and optimality at a probe proves its intercept is minimal for that depth. Their successive intersections are the increasing thresholds $P_i - P_0$, giving the hinge formula. The center-root recurrence gives the two-sided identity. □

The factorization extends to the entire quadrant. Set

$$W = R + 2S + 2m + 2, \qquad V = W - P_0 - 1,$$
$$L = C_0 - mP_0 - m - 1, \qquad J = R + 2(S - P_0) + 2m + 2L.$$

Here L is the fixed cost of the remaining one-sided keys $1, \ldots, m-1$, and J is the fixed inner cost after both outer keys and outer gaps are removed.

Theorem 59 (Full-quadrant factorization)**.** *For the geometric family and every $x, y \geq 0$,*

$$\begin{aligned}\mathcal{R}(x,y) = W + x + y + \min\{&H_m(x) + H_m(y),\\ &V + y + L + H_m(y),\ V + y + J,\\ &V + x + L + H_m(x),\ V + x + J\}.\end{aligned}$$

This function has exactly $m^2 + 2m + 2$ affine pieces. It admits $O(m)$ arithmetic preprocessing and storage, $O(\log m)$ value queries, and $O(m)$ complete tree construction per query.

Proof. Any root other than keys $0, m, 2m$ gives both added endpoint weights depth at least two and the central key comparison length at least two. A center-root candidate with the two outer keys as children also puts both endpoints at depth two, with base cost at most $R + n(W - R) < 2R$. It therefore strictly beats such a root for all x, y. If root 0 is used, its remaining one-sided problem similarly has only the center or the other outer key as a possible root; symmetrically for root $2m$. Expanding these five cases gives the formula. The fixed remaining intervals are geometric chains on either side of a forced central root, giving the stated L, J.

The center-root term supplies m^2 slopes, all exposed by the original probes. For root 0, the same $2R$ argument forces the center in the right problem throughout the original probe range, exposing all m pieces inherited from $H_m(y)$. As y grows unbounded, one final piece appears when the outer key becomes its root. Taking x sufficiently large forces root 0, exposing these $m + 1$ pieces globally. Their slopes are $(1, 2), \ldots, (1, m + 2)$. Reflection gives another $m + 1$ pieces with second slope one. All slopes are distinct. The formula has no further candidate slopes, proving the count.

Prefix sums of the ordered thresholds let two binary searches evaluate $H_m(x)$ and $H_m(y)$. Four comparisons select the outer minimum. The selected case and one-sided chain indices reconstruct exactly $2m + 1$ keys in linear work. □

Thus $2(m - 1)$ hinge terms and five alternatives represent a quadratic number of affine regions. Endpoint-dominance shortcuts already solve this family. It is not a hard-query example, and the facet theorem is not an expression-size lower bound. A lossless factorization also does not repair Theorem 56: the value function itself omits information needed by some surrounding contexts.

A favorable bound independent of the added weights

Lemma 60 (Endpoint mass-ratio bound)**.** *Let the original total mass be W, with $q_0, q_n > 0$. The entire response for arbitrary $x, y \geq 0$ has*

$$O(\log(2 + W/q_0) \log(2 + W/q_n))$$

affine pieces, also bounded by $O(n^2)$. Polynomial endpoint mass ratios therefore give $O(\log^2 n)$ pieces.

Proof. If an optimum has left endpoint depth $a \geq 2$, its last internal ancestor is at depth $a-2$ in the root's left subtree. This subtree is optimal and excludes the opposite endpoint and its added weight y. Its mass is at most $W + x$, while the last ancestor's subtree has mass at least $q_0 + x$. Lemma 2 gives

$$F_{a-1} \leq \frac{W+x}{q_0+x} \leq \frac{W}{q_0}, \qquad F_1 = F_2 = 1.$$

The symmetric bound is $F_{b-1} \leq W/q_n$ for right depth $b \geq 2$. Fibonacci growth bounds the number of possible slopes in each direction; only a least intercept per pair can be required. □

The added parameters do not occur in this bound. The quadratic example has exponential original mass ratio, so the two results are consistent. This lemma bounds output size, not the work to discover the pieces. Moreover, the composition counterexample has just five pieces: a small response need not be a sufficient interface.

5.5 Ruler patterns and a dense-core representation barrier

The geometric-weight control has a dense core whose entries are constant along each row. This motivates asking whether another finite difference could restore sparse support. The answer is negative even for a smooth, positive, polynomial-weight family that has an exact compact formula. Before proving that barrier, we isolate the ruler pattern underlying both the penalty layers and a new sufficient exchange condition. Priority of these identities and conditions is unestablished.

5.5.1 Saturated layers as weighted subsequences

Write $c_h = 2^h - 1$ and let $\nu_2(t)$ be the exponent of two dividing the positive integer t. Use the penalty recurrence above with $M > HW$. For any interval of length $k = j - i \geq c_h$,

$$A_h(i,j) = M(k - c_h) + hW(i,j) - \max_{i \leq s_1 < \cdots < s_{c_h} < j} \sum_{t=1}^{c_h} \nu_2(t) p_{s_t}. \tag{30}$$

Indeed, the penalty forces exactly c_h exposed keys: losing one exposure costs more than any possible reduction in the ordinary weighted cost. Every position in the first h levels is occupied. The exposed skeleton is perfect, all gaps and unexposed keys pay h comparisons, and its tth inorder key has comparison length $h - \nu_2(t)$. This proves the identity. The coefficient sequence is $0, 1, 0, 2, 0, 1, 0, 3, \ldots$.

At $h = 2$ the maximum is simply $\max_{i+1 \leq r \leq j-2} p_r$. At $h = 3$, greedy selection of the largest eligible root fails: for $p = (0, 9, 0, 9, 10, 9, 0, 0)$ and all nine gap weights equal to one, root 3 permits score 36, while the larger key at root 4 permits only 29. Adding one to all key weights adds four to every feasible score, giving the same failure with strictly positive weights. For nondecreasing p, the rightmost c_h keys maximize the score coordinatewise; for nonincreasing p, the leftmost keys do. Thus their saturated-layer density vanishes whenever $j - i \geq c_h + 1$, independently of the gap weights. This does not discard the gaps or solve the full problem: shorter intervals have nonuniform gap depths, and their band becomes quadratic at large h. The weighted-subsequence reformulation by itself supplies no new efficient min-plus product.

5.5.2 A positive family with dense differences and compact costs

Set

$$B = 4(n+1)^3, \qquad p_r = B + r \ (0 \leq r < n), \qquad q_r = B + r \ (0 \leq r \leq n). \tag{31}$$

All weights use $O(\log n)$ bits and the deviation above their minimum baselines is exactly $R = n^2$.

Theorem 61 (Dense support does not force expensive construction)**.** *For the family (31), the optimal interval-cost matrix has $\Omega(n^2)$ nonzero Monge densities and $\Omega(n^2)$ nonzero mixed fourth differences. It also has quadratically many rowwise density changes. Nevertheless, its exact tree is constructible in $O(n)$ time, and any interval value can be queried using $O(\log n)$ arithmetic operations after linear recognition of the family.*

Proof. First force the shape class. An interval of k keys starting at i has constant baseline $B+i$. A tree of internal path length I has total key and gap comparison length $D = 2I + 3k \le k^2 + 2k$. All relative weights lie between zero and k, so their total contribution is at most $n(n^2+2n)$. Increasing I by one adds at least $2B$, which exceeds this entire bound. Hence every optimum has minimum I. Its adjacent exchange cost (27) is $-3 - \nu_2(g+1) < 0$, so its last level is filled from the left.

Here is an exact value formula. For $k = j - i > 0$ put $h = \lfloor \log_2(k+1) \rfloor$, $c = 2^h - 1$, $s = k - c$, and define

$$\begin{aligned} I(k) &= (h-2)2^h + 2 + hs, & D(k) &= 2I(k) + 3k, \\ J(c) &= \sum_{t=1}^{c} (t-1)\nu_2(t), & T(c,s) &= \sum_{t=1}^{c} \min\{s,t\}\nu_2(t), \\ E(k) &= hc^2 - J(c) + s(2hc + h + 1) \\ &\quad + (h+3)s(s-1) - T(c,s). \end{aligned}$$

Then

$$C(i,j) = (B+i)D(k) + E(k), \qquad C(i,i) = 0. \tag{32}$$

To verify E, the perfect skeleton has linear-weight cost $hc^2 - J(c)$. For an arbitrary set S of expanded slots, each skeleton key at r shifts by the number of selected slots at most r, and each skeleton gap at g shifts by the number strictly below g. Summing these index changes and the new key/gap contributions gives

$$\begin{aligned} E(S) = hc^2 - J(c) + \frac{(2h+3)|S|(|S|-1)}{2} \\ + \sum_{g \in S} \left(2hc + 3g + h + 1 - \sum_{t=g+1}^{c} \nu_2(t) \right). \end{aligned}$$

Taking $S = \{0, \dots, s-1\}$ gives (32). For each $\ell = 1, \dots, h-1$, let $d = 2^\ell$, $m = \lfloor c/d \rfloor$, and $u = \min\{m, \lfloor s/d \rfloor\}$. The contributions to J and T are, respectively,

$$\frac{dm(m+1)}{2} - m, \qquad \frac{du(u+1)}{2} + (m-u)s.$$

This evaluates the formula in $O(\log n)$ steps, using $\nu_2(t) = \sum_{\ell \ge 1} [2^\ell \mid t]$. The tree itself needs one recursive visit per key: its left subtree has $2^{h-1} - 1 + \min\{s, 2^{h-1}\}$ keys. Recognition scans the given input; the interval oracle then retains only n and B.

Now examine support. Write

$$d(i,j) = C(i,j+1) + C(i+1,j) - C(i,j) - C(i+1,j+1).$$

Inside a fixed h band, for $1 \le s < 2^h$,

$$d(i,j) = 3 + \nu_2(s), \qquad s = (j-i) - (2^h - 1). \tag{33}$$

Indeed, D has slope $2h+3$, while E has second difference $2h+6+\nu_2(s)$. The endpoint at $s=2^h$ is the next perfect tree, so the calculation remains valid for $s=2^h-1$. For $2\le s\le 2^h-2$ the mixed fourth difference is therefore

$$\begin{aligned} &d(i,j+1)+d(i+1,j)-d(i,j)-d(i+1,j+1)\\ &\quad=\nu_2(s+1)+\nu_2(s-1)-2\nu_2(s). \end{aligned} \tag{34}$$

It is strictly positive for odd s and strictly negative for even s. For $n=2^m\ge 32$, choose $h=m-2$ and $2\le s\le n/4-2$. There are $n/4-3$ lengths and at least $n/2+2$ valid placements per length, giving at least $(n/4-3)(n/2+2)=\Omega(n^2)$ nonzeros. For arbitrary sufficiently large n, use $h=\lfloor\log_2 n\rfloor-2$; both the band width and number of placements remain $\Theta(n)$. The same bands prove quadratic second-difference support and rowwise density changes. □

Consequently any method explicitly listing either support has quadratic output work on this family. This is a representation lower bound; the linear constructor rules out interpreting it as a computational lower bound for the family. The example also reaches the $R=\Theta(n^2)$ boundary of the centered analysis with smooth, very nearly equal relative weights. Symbolically representing the dyadic baseline avoids the dense list, but subtracting it need not leave a Monge residual. Efficient product closure with arbitrary residuals is still missing.

5.6 Joint profile coupling and a strict merge obstruction

The separate key and gap minima need not coexist. Let $J(p,q)$ relax only the key positions, keeping the gaps alphabetic. With key weights $p_{[r]}$ decreasing and key comparison lengths $\ell_{(r)}(T)$ increasing,

$$J(p,q)=\min_T\left\{\sum_r p_{[r]}\ell_{(r)}(T)+\sum_i q_i d_i(T)\right\}.$$

Writing $A(q)$ for the exact alphabetic gap optimum gives $K(p)+G(q)\le K(p)+A(q)\le J(p,q)\le \mathrm{OPT}(p,q)$. If $P=\sum p_i$, S_j sums the j largest key weights, and $a_t(T)$ counts keys of comparison length at most t, then

$$\sum_r p_{[r]}\ell_{(r)}(T)=P+\sum_{t\ge 1}(P-S_{a_t(T)}).$$

Each key is counted once plus once per additional level. The function $P-S_j$ is decreasing and discretely convex. This standard rearrangement and tail-sum identity describes global depth-profile coupling; it does not give a fast minimizer over alphabetic trees.

5.6.1 A precisely specified rejected extension

Start with ordered original gap leaves. Active items are compatible when no unmerged original leaf lies strictly between them. Merge a compatible pair of minimum total mass, adding the smallest unused key weight to its mass; replace its left item and delete its right item. Recover original leaf depths, reconstruct an alphabetic tree, and optimally reassign the key multiset to its internal depths. This is a proposed mixed extension of compatible merging, not the established alphabetic algorithm of Hu and Tucker [15].

For the full ordered instance

$$p=(7,1,5,8,6,4),\qquad q=(18,2,13,3,17,1,2),$$

the next-key/pair/new-mass triples are

$$(1;1,2;4),\ (4;2,13;19),\ (5;3,17;25),\ (6;18,4;28),\ (7;19,25;51),\ (8;28,51;87).$$

The margins over the next compatible pair are $12, 1, 1, 1, 3$, followed by the sole remaining pair. Thus all choices are strict, all weights positive, and all key weights distinct. The charges sum to 214.

Quantity	Greedy profile	Better feasible tree
Sorted key lengths	$1,2,2,3,3,3$	$1,2,2,3,3,4$
Ordered gap depths	$2,3,3,3,3,3,3$	$2,4,4,3,2,3,3$
Optimally assigned key cost	64	65
Gap cost	150	148
Total	214	213

The better tree has root 3, left root 0 with right child 2 and 2's left child 1, and right root 4 with right child 5. Its inorder key lengths are $(2,4,3,1,2,3)$, so the displayed ordered weights attain the relaxed assignment. Enumeration of all 132 shapes gives $J = 213$; independent cubic and Knuth recurrences give OPT $= 213$. Here $K = 64$, $G = 144$, and $A = 148$, leaving deficits even for $K + A = 212$. The value 214 is a relaxed candidate cost: the greedy reconstructed shape with the original ordered p costs 220. Thus the failure persists even after freeing all key positions, and 214 is unsafe as a lower bound.

Theorem 62. *This specified merge extension fails on a positive open box, has arbitrarily large additive error under scaling, and fails for every number of keys $n \geq 6$ with positive weights, distinct keys, and strict pair choices. The latter family uses weights with $O(n)$ bits.*

Proof. Scaling multiplies 214 and 213 by the same factor. Perturb all thirteen weights independently by at most $\epsilon = 1/1000$. Key ranks are unchanged. Each active mass is a sum of original outcomes, so each pair-score difference changes by at most $26\epsilon < 1$. Induction preserves every strict merge. Candidate minus displayed witness cost is $q_4 - q_1 - q_2 - \min p$, at least $1 - 4\epsilon > 0$. No optimality assumption on the perturbed witness is required.

For padding, let an old full instance have mass W and known optimum U. Put $B = W + U + 1$, and prepend a key of weight $2B$ and a gap of weight B. The tree rooted at the new key, with the old tree on the right, costs $3B + W + U = 4B - 1$. Any other root puts the new key at comparison length at least two, costing at least $4B$. Therefore $\mathrm{OPT}' = 3B + W + \mathrm{OPT}$. All old compatible pair masses are at most W, whereas any pair involving the new gap exceeds $B > W$. The new key is largest. All old merges occur first and the final charge is $3B + W$. Alphabetic reconstruction raises all old depths by one, so the relaxed candidate gains the same amount and its excess of one persists. The recurrences $W' = W + 3B$, $U' = 4B - 1$, $B' = W + 7B$ imply $7B \leq B' < 8B$, and hence linear bit length. This proves failure of the rule, not a computational lower bound for the original problem. □

5.6.2 Uniform-transfer invariance and exact controls

Every extended binary tree satisfies $\sum d_i(\text{gap}) - \sum \ell_i(\text{key}) = n$. For $-\min q \leq a \leq \min p$, set $p' = p - a$, $q' = q + a$. Every full tree cost, and also J, increases by an. If $M(n)$ is the minimum unit-key comparison-length sum, $E(n+1) - M(n) = n$. The key bound decreases by $aM(n)$, while G increases by $aE(n+1)$: its variation term is translation invariant and its rank coefficients sum to $E(n+1)$. Consequently $K(p') + G(q') - an = K(p) + G(q)$. For the rejected merge rule every

active cluster has one more original gap than consumed key. Its mass increases by a, so all pair scores increase by $2a$, all choices and margins are unchanged, and the sum of charges increases by an. Uniform transfer cannot repair that rule or the $K + G$ deficit. The exact alphabetic bound behaves differently below.

5.7 A four-key barrier for every admissible transfer

Theorem 63. *For $p = (1, 1, 2, 1)$ and $q = (1, 2, 1, 1, 1)$, the full optimum is 24, while the largest $K + A$ lower bound over all nonnegative additive transfers preserving relative tree costs is exactly 23.*

Proof. The all-unit full instance costs at least 20. The two additional units add $\ell_2 + d_1$. If key 2 is root, gap 1 has depth at least three; otherwise $\ell_2 \geq 2$ and the internal gap has depth at least two. The added cost is always at least four. Root 1, left child 0, and right child 2 with right child 3 attains 24.

On the maximal face $a_* = 1$, $L = R = 2$ and

$$p' = (x, 0, 1, y), \qquad q' = (2 - x, 3, 2, 2, 2 - y).$$

The capacity assignment $(2, 3, 1, 2)$ gives $K(p') \leq 2x + 1 + 2y$. The feasible gap depths $(2, 2, 2, 3, 3)$ give $A(q') \leq 26 - 2x - 3y$. Hence $F(x, y) \leq 23 - y \leq 23$ throughout the box. At $x = y = 0$, $K = 1$ and $A = 26$: five gaps of weight two cost at least $2E(5) = 24$, and the additional unit on the internal gap costs at least two; the displayed depths attain equality. Theorems 47 and 48 extend the bound to every admissible transfer. □

Integer rounding does not close this gap. Only one residual interior key has positive weight after normalization. The theorem limits this separable certificate class; it does not rule out a richer joint representation or a faster full algorithm.

The failure persists on a positive open box. Perturb each of the nine weights by at most $\epsilon = 1/100$. The interior minimum remains p_1. The same assignment and gap tree bound the maximum by

$$2p_0 + 2p_1 + p_2 + 3p_3 + 2q_0 + 2q_1 + 2q_2 + 3q_3 + 3q_4 \leq 23 + 20\epsilon.$$

Every four-key tree has unit-weight cost at most 24, so every perturbed original tree costs at least $24 - 24\epsilon$. Thus the gap is at least $1 - 44\epsilon > 0$ throughout the box.

A failed transfer of the sharper mass constant. The initial implementation used $2W/F_{d+3}$ as the maximum internal subtree mass at edge depth d, following a literal transfer of the Fibonacci formula in [5]. Its first full-input check failed: the resulting class could be empty. The positive two-key example

$$p = (1, 1), \qquad q = (1, 100, 1)$$

explains the failure. Both trees cost 206; the total mass is 104 and the internal child has mass 102. Thus $102 > 2 \cdot 104/3$ at depth one. Charging an additional comparison at every gap is a tree-independent constant and does not remove this violation. We preserve the counterexample and use the following independently justified, weaker constant in our model. This does not refute the KLR architecture or its asymptotic guarantees.

Partial-tree height caveat. An earlier proposed iteration started from the additive singleton lower bound. Its attempted rotation proof of logarithmic convergence was invalid: a rotation can lift an unfinished interval, requiring extra refinement to restore the prescribed number of levels. Small passing tests do not remove that issue. The upper-penalty proof above avoids that argument.

6 Core-size tightness and further questions

Theorem 6 rules out an $O(n \operatorname{polylog} n)$ bound for all intermediate cores at linear integer mass. The remaining size question is the $\sqrt{\log n}$ factor between this lower bound and $O(n\sqrt{WL})$ when $W = n$. A stronger upper bound must respect the explicit odd-ramp density formula, or a stronger lower bound must force additional nonzero cells. The family arose from the adversarial search; its theorem follows from the deletion lemma and the closed-form density, independently of the search outcomes.

The bound concerns an explicit density representation. The unrestricted subquadratic construction problem remains open, as does the possibility of computing with a more compact representation of these structured dense cores. The quadratic ruler example already shows why representation size alone cannot establish a general algorithmic lower bound.

A Independent verification and operation counts

The theoretical statements above do not depend on these finite tests. Independent cubic recurrences, literal tree enumeration, direct original-cost evaluation, and operation counters provide falsification controls. Reproduction code and source-hashed records are available in the accompanying repository. The filenames below identify those records; they are not definitions needed by the proofs.

A.1 Implemented condensed constructors

The compact path is now implemented in Python and independently in C++. The product algorithm is the published compression, balanced core split, staircase merge, and decompression method of [10]. We credit that primitive separately from the OBST reduction. A matrix stores its top row, left column, and positive density triples. Two linear bucket passes canonicalize the triples. Boundary extraction uses one-dimensional sweeps; the product merge uses adjacent-entry recomputation along its staircase. Each relevant core row or column is scanned only a constant number of times per merge. None of these operations enumerates the full interval table.

Exact early stopping. The constructors compare consecutive completed layers by their top rows and canonical core triples; the left column is fixed. Equality is therefore equality of the whole table. A fixed point satisfies the ordinary OBST recurrence with zero diagonal. Induction on interval length proves that this recurrence has a unique solution, namely C. Stopping at such a fixed point is exact. Equality of the single full-interval cost is not used. Retained layer witnesses reconstruct a complete optimum: any unfinished nonempty terminal would contribute at least $M > HW$, exceeding the cost of an optimal tree of height at most H. All charges are nonnegative. Reconstruction uses an explicit stack and visits every original key.

Implementation costs and arithmetic. Both implementations retain ordinary integer coordinate maps and recursive witness nodes. Their witness storage is bounded by $O(H(n+K)\log(n+1))$ words, where $K = O(n + \min\{n^2, n\sqrt{WL}\})$ bounds the completed cores. This can exceed the succinct storage bound in Theorem 1 by a logarithmic factor. Compression bounds recursive dimensions by core size, and the core count halves along a recursive branch, so the time bound is preserved. No succinct rank/select implementation is claimed.

The selected perturbation sweep certified amplitude 8 around 1,000 at $n = 256$, amplitude 4 at $n = 512$, and amplitude 1 at $n = 1,024$; larger tested amplitudes at those sizes failed. These are

seeded observations, not universal thresholds. Ordinary random inputs failed the certificate. Sizes 384, 768, and 1,000 deliberately skip the shortcut because their minimum-I shape space exceeds the logarithmic width budget.

A.2 Exact bounded-excess construction by dyadic intervals

Validation, failed precursor, and prior work. The implementation passed 1,200 restricted-optimum comparisons against all shapes through eight keys with budgets $0, \ldots, 4$, and 3,000 full mixed-weight checks against cubic DP. It separately checks direct tree costs, both residual-bound implementations, and exact fallback. An additional 104 family checks include zeros, geometric weights, spikes, and key-only/gap-only controls. The seeded size sweep is retained. A precursor substituting ordered depth bounds into the old certificate passed 480 exhaustive checks but rescued none of 24 near-uniform fixtures at its 32-iteration budget. This negative experiment did not prove optimal multiplier search useless; its cost and the failed perfect-tree hypothesis motivated expanding the candidate class.

On the 1,023-key periodic fixture with weights 16 and 17, the excess-two DP finds cost 312,154 and excludes other shapes with bound 312,185. It uses 27,570 states, 137,835 transition checks, 6,052 sorting comparisons, and 1,022 merge comparisons, versus Knuth's 523,776 states and 1,046,532 root candidates. At $n = 1{,}024$, a seeded amplitude-four perturbation uses 48,003 states and 240,000 transition checks, plus 13,784 sorting and 1,264 merge comparisons. It certifies cost 19,477,996.

The dyadic/prefix-code viewpoint and sequential frontier DP are established ideas. Becker [6] gives a sequential mixed-weight DP for a near-minimal *height* restriction in quadratic time. The band above uses excess *total unweighted path length*, a different restriction; it does not improve Becker's stated problem. Priority of this particular potential, band bound, and exclusion combination remains unestablished.

A.3 A split-height construction covering every input size

Exact verification and operation evidence. The harness passed 450 exhaustive weighted height-class checks through eight keys, 44 exhaustive tall-path minima, 2,000 cubic-checked mixed inputs with complete fallback (917 successful certificates), and 104 additional adversarial-family inputs. An independent all-root height recurrence checked 520 candidates and 261,482 stored short-table entries; the same 520 cases also compared SMAWK with a retained divide-and-conquer matrix search. The larger sweep has 36 fixtures at three height budgets each. Every near-uniform fixture certified, including previously skipped sizes 384, 768, 1,000, and 1,536. All random, geometric, and gap-spike fixtures failed the certificate, sometimes despite an optimal restricted candidate.

At $n = 1{,}000$, the minimum-height candidate and full optimum are both 18,973,413, below the exclusion bound 18,974,193. The optimizer uses 117,453 short states, 220,483 short root candidates, 84,937 genuine matrix queries, 10,268 padding queries, 16,016 prefix scans, and 5,630 coarse transitions. The bound adds 10,648 sorting and 1,335 merge comparisons, plus 1,000 contractions; all keys are reconstructed. Knuth uses 500,500 states and 1,002,162 root candidates. These distinct operation types are reported separately, without treating them as equal-cost instructions.

A.4 A variable interval frontier for exact full construction

Independent verification. The harness passed 960 weighted exhaustive-shape comparisons through eight keys on arbitrary nondecreasing contours, including infeasible classes. It checked 914 interval costs independently and 90,193 top minima by brute search. Another 2,400 full solves,

using both modes on 1,200 inputs through 40 keys, matched cubic DP and reconstructed costs. All 24,600 internal subtrees in the cubic-optimal trees satisfied Lemma 2. Families include zeros, ties, geometric variants, spikes, and key-only and gap-only controls. A further 56 fixtures through 2,048 keys matched Knuth in both modes, including ordinary random inputs that failed the preceding height certificates.

At 2,048 random keys, the adaptive version chose depth six and returned cost 21,053,524. It used 309,261 interval cells and 616,642 root candidates, versus Knuth's 2,098,176 cells and 4,215,527 candidates. Additional work included 24,035 contour mass checks across seven parameter depths, 1,245,431 genuine matrix queries, 149,319 padding queries, 131,136 scans of each row/column type, and 107,859 coarse transitions. The uniform version stored 337,550 cells. On log-uniform weights the corresponding cell counts were 308,487 versus 431,951; on a single heavy key they were 1,048,576 versus 1,573,376. Geometric inputs selected the full table. All operation categories and parameter levels are retained separately.

A.5 Dominant-gap separators and local mass frontiers

Verification and complete work accounting. The harness exhaustively checks 22,140 weight assignments in $\{0, 1, 2\}$ through four keys against every tree shape. It checks 3,211 qualifying gap certificates and their cost identities, 2,036 endpoint certificates, and 380 key certificates. A further 3,280 full solves, using both frontier and Knuth terminal strategies on 1,640 inputs through 40 keys, match independent cubic costs and direct tree evaluation. Inputs include empty, zero, tied, geometric, spike, key-only and gap-only controls. Both strategies and the direct global frontier also match Knuth on 40 fixtures through 2,048 keys.

At $n = 2048$, the direct and local frontier interval counts are:

Fixture	direct frontier	local frontier	terminal components
Random	308,115	308,115	1
Near-uniform	309,290	309,290	1
Geometric	2,098,176	0	0
Heavy key	1,048,576	153,660	2
Heavy gap	1,047,552	153,582	2
Random background, heavy gap	1,047,552	153,362	2
Hierarchical gaps	522,242	22,333	64

Cell counts do not include all work. On hierarchical gaps, 63 gap separators require 254 endpoint checks, 6,142 index-build comparisons and 1,476 range query comparisons. Terminal solves use 42,885 root candidates, 39,284 genuine top-matrix queries, 6,261 padding queries, 8,122 contour mass checks, 6,952 scans of each row/column type, and 1,792 coarse transitions. Their 1,922 keys are reconstructed and relabeled. On the heavy-gap fixture, 305,118 terminal root candidates and 1,214,092 genuine top queries accompany the reduced interval table. All counter categories are retained; metadata such as local selected depths and actual heights are summed over terminals, not reported as a global tree height. Geometric input uses 2,048 endpoint separators and 4,095 endpoint checks, with no index or terminal work.

A.6 Joint heavy-outcome cores

Independent checks and operation counts. Every nonempty selected-outcome subset and every tree shape through four keys were examined. In all 1,937 non-core-closed cases a core-closed tree has componentwise no-greater selected depths and at least one strict improvement. On 22,140

inputs with weights in $\{0, 1, 100\}$, all 17,054 qualifying nontrivial certificates match the optimum over every shape and the composition identity. Another 2,640 full solves on 1,320 inputs through 32 keys, with both kernels, match independent cubic DP and reconstructed costs. On 520 small inputs the threshold sweep matches direct enumeration of every threshold and payload partition. Both kernels also match Knuth and the previous separator constructor on 30 larger fixtures through 2,048 keys.

At $n = 2048$, the following cell counts include both core and payload solves, or the complete fallback when certification fails:

Fixture	previous separator	joint core/frontier	core keys
Random	309,201	309,201	—
Near-uniform	309,290	309,290	—
Two heavy gaps, unit background	2,098,176	10	4
Two heavy gaps, random background	2,098,176	306,506	4
45 heavy gaps	304,486	23,965	90
45 moderate gaps	304,486	23,695	90
45 mixed heavy outcomes	305,421	24,016	68

The moderate-gap fixture sets each selected weight to exactly $(H + L)W_0 = 42{,}395{,}178$, with $H = 9$, $L = 12$, and $W_0 = 2{,}018{,}818$. The old size-based condition would require $(90 + 12)W_0$ and fails. Work includes 63,549 sorting comparisons, 2,047 Fibonacci additions, 3,024 height checks, 1,958 activations, 1,912 unions, and 1,511 find steps. Kernels use 45,365 root candidates, 116,338 genuine top queries, 23,904 padding queries, 12,336 contour checks, and 17,488 scans each for rows and columns. All 2,048 keys are reconstructed. Local kernel metadata such as heights and chosen depths are sums across calls, not global tree heights. The uniform two-gap case still pays selection, 2,044 payload-node constructions and relabelings, and ten core DP cells; it does not perform only ten total operations.

A.7 Joint heavy-outcome cores

Full construction beyond the quadratic-table scale. On the unit-background two-heavy-gap family, full trees were constructed at $n = 512, 2048, 8192, 32768, 65536$. Each exact cost matched an independent cubic solve of the four-key instance

$$p = (1, 1, 1, 1), \qquad q = (1, n^3, 2n - 7, n^3, 1),$$

plus $2I_*(n - 4) + 3(n - 4)$ for the uniform payload, and direct evaluation of the full reconstructed tree. The uniform closed form was separately checked against cubic DP through 16 keys. At $n = 65536$, the exact cost is 1,407,374,885,846,941. The construction used ten core DP cells and fifteen core root candidates. It also used 196,799 sorting comparisons, 65,535 Fibonacci additions, 65,535 height checks, 65,532 activations, 65,531 unions, 65,529 find steps, and 196,609 uniformity comparisons. All 65,532 payload keys were constructed and relabeled, and the four core keys were reconstructed.

A.8 Residual-cost certificates for heavy-outcome cores

Independent exact checks. On all 22,140 assignments in $\{0, 1, 4\}$ through four keys, every one of 17,054 nontrivial threshold cases passed the excluded-class lower-bound check against all tree shapes. There were 13,669 passing witness tests and 15,181 passing exact-candidate tests, including 1,512 cases rescued by refinement. Another 14,912 excluded-tree checks used arbitrary

selected subsets, weights in $\{1, 3\}$, and up to three keys. These latter selections need not be weight thresholds.

Both kernel choices and both refinement settings passed 6,560 full solves on 1,640 inputs through 40 keys against independent cubic DP and direct tree evaluation. All 1,640 candidate portfolios matched direct enumeration of the two representatives in each core-size band. Forty-eight larger fixtures and nine additional fixtures with independently varying positive key and gap weights matched Knuth through 2,048 keys. The varying backgrounds are integers in $[1, 1000]$ and their selected outcomes have weight $4000n$; the unit-background success rows below use weight $4n$. These are finite tests, not an assertion that every such random instance passes certification.

At $n = 2048$, selected operation categories are:

Fixture	Previous intervals	New intervals	Protected proof cells
Two linear-weight gaps, unit background	568,350	10	10
45 linear-weight gaps, unit background	304,486	1,394	4,095
45 mixed outcomes, unit background	305,421	1,250	2,346
Two gaps, varying background	951,851	307,541	30,941
45 gaps, varying background	304,486	23,757	22,519
45 mixed outcomes, varying background	305,421	24,163	27,558
Random control	308,860	308,860	33,380

New intervals count all executed full-weight kernel cells, including fallback, but exclude the separately displayed protected-core proof cells. The proof column sums all attempted cores. Selected-state metadata is not counted again as executed cells. Sorting, profile scans, lower-bound contractions, witness construction, kernel matrix work, and reconstruction have separate raw counters; this table is not a total instruction count. The random control fails all ten witnesses and pays additional proof work before fallback.

A.9 Residual-cost certificates for heavy-outcome cores

Full large constructions and a remaining certificate gap. For every $n = 2^h$, $3 \le h \le 16$, the full $\theta = 4n$ construction matches Equation (14), direct evaluation of the entire output tree, and an independent four-key cubic core plus the uniform payload formula. At $n = 65,536$, the optimum is $3,604,381$. The proof core and full-weight core each use ten cells and fifteen root candidates. Other counts include 196,799 selection-sort comparisons, 131,204 bound-sort comparisons, 65,536 bound contractions, 65,535 two-queue comparisons, 65,532 activations, 65,531 unions, and 65,529 find steps. All 65,536 witness keys are constructed and evaluated, then all output payload and core keys are constructed. The lower bound is $3,604,482$, giving a certificate margin of 101. The previous method was only planned at this size: it rejects the core and selects 146,498,625 frontier cells at depth 11.

Failure remains informative. At $n = 2048$ and selected gap weight $2n$, the exact core-closed cost $71,611$ equals the unrestricted optimum, but the exclusion lower bound is only $67,586$; even refinement cannot certify it. At weight n, the closed candidate costs $61,371$ while the unrestricted optimum is $59,332$. Thus some failures reflect a weak certificate and others reflect a genuinely suboptimal proposed structure. Improving the bound must preserve this distinction.

Raw results and source hashes are in `core_exclusion*.json` under `obst/results`. An initial full-harness failure was a metadata bug: list-valued fallback fields were passed to arithmetic counter update. Direct assignment fixed it, and the final full harness was rerun. A completed single-threshold experiment was superseded after it missed several useful cores; the published selector and every final result use the bounded portfolio. The general subquadratic objective remains unresolved.

A.10 Coupled two-gap skeleton certificates

Verification and measured limits Independent forest enumeration passed 11,880 checks. Exhaustive small weight inputs numbered 24,188, with 51,654 pair-exclusion checks and separate checks for both first and tail lower bounds. Among these are 3,288 exact closed candidates accepted by the coupled bound but rejected by B_{basic}. Complete construction passed 6,560 comparisons with independent cubic DP, covering both payload kernels and refinement modes. Larger tests at $n = 128, 512, 2048$ include random, near-uniform, geometric, single-spike, adjacent-pair, interior-pair, and varying-weight inputs. Neither favorable examples nor these exhaustive finite checks replace the preceding proofs. Raw counts and source hashes are stored in `pair_skeleton*.json`.

The nonuniform-family harness checks 31 placement/size combinations, including clustered, spread, and random placements; 28 are compared with Knuth and eight with cubic DP. All satisfy the exact residual formula and the proved certificate margins. Fourteen unit-background cases extend through $n = 65,536$. At $n = 2048$ with unit keys and selected gap weight $2n$, the earlier bound $67,586$ is replaced by $71,619$, which certifies the exact cost $71,611$. At gap weight n, the closed candidate remains suboptimal and is correctly rejected.

At $n = 16,384$ the seeded nonuniform instance has 1,023 doubled key weights. Full construction returns cost $767,240$, with witness $768,946$, first bound $780,208$, and tail bound $795,651$. Its 7,758,912 short states, 3,922,465 top states, and 20 total core states give 11,681,397 DP states versus 134,225,920 in a full Knuth table. Parameter-selection metadata is not counted a second time. There are also 42,351,013 top-matrix queries, 15,502,473 short-root candidates, 261,181 forest sorting comparisons, 49,143 forest merges, and 256,673 contour mass checks. The complete tree is reconstructed and its cost evaluated directly. The largest instance is certified by the theorem and construction, not compared with an independently executed quadratic table.

The chief unresolved issue is how to retain useful path-union information when no constant-size core can be certified. The present method does not bound the number of necessary boundary states for arbitrary mixed input. Its general worst-case construction remains quadratic.

A.11 An ordered depth-chain lower bound

Preliminary strength and cost. At $n = 256$, normalized gaps $(\text{OPT} - \text{LB})/W$ for the chain bound versus order-free Huffman were 0.15003 versus 0.22080 on a random mixed instance, and 0.00873 versus 0.49294 on an alternating instance. However, 300 ascent iterations used 1,082,368 key-depth cells including initialization, compared with 32,896 interval states for Knuth. The unique-root variant uses two passes and still leaves 254 of 256 potentially improving roots on the random instance, and 255 on the alternating instance. Only two survive on the gap-spike instance. These results identify a lower-bound component and its limitations; they do not establish a faster general exact solver.

A.12 Exact depth ordering and an arbitrary-window obstruction

Independent verification. The harness enumerated all 873,612 sequences in $\{1, \ldots, n\}^n$ for $1 \le n \le 7$. The characterization accepted exactly the independently generated BST depth sequences: respectively 1, 2, 5, 14, 42, 132, and 429. The stack recognizer agreed with direct interval scans on every sequence. Width3 and exact flow already admit four invalid sequences at $n = 6$ and 34 at $n = 7$. The positive family was checked at $t = 16$ for all 127 widths $2 \le w \le 128$: exact costs, window validity, zero flow, and the tree-cost lower bound all passed. Knuth and direct tree evaluation checked every instance; independent cubic DP also checked $w \le 24$. Another 120 cases

combine $2 \le w \le 16$ with forced prefix depths $1 \le D \le 8$. Cubic DP, Knuth, direct costs, and inspection of the optimal prefix verify the exact additive identity and preserved strict gap.

A.13 Endpoint responses: insufficiency and representation size

Independent checks and remaining question. The exact-arithmetic harness checks 875 response queries against cubic DP on 175 inputs through seven keys. Rational half-plane clipping checks exposed pieces; 725 retained planes with positive endpoint weights satisfy the Fibonacci bound. The geometric family passes 408 hinge queries, including thresholds and neighboring integers; 1,496 two-parameter probes through 33 keys; 204 full cubic comparisons; 30 all-shape checks that every optimum has the prescribed depth pair; and 680 strict endpoint inequalities. The product checks execute 887,808 Knuth root candidates. Another 304 queries expose all outer-root pieces, and 384 queries test arbitrary parameters and the global transition threshold. All 2,184 factored complete constructions agree with Knuth and direct evaluation, and construct exactly n keys. Full-quadrant envelopes through nine keys have 5, 10, 17, and 26 pieces, agreeing with the proved formula. The factored constructions use 14,492 threshold comparisons, 8,736 outer-minimum comparisons, and construct 52,232 keys in total; these categories exclude the independently executed reference solvers.

The collision search found an initial surrounding context with outer weights 16; reducing them to seven preserves the one-unit optimum difference. The published certificate independently checks domination of discarded planes and enumerates both full eight-key tree sets. It does not depend on the clipping implementation or sampled parameters. Code, raw results, and the harness source hash are in `boundary_response.py/json`.

A richer representation must preserve some information about boundary path decisions or internal alternatives discarded by the isolated value response. The present results rule out two specific shortcuts; they do not resolve or replace the full mixed-weight subquadratic objective.

A.14 A core-sparse Monge reduction with terminal penalties

Independent evidence. The condensed matrix harness passed 1,600 random rectangular products, checking 251,779 output values and exact leftmost witnesses by enumeration, and 1,600 arbitrary submatrix selections. Sparse distance-matrix products at dimensions 128, 512, and 2048 each retain seven nonzero output core entries from seven entries in each input. Core visits remain 673 in these three checks, while dimension scans grow. This exercises nontrivial sparse output, not just products with empty cores.

The complete Python constructor passed 1,134 solves, including 682 exhaustive binary-weight cases through four keys (including two empty-key inputs), 838 cubic checks, and 32 fixed-budget controls that disable early stopping. On 78 layers of the small controls, all 3,367 interval values and leftmost witnesses agree with explicit all-root enumeration. Full tree audits check 15,544 reconstructed keys. Geometric and folded weights through 64 keys remain adversarial controls; the geometric input requires all 64 layers.

The C++ address/undefined-behavior sanitizer run passed 1,278 complete solves, including 826 independent cubic checks, 12 Python compact comparisons, random weights through 48 keys, geometric/folded inputs through 64 keys, and all-zero and single-active-gap controls. It checked 18,248 reconstructed keys. The release run passed 1,362 solves and 88,904 reconstructed keys, including 1,302 independent Python Knuth comparisons. All release cases also compare against the separately implemented C++ Knuth constructor. The independent parent-array audit verifies inorder validity, unique key coverage, and direct mixed weighted cost. Three malformed/out-of-domain inputs are explicitly rejected by each binary. No sanitizer error was reported. Results include source hashes, compiler flags, and seeds.

A.15 A centered core bound for large integer baselines

Independent tests and remaining limitations The dense falsification harness `centered_layers.py` passed 1,100 complete solves, 824 cubic comparisons, 3,317 all-root cells, 1,538 baseline all-shape checks, 99,087 baseline-density checks, 685,288 tile checks, and 2,327 scale checks. It explicitly computed 4,153,053 dense states and 8,223,340 root candidates and reconstructed 11,656 keys. Baselines include 2^{256} and 2^{200} independently for keys and gaps. Zeros, concentrated gap mass, and geometric weights remain in the suite.

At 256 keys, three large-baseline fixtures have deviations 501, 505, and 504, normalized height budgets 14, 14, and 12, and maximum interior cores 6,428, 6,386, and 6,351. The earlier numerical-mass budgets were all 256. These are budget comparisons, not measured speedups: the original constructor already stops at exact fixed points. The 64-key geometric control still has $H = 64$ and all 2,016 interior density cells nonzero.

The actual condensed variant `centered_obst.py` passed 999 complete solves, 790 cubic comparisons, 32 fixed-budget controls, 1,001 intermediate values and leftmost witnesses, and 12 original-compact controls. It checked 9,992 reconstructed keys. Its large-baseline 256-key fixtures finish in ten layers with completed cores 6,624, 6,540, and 6,496. The earlier solver and its source-hashed compiled performance controls remain unchanged.

A.16 Ruler patterns and a dense-core representation barrier

Exact checks, counts, and controls The dense `ruler_layers.py` diagnostic checked 921 layer solves, 70,100 weighted-subsequence interval identities, 22,437 range maxima, 20,432 all-root cells, and 2,168 all-shape layers. It explicitly computed 5,708,220 layer states and 11,108,348 candi-

dates. Its separate full-table controls used 1,422,592 states and 2,692,718 candidates, with 96 cubic comparisons. These are diagnostic costs, not an implementation of a new fast layer constructor.

The family audit checked 68 sizes through 512, 221,284 interval/formula equalities, 202,127 instances of (33), 179,486 instances of (34), 2,974 enumerated shapes, and 13 cubic controls. At $n = 512$, all 130,816 ordinary core cells and all 129,795 valid fourth-difference cells were nonzero. The theorem uses only its explicitly proved quadratic subset. Formula and direct tree-cost checks continue through 65,536 keys without dense interval tables.

The full exchange/gate/fallback constructor passed 1,210 solves, 826 cubic comparisons, 658 checks against the prior minimum-path-length shape DP, and 386 independent enumeration checks over 950 minimum-path-length shapes. It certified 349 inputs; 512 failed the exchange condition and 349 passed that condition but failed the global gate. All fallback costs were checked. Both orientations and nonmonotone compensated key fixtures are included, alongside ordinary random and geometric failures.

A.17 Exact transfers for joint key/gap signatures

Independent checks and complete work counts The identity harness preceded the implementation. It checked all forty boundary-count intervals through block length ten, 1,365 direct costs using independent slot-expanded tree shapes, and 228 block path-independence identities. All 120 feasible short boundary/count reconstructions were checked. Another 1,000 forward/backward transfer vectors were compared with explicit word enumeration, including negative profile values and unreachable entries. The oracle does not reuse the production deque.

The final source-hashed M4 suite passes 1,023 standalone full solves, 778 cubic comparisons, 1,023 unrestricted shape-DP controls, 762 independent shape enumerations, and 1,023 independent checks of multiplicity sorting. There are 366 certificates and 657 gate rejections, including 411 in preflight. Another 1,023 portfolio solves return the same exact costs: 912 select signatures and 111 select the affine route. Eight large portfolio controls agree with the standalone constructor. Inputs include zeros, ties, large integers, geometric weights, arbitrary block values, and one-key blocks. Both scaling families at 1536 keys also match Knuth and the unrestricted shape optimizer.

Table 1 includes every block transfer performed during traceback, not just the initial forward sweep. The four-block 98,304-key solve also scans 2,228,224 source positions, pushes 311,317 deque entries, and performs 311,283 deque value comparisons, 262,146 traceback split checks, 98,304 residual contractions, and 110,590 bound-queue comparisons. Its certificate sorts thirteen multiplicity records using 26 comparisons and expands all 196,609 input weights. The sixteen-block solve uses forty-nine records, 167 sorting comparisons, 2,207,873 deque pushes, 2,415,139 deque value comparisons, and 524,310 traceback split checks. Both construct and directly evaluate all 98,304 keys. These are distinct operation categories, not a machine-instruction total. Maximum profile width counts ranks and is not a measured peak-memory byte count.

A.18 Global alternating-gap bounds and linear profile construction

Independent evidence and full-construction counts The exact word/tree harness preceded the production constructor. Recognition matches literal word enumeration on all 1,022 nonempty binary patterns through nine keys; 865 eligible patterns additionally match independent minimum-path-length tree enumeration. A separate exhaustive oracle checks 13,834 parity/tree inequalities and 69,170 mixed-weight/tree inequalities through nine keys, including exact equality of each size/parity minimum with the gap formula. The perfect-size counterexample is retained.

The full suite passes 3,157 standalone solves against Knuth, including 2,847 cubic controls and

Keys	Blocks	Window queries	Potential terms	Max. profile width
1,536	4	40,968	10,242	1,024
1,536	16	188,552	47,138	1,024
6,144	4	163,848	40,962	4,096
6,144	16	753,800	188,450	4,096
24,576	4	655,368	163,842	16,384
24,576	16	3,014,792	753,698	16,384
98,304	4	2,621,448	655,362	65,536
98,304	16	12,058,760	3,014,690	65,536

Table 1: Actual full-constructor transfer counts on certified non-affine mixed signatures, including linear-space traceback recomputation. Larger-than-1536 cases use the proved certificate and direct output-cost checks, rather than executing the dense comparison tables.

1,325 class-candidate comparisons against the unrestricted shape DP. Another 3,157 routed solves agree. There are 1,201 accepted full solves: 1,083 use the global profile equality, 116 use the older mixed gate on perfect shapes, and two are empty instances. The remaining 1,956 route to exact fallback: 86 have unsupported key alphabets, 600 unsupported gaps, 241 unsupported high counts, 905 fail profile attainment, and 124 fail the perfect-shape mixed gate. Zeros, ties, huge integers, random, geometric, near-uniform, arbitrary binary patterns with and without a baseline, and earlier affine inputs are included.

Both scaling families now have no added baseline. At 393,216 keys, a nonperiodic witness-generated pattern has 262,111 joint-signature runs but only 196,305 interval transitions; strict alternation has 393,216 runs and 196,608 transitions. They require 212,793 and 262,144 membership queries respectively, and one traceback step per stored high boundary. Each orders two multiplicity records, expands 393,216 key weights, constructs every key, and directly checks the full mixed cost. Neither uses a residual-Huffman contraction or a dense shape state. These are operation categories, not machine-instruction totals or measured peak bytes. Ten scaling portfolio controls agree; the two 1536-key cases also match Knuth and the unrestricted shape DP. Larger cases use the proved global bound and direct output-cost verification.

A.19 Threshold runs and a linear Cartesian certificate

Independent evidence and complete operation counts The threshold probe preceded the production constructor. Across all 1022 binary gap patterns through eight keys it checks 862118 indicator/tree inequalities and equality with the exhaustive optimum for every pattern. It checks 20560 weighted tree inequalities and key equality characterizations, plus 1200 mixed cubic-DP controls.

The full constructor harness checks 3000 cases against independent cubic and Knuth recurrences, including 2610 literal exhaustive controls, and another 3000 portfolio solves. The Cartesian attempt accepts 885, rejects 927 key profiles, and rejects 1188 gap profiles. The harness also checks 21844 gap vectors against explicit threshold and sorted-rank sums, 36 positive multilevel fixtures against Knuth, six larger family instances, and a 20000-key linear-height tree. Twenty independent selection scaling controls cover random, ascending, descending, duplicate, and organ-pipe weights through 100000 gaps.

At $n = 327679$, there are 327679 distinct key weights and 65538 gap levels. The exact full cost is 253331707448885247. The attempt uses 655322 Cartesian comparisons, 327679 key visits, 327680 gap visits, 327680 gap-bound scan items, and reconstructs all 327679 keys. The checked

extrema formula avoids selection on this family. There are no interval-DP cells. For the general selection controls at $N = 100000$, the sum of outer rank-level sizes is 231070; all recursive selection, partition, copy, sum, and comparison counts are separately retained in `threshold_complete.json`. These counters describe logical work, not measured peak bytes or machine instructions.

A.20 Shifted alphabetic certificates beyond key-capacity equality

Independent checks, scaling, and limitations The independent probe precedes the optimized constructor: 960 cubic controls, 3629 scalar-transfer checks, and 37500 literal tree inequalities. It certifies 138 candidates, including 24 strictly above the old bound. The constructor checks all 9840 ternary gap vectors through seven keys against cubic values and the literal compatible-pair implementation, including identical ordered gap depths. Another 960 full mixed solves match cubic and Knuth, including 560 exhaustive controls: 336 certificates and 624 fallbacks, with all 279 constant-key cases accepted. Random, near-uniform, geometric, binary, and zero inputs are included. Eight family sizes through 895 keys match Knuth; the four smallest also match cubic. A 1199-key skewed tree tests iterative traversal.

Five large constructions reach $n = 229375$, with 196608 distinct key weights, capacity deficit 32768, and exact full cost 2029411090432. The independent explicit family tree, constructed tree, and bound agree. The largest run uses 229375 compatible merges, 458749 priority pushes and pops, 6885341 priority comparisons, 5136371 heap-item comparisons, 1473827 key-sort comparisons, and 5021690 meld calls. It reconstructs every key and uses no interval-DP cells. The solver receives only p, q.

A.21 Endpoint prices and the limit of additive transfers

Independent evidence and measured costs The independent harness precedes the optimizer. It checks 2052 literal transfer identities and exact cost-difference ranks through eight keys: rank $2n - 2$, nullity three. For 60 three- and four-key instances, an independent arrangement of all alphabetic-tree and key-rank breakpoints evaluates 5111 rational points; every maximum agrees with the price search. On 600 mixed cases through 64 keys, 1712 individual oracle values match independent cubic values, and small cases also match literal trees. All 600 bounded full solves match cubic, Knuth, and direct cost: 220 certify, including 152 previous rejections, and 380 fall back. The uncapped diagnostic certifies 228, rescues 160, and completes every maximum within ten calls in this suite. There are 358 strict bound gaps and 600 clamped-transfer monotonicity controls.

Nineteen exact reduction controls produce the four-key obstruction. A further 6000 seeded positive cases find no nonintegral price maximum; this is not an integrality theorem. Intermediate prices are fractional: the obstruction search queries $y = 1/3$ and obtains $F = 68/3$. The code uses exact rationals and does not assume integral maxima.

A.22 Root/nonroot coupling beyond the transfer barrier

Independent controls, regressions, and a retained conjecture The independent case-partition harness precedes the optimized constructor. Across 840 mixed instances through eight keys, it checks 246480 literal conditioned inequalities. The bound strictly improves $K + A$ on 373 cases and is exact on 324. Random, near-uniform, geometric, and zero/binary inputs are included.

The normalized full implementation matches cubic, Knuth, and reconstructed original cost on 1040 instances through 64 keys; 720 also match literal tree enumeration. There are 800 independent bound controls and six additional family checks. The route certifies 505, including all 240 constant-size base cases, and falls back on 535. It exceeds the completed maximum transfer bound in 84 cases and rescues 11 price rejections, but loses 47 price certificates. These finite counts are not distributional guarantees, and the fixed endpoint choice is not universally superior to price optimization.

A separate diagnostic retains an unproved marked-boundary event family. It inserts a barrier at one weighted interior key, permits ordinary compatible merges on either side, and charges the key weight when a pair first consumes that barrier. Greedily taking the cheapest event fails: for $p = (0, 4, 0)$, $q = (8, 10, 2, 18)$, it returns 82 instead of 80, with strict primary choices. Inverting a unit uniform transfer gives the positive instance $p = (1, 5, 1)$, $q = (7, 9, 1, 17)$, with costs 79 and 77 respectively.

The alternative probe tries the cheapest crossing pair after every prefix of ordinary merges, completes compatible merging, and evaluates reconstructed trees with the original positional key cost. It matches all 17769 checked instances, including 15769 literal-tree controls; 4134 controls also enumerate every crossing choice without changing the best result. This is a conjectural candidate family, not an exact solver theorem. There are 20800 invalid depth reconstructions across the attempts, which are rejected. Merge charges are not assumed to equal positional costs.

Its candidate costs are not unimodal in the activation prefix. With gaps $(17, 83, 93, 11)$ and weight 92 on key 1, the ordinary merges cost 100 and 104; activation before, between, or after them gives costs 500,681,500. Thus a single-valley search is unjustified even with positive gaps and strict ordinary priorities. The full probe records 6759 nonunimodal sequences, 116 greedy failures, and 65 charge/cost disagreements. No subquadratic complexity is claimed for the probe. The proved root/nonroot certificate does not depend on this conjecture.

A.23 Exact checks and preliminary measurements

The code uses nonnegative integer weights, an independent cubic recurrence, Knuth's recurrence, and direct evaluation of reconstructed trees. The basic harness additionally enumerates all tree shapes for small sizes. CPU experiments ran on the sponsor's M4 Mac mini. The reported quantities are algorithmic counts, not inferred asymptotic bounds from elapsed time.

- Single-key contraction: 22,143 exhaustive weight assignments in $\{0, 1, 2\}$ through $n = 4$, and 6,000 seeded random instances through $n = 20$. Every cost and reconstructed tree matched the cubic oracle.
- Block theorem: 19,486 qualifying interval contractions across 4,500 seeded instances through $n = 16$ matched independent cubic costs.
- Block algorithm: 3,000 random instances through $n = 20$, including fixed and adaptive thresholds, matched the cubic oracle and tree evaluation. Additional small cases check threshold selection against every threshold.

In the heavy-separator instance all gaps and ordinary keys have integer weights in $\{1, 2, 3\}$, and 22 randomly selected keys have weight 10^6. The block algorithm retains exactly 22 keys. Its state reduction is about 12.7-fold, but this is not a claim of a 12.7-fold runtime speedup. On ordinary random mixed weights the gain is small. The single-key pass contracts every key for the tested sequence $p_i = 4^i, q_i = 4^i$, but not for most of the ratio-two sequence. Geometric integer inputs also illustrate why bit complexity must be distinguished from arithmetic-operation complexity.

A.24 Additional shape and contraction controls

The recursive constructor passed 3,000 random mixed inputs through thirty keys against cubic recurrence and direct tree evaluation. The minimum-path- length optimizer matched literal shape enumeration on 360 weighted inputs through eight keys. Its full fallback variants matched 3,000 further inputs. The affine-corridor audit includes independent candidate, regret, and complete construction checks; its parameter-dependent state bound is proved in the main text. Source-hashed records retain the individual operation categories.

A.25 Additional recurrence and certificate controls

Reference checks and implementation boundary. The dense audit `penalty_layers.py` passed 1,092 complete solves: 680 exhaustive binary-weight inputs through four keys, 384 random inputs through 32 keys, and 28 size/family fixtures through 256 keys. These include zeros, near-uniform, alternating, varying mixed, and geometric weights. All complete values and trees match Knuth and direct cost evaluation. There are 7,972 independent cubic interval checks, 8,048 all-root cell checks, 792 explicit padded-product checks, 792 all-shape truncated-layer checks, and 139,992 verified dyadic tiles. The geometric family correctly needs all 64 layers at 64 keys; no logarithmic-in-n claim is made for its exponentially large mass.

At 256 keys, the maximum interior core over all layers was 3,857 for mixed weights in $\{1, 2\}$, 6,196 for weights in $\{100, 101, 102\}$, and 5,008 for mixed weights in $\{1, \ldots, 16\}$, against 32,640 possible interior positions. These are representation counts, not runtime speedups. The audit actually computed 6,318,862 dense states and examined 12,381,033 root candidates across its full suite. It constructed 11,912 keys. Source-hashed raw data is in `penalty_layers.json`.

The lexicographic oracle matched 400 exhaustive shape-profile comparisons through eight keys. Uncapped search matched exhaustive evaluation of all allowed integer prices on 1,200 inputs. On 3,000 additional full-OBST inputs through thirty keys, both capped fallback versions matched cubic DP and direct tree costs; 87 previously unsuccessful unrestricted certificates were recovered. A separate seeded sweep recovered the $n = 1,024$, amplitude 2 input with two additional oracle calls, and the $n = 2,048$, amplitude 1 input with three. Larger tested perturbations usually exceeded the profile ceiling. This applies standard concave-price optimization to the certificate; it is not a general subquadratic result.

In the seeded 2,048-key case, the candidate costs 43,032,446; the bound rises from 43,031,347 at $\delta = 0$ to 43,032,459 at $\delta = 125$. The three extra passes use 6,144 contractions and 6,141 pair comparisons, in addition to 4,096 shape states, 6,142 transitions, 20,784 sorting comparisons, and 2,752 original merge comparisons. Knuth uses 2,098,176 interval states and 4,198,710 root candidates on that input.

Twelve full constructions reach $n = 262143$, with 262141 distinct key weights, 262144 distinct gap weights, and cost 92530765201355. The best preceding transfer bound is 91912294498251, a deficit of 618470703104. The largest run uses three alphabetic calls, 524285 compatible merges, 13894100 priority comparisons, 17365446 heap-item comparisons, and 1482337 key-sort comparisons. All scans, heap events, rebasing, and output visits are recorded; there are no interval-DP cells. Seven sizes through 511 keys match Knuth and three match cubic. Larger sizes match a separately constructed perfect tree and the closed form. Six smaller sizes additionally complete the exact endpoint price search and verify the full-class deficit independently.

Both oracles passed 50 comparisons against exhaustive length-sequence minimization, 5,000 arbitrary-multiplier bound checks against cubic OBST, and 1,050 direct comparisons of the full and shortened oracles. Forced-root marginals were additionally checked against exact root-constrained costs.

The initial probe checks 1092 alphabetic controls, 2400 mixed controls, 616500 literal tree evaluations, and 42563 pair evaluations: 2385 candidates are exact and 15 are suboptimal. Exhausting key multisets and ordered gaps over $\{0, 1\}$ and $\{1, 2\}$ through seven keys gives 7172 cases, including 14 binary failures and no $\{1, 2\}$ failures; the latter finite observation is not a theorem. There are 2036 constant-key controls, 6918 profile/depth identities, and 1421 transfer controls. All 8192 integer corners of $1000(p, q) \pm 1$ retain a gap at least 996. Padding is checked by Knuth through 256 keys (largest weight 732 bits), with cubic checks through 16.

All 512 integer corners of $100(p, q) \pm 1$ are checked; the smallest observed gap is 94, above the proved lower bound 56. With all nine weights distinct, the example

$$p = (996, 997, 1998, 999), \qquad q = (1000, 2001, 1002, 1003, 1004)$$

has full optimum 23999 and maximal transfer bound 23004. A constant supporting plane certifies that maximum in one call.

B Adversarial core search and the odd-ramp proof audit

The search fixes nonnegative integer total mass at $W = n$ and maximizes

$$\frac{\max_{0 \le h \le H} \delta(E_h)}{n\sqrt{W \lceil \log_2(n+1) \rceil}}.$$

The core includes the n finite-completion diagonal entries. All layer costs and density counts use exact integers. A whole-table fixed point, when reached, justifies repeating the same count for

subsequent layers. Single-unit and larger mass transfers, swaps, and reversals preserve the mass constraint. Simulated annealing affects only search acceptance; floating point is not used to compute costs or cores.

There are 48 starts over twelve sizes from 16 through 1,024, including non-power-of-two sizes, and 24 additional concentrated starts at six sizes. They perform 145,155 candidate evaluations in total. The first batch uses spread mixed weights, key-only weights, and uniform initial keys; the second uses spaced heavy keys, spaced ramps, clustered equal weights, and random sparse positions. Each batch first checks 232 complete layer vectors against the independent recurrence, with all-root and literal-tree checks on small inputs; selected winners receive further comparisons. Search budgets, seeds, full winning vectors, all layer counts, and operation counters are recorded. The table reports the best result at each displayed size across these starts.

$n = W$	Largest completed core	Normalized ratio
16	69	0.482152
32	185	0.417226
64	463	0.341792
128	1162	0.283691
256	2724	0.221680
512	6470	0.176604
1024	14582	0.134175

A separate diagnostic evaluates eight explicit patterns at fifteen sizes, for 120 complete layer vectors through 4,096 keys. These include a contiguous ramp, a contiguous random block, a spaced ramp, isolated key and gap mass, binary blocks, ruler-weight sampling, and random unit placements. The contiguous ramp gives 13,461, 36,276, and 98,168 completed core entries at 1,024, 2,048, and 4,096 keys, respectively. These measurements motivated the odd-ramp family with exact mass $W = n$ in Theorem 6. The proof uses the separately specified odd weights, not a fitted formula for the diagnostic's initial ramp.

The decreasing ratios in the first table alone would have been misleading as evidence for a universal near-linear bound. The actual conclusion follows from the theorem's explicit density coordinates. Search supplies candidate patterns and opportunities for falsification; it supplies neither an upper bound nor an asymptotic tightness certificate.

Independent proof controls. The new mathematical audit checks 2,652 straight and 2,652 bent rotation identities, 894 dyadic partitions through $n = 129$, and 4,277 scale bounds. The odd-ramp audit compares its subsequence recurrence with 445 literal subset enumerations and checks 388 perfect-slot embeddings against an all-root interval recurrence. There are 42,748 independent checks of the rightmost-zero formula and 17,688 checks of the closed-form density against subsequence DP. At $n = 1024$, all 1,896 cells in the proved set also match the original penalty-layer recurrence. That original-layer audit and the smaller controls evaluate 4,736,120 dense states and 8,852,066 root candidates. They are diagnostics, not operations of the sparse construction algorithm.

$n = W$	Proved positive cells	Subsequence DP	Original-layer DP
1024	1896	checked	checked
4096	15792	checked	—
16384	128736	formula only	—
65536	1039296	formula only	—

The last two rows evaluate the proved coordinate count and densities; they are not additional independent full-layer computations. The lower bound holds for every $n = 2^{2t}$, $t \geq 5$, by its proof. These records are in `core_size_search*.json`, `core_size_patterns.json`, `core_size_lemmas.json`, and `ramp_core_tightness.json`. Each records hashes of the exact sources used.

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*Dedication

Glory first be to The Holy One, Blessed be He (הַקָּדוֹשׁ בָּרוּךְ הוּא), from whom all good things come. To Him we attribute the ability that has provided for all good to come and be demonstrated anywhere in our work, and declare ourselves unworthy servants. If in any way we have erred, or our work is unsatisfactory, we ask mercy and kindness, for our effort is good and earnest.